The Chinese University of Hong Kong

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Number Theory in Function Fields

Undergraduate Research Opportunity Program Report

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Declaration

I hereby declare that this UROP Project titled **"Number Theory in Function Fields"** is the result of my own work, except where explicit reference is made to the work of others or course materials.

This work has not been submitted, either in whole or in part, for any other degree or professional qualification.

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Abstract

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Chapter 1

Back ground

1.1 some Group Theory

Definition 1.1. Let H, G be groups. Define a binary operation on $H \times K$ by

$$((h,k),(h',k')) \mapsto (hh',kk')$$

Then $H \times K$ is a group, called the direct product of H and K.

Proposition 1.1. *Let $G = H \times K$. Then $\bar{H} = \{(h,e)|h \in H\} \triangleleft G, G/\bar{H} \cong K$. Similarly, $G/\bar{K} \cong H$*

Proof. Consider the homomorphism $\pi_1 : G = H \times K \hookrightarrow K$, by $(h,k) \mapsto k$, so we have π_1 is sujective then $\mathrm{im}\pi_1 = K$, and also we have $\ker \pi_1 = \bar{H}$, so by First Isomorphism Theorem, we have $G/\bar{H} \cong K$, same proof for the second isomorphism. □

Definition 1.2. Let $G_1, ..., G_n$ be groups. Define the binary operation on

$$\prod_{i=1}^{n} G_i \times \prod_{i=1}^{n} G_i \to \prod_{i=1}^{n} G_i$$

by

$$((a_1, ..., a_n), (b_1, ..., b_n)) \mapsto (a_1 b_1, ..., a_n b_n)$$

, then $\prod_{i=1}^{n} G_i$ is a group called the direct product of $G_1, ..., G_n$.

Remark. However, if $(G_i, +)$ is abelian for any i, then we called $\prod G_i$ the direct sum instead, and denoted as

$$\bigoplus_{i=1}^{n} G_i = G_1 \oplus .... \oplus G_n$$

Proposition 1.2. *Given $N_i \triangleleft G_i, i = 1, ..., n$, Then*

$$\prod_{i=1}^{n} N_i \triangleleft \prod_{i=1}^{n} G_i \qquad \prod_{i=1}^{n} G_i / \prod_{i=1}^{n} N_i \cong \prod_{i=1}^{n} (G_i/N_i)$$

Proof. induction. □

Proposition 1.3. *Consider the group $\mathbb{Z}_m \times \mathbb{Z}_n$. The order of the element $(1,1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ is given by* $\text{lcm}(m,n)$.

Proof. let $k = |(1,1)|$, the order of $(1,1)$. Then $k(1,1) = (0,0)$ in $\mathbb{Z}_m \times \mathbb{Z}_n$. Hence $m|k, n|k$, so $\text{lcm}(m,n)|k$.

OTOH, we also have $\text{lcm}(m,n)(1,1) = (0,0) \in \mathbb{Z}_m \times \mathbb{Z}_n \implies k|\text{lcm}(m,n)$ □

Corollary 1.4. *$\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic $\iff$ m, n are relatively prime.*

Proof. ($\Leftarrow$) by proposition above and the fact that $\gcd(m,n)\text{lcm}(m,n) = mn$,

so order of $(1,1) = mn/\gcd(m,n) = mn = |\mathbb{Z}_m \times \mathbb{Z}_n|$.

($\Rightarrow$) $\forall p, q \in \mathbb{Z}_m \times \mathbb{Z}_n, \text{lcm}(m,n)(p,q) = (0,0) \implies |(p,q)||\text{lcm}(m,n)$.

In particular, $mn = |(p,q)| \leq \text{lcm}(m,n)$, which is less than mn if m,n are not relatively prime. □

Proposition 1.5. *Let $(a_1, ..., a_n) \in \prod G_i$. Suppose that $|a_i| = r_i$, then*

$$|(a_1, ..., a_n)| = \text{lcm}(r_1, ..., r_n)$$

Theorem 1.6. *Every finitely generated abelian group G is isomorphic to a direct product of cuclic group of form*

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{p_1^{n11}} \times \cdots \times \mathbb{Z}_{p_1^{n_1 l_1}} \times \mathbb{Z}_{p_2^{n_{21}}} \times \cdots \times \mathbb{Z}_{p_2^{n_2 l_2}} \times \cdots \times \mathbb{Z}_{p_k^{n_{k1}}} \times \cdots \times \mathbb{Z}_{p_k^{n_k l_k}}$$

Where $p_1 < ... < p_k$ are primes and $\{n_{ij}\}_{j=1,...,l_i}$ is a decreasing sequence of positive integers. The direct product is uniquely determined.

In particular, if the exponent of the finite abelian group G is p, and $|G| = p^d$ which means $\forall g \in G$, the order of g:$|g| = p$, then

$$G \cong \prod_{i=1}^{d} \mathbb{Z}_p$$

.

Proof. we only show the particular case: since the exponent is p. So $d_r = p$. note that $d_1|d_2|\cdots|d_r = p$ and since p is a prime, then d_i is either 1 or p, so we conclude that $d_i = p$ for any $i = 1, ..., r$. And since G is finite, there is no free part. So we can express G as $G \cong \prod_i^d \mathbb{Z}_p$ for $(G, \cdot)$, and $G \cong \bigoplus_i^d \mathbb{Z}_p$ for $(G, +)$. □

Remark. The non negative integer r is called the rank of G.

The prime powers $p_i^{n_{ij}}$ are called the elementary divisors of G.

Note that $|G_{tor}| = \prod_{i=1}^{k} \prod_{j=1}^{l_i} p_i^{n_{ij}}$

Another way to formulate the isomorphism is as

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_s}$$

, where $1 < d_1|d_2|...|d_s$, which this expression is also uniquely determined and the positive integers d_i are called invariant factors of G.

Note that $|G_{tor}| = \prod_{i=1}^{s} d_i$.

Corollary 1.7. *Let F be a field and let $F^* = F\backslash\{0\}$ be the units of F. If $G < F^*$ is a finite group, then G is cyclic.*

In particular, F^* is cyclic if F is a finite field.

Proof. By structure Theorem of Finite generated Abelian Group, we can express G as follows:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_r}$$

, where $d_i = p_i^{s_i}$ is a prime power for $i = 1, ..., r$, **Note that the prime could be non-unique!**. Let $m := \text{lcm}(d_1, ..., c_r) \leq |G| = \prod d_i$. Then, since G is a subgroup of F^*, and since G each element $\alpha \in G$ have order s.t. $|\alpha|^{\text{lcm}(d_1,...,c_r)} = 1_G$, $\forall \alpha \in G$ by above proposition 1.4. So, consider the polynomial $x^m - 1$, As F^* is a field, by factor theorem, the polynomial has at most m distinct roots. So we must have at most m such $\alpha \in G$ that satisfy the polynomial. Hence $|G| \leq m$. Since we have $m \leq |G|, |G| \leq m$, we conclude that $m = |G| = d_1 \cdots d_r \implies p_1, ..., p_r$ are all distinct primes and $G \cong \mathbb{Z}_m$ and by corollary 1.1 , G is cyclic.

Now, if F^* is finite, then the group $(F^*, +)$ is a finite subgroup of itself. By above, $(F^*, +)$ is a cyclic group. □

1.2 some Ring and Field Theory

Definition 1.3. An additive subgroup I of a ring R satisfying

$$aI \subseteq I, Ib \subseteq I \ \forall a, b \in R$$

is called an **ideal** of R.

Remark. (i) An ideal $I \subseteq R$ is in particular a subring of R

(ii) An ideal $\{0\} \subsetneq I \subset R$ is called nontrivial

(iii) An ideal $I \subsetneq R$ is called proper

Definition 1.4. Let R be a commutative ring, and $a \in R$, then

$$\langle a \rangle := \{ra | r \in R\}$$

is called **principal ideal generated by a**.

Where more generally, if $A \subset R$ is nonempty, then

$$\langle A \rangle := \{r_1 a_1 + ... + r_k a_k | n \in \mathbb{Z}_{>0}, r_i \in R, a_i \in A\}$$

is called the **ideal generated by A**

Although ideal could be complicated, principal ideal is generated by only 1 element from using "$a \cdot r$".

Note that we can think of ideal of ring as the absorbing set, where it stays in the ideal for any arithemetics on the ring R, we can think of it as the kernel of ring homomorphism too.

Proposition 1.8. *Let $\phi : R \to R'$ be a ring homomorphism, then,*

1. *If I is an ideal of R, then $\phi(I)$ is an ideal of* $\mathrm{im}(\phi)$
2. *if I' is an ideal of R', then $\phi^{-1}(I')$is an ideal of R.*

Proof. 1) suppose that I is an ideal of R, then we have , by homomorphism, that $\phi(I)$ is the addition group of $\phi(R) = \mathrm{im}(\phi)$. Now, pick arbitrary $r \in R, a \in I$, as I is ideal of R, we must have $ar \in I, ra \in I$, then, $\phi(ar) \in \phi(I), \phi(ra) \in \phi(I) \implies \phi(a)\phi(r) \in \phi(I), \phi(r)\phi(a) \in \phi(I)$, since we know that $\phi(a) \in \phi(I), \phi(r) \in \phi(R)$ arbitrarily, we conclude that $\phi(I)$ is an ideal of $\phi(R)$

2) suppose that I' is an ideal of R', then $\phi^{-1}(I')$ is an additive subgroup of R(**as pull back of homomorphism is still a homomorphism which preserve addition and multiplication of rings**).we only need to show that it satisfy the property: pick arbitrary $r \in R, a \in \phi^{-1}(I')$, then $\phi(a) \in I'$, and as I' is ideal, we have $\phi(ra) = \phi(r)\phi(a) \in I'$ so we have $ra \in \phi^{-1}(I')$, similarly, we have $ra \in \phi^{-1}(I')$. So $\phi^{-1}(I')$ is an ideal of R. □

Corollary 1.9. *If $\phi : R \to R'$ is a ring homomorphism, then* $\ker(\phi)$ *is an ideal of R.*

Proposition 1.10. *A nonzero commutative ring R is a field iff it has no proper nonzero ideals*

Proof. suppose that R is a field. Consider any proper nonzero ideal I, note that it is nonzero, we must have $\exists a \in R$ such that $a \in I$.Since I is an ideal, then pick $a^{-1} \in R$, we must have $1 = a^{-1}a \in I \implies I = R$.

Conversely, let R be nonzero commutative ring whose only ideals are $R, \{0\}$.Given a nonzero element $a \in R$, we have the principal ideal (a) contains a hence nonzero.By hypothesis, we have (a) nonzero, so $(a) = R$. as $1 \in R$, then $1 = ar' = r'a \in (a)$ $\exists r' \in R$. which we then conclude that r' is the multiplicative inverse of a.

□

Proposition 1.11. *Let F be a field, and R is z nonzero ring. Any ring homomorphism $\phi : F \to R$ is necessarily injective.*

Proof. Since R is not z zero ring, it contains $1 \neq 0$ and also by corollary, we must have $\ker \phi$ the ideal of Field F. Also note that $\phi(1) = 1 \neq 0$, so $\ker \phi$ is proper ideal of F. As F is field, $\ker \phi = \{0\}$, which follow that ϕ must be injective. □

Definition 1.5. Let R be a ring, I be an ideal of R. Then denote R/I the set of all cosets of I in $(R, +)$, namely the set of all elements of the form

$$\bar{r} := r + I = \{r + a : a \in I, \ \ r \in R\}$$

, called **the Quotient ring of R by I** if it is equipped with the operation

$$(a + I) + (b + I) = (a + b) + I$$

$$(a + I) \cdot (b + I) = ab + I$$

Remark. we already know that there is a well-defined addition on R/I is done by the showing that I is a normal group(as a ring R must be abelian).

Theorem 1.12. *Let $I \subset R$ be a subring. Then the multiplication on cosets*

$$(a + I)(b + I) = ab + I$$

is well defined iff I is an ideal.

Proposition 1.13 (canonical projection)**.** *The map $\pi : R \twoheadrightarrow R/I$, defined by*

$$\pi(r) = r + I, \ \forall r \in R$$

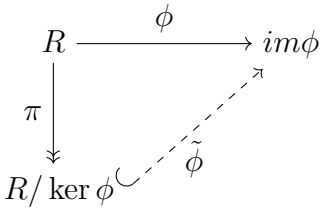
is a surjective ring homomorphism with kernel $\ker \pi = I$

Proof. first, we show the homomorphism:

1. first note that $\pi(1_R) = 1_R + I$
2. $\pi(a + b) = (a + I) + (b + I) = (a + b) + I = \pi(a + b)$
3. $\pi(a)\pi(b) = (a + I)(b + I) = ab + I = \pi(ab)$

And to show that it is surjective, consider any $r \in R$,it is surjective based on construction of map. For the kernel, by definition we have $\ker \phi = \{r \in R : \pi(r) = 0\} = \{r \in R : I = -r\} \implies r \in I \implies \ker \phi = I$ □

Theorem 1.14 (First Isomorphism Theorem)**.** *Let $\phi : R \twoheadrightarrow R'$ be ring homomorphism. Then the map $\bar{\phi} : R/\ker(\phi) \to \text{im}\phi$ defined by $\bar{\phi}(\bar{r}) = \phi(r)$ is a isomorphism such that $\phi = \bar{\phi} \circ \pi$*



which is the case that $R/\ker\phi \cong \text{im}\phi$, if we want to have $R/\ker\phi \cong R'$, ***then we MUST have ϕ be surjective***

Remark. $\forall n > 0$, we have the remainder or mod n map $\phi : \mathbb{Z} \to \mathbb{Z}_n$ induced the ring isomorphism $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$

Definition 1.6. the characteristic of a field F denoted as char(F) is defined by the smallest positive integer n such that

$$n \cdot 1_F = 1_F + ... + 1_F = 0_F$$

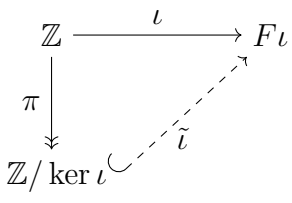
. If such positive integer n exists, the smallest such n is always a prime number p. else,0.

Corollary 1.15. *Let F be a field.*

1. *if char(F)=0, then $\exists \iota : \mathbb{Q} \hookrightarrow F$*
2. *if char(F)=p, then $\exists \iota : \mathbb{Z}_p \hookrightarrow F$*

Proof. Consider the map $\iota : \mathbb{Z} \to F$ by $n \mapsto n \cdot 1_F$. Clearly this is a ring homomorphism, and the kernel is defined by

$$\ker \iota = \begin{cases} 0 & \textbf{if } char(F) = 0 \\ p\mathbb{Z} & \textbf{if } char(F) = p \end{cases}$$



Since ι is surjective by construction, by first isomorphism theorem, we have

which ι induced $\begin{cases} \iota : \mathbb{Q} \hookrightarrow F & \text{since } \mathbb{Q} \text{ is field of quotients of } \mathbb{Z} \\ \iota : \mathbb{Z}_p \hookrightarrow F \end{cases}$ □

Recall that 'A commutative ring with unity is a field iff it has no proper nontrivial ideals'

Definition 1.7. A maximal ideal of R is a proper ideal $M \subseteq R$ s.t. $\nexists$ a proper ideal $N \subseteq R$ s.t. $M \subsetneq N \subsetneq R$

Recall also that
‘ Let $\phi : R \to R'$ be a ring homomorphism, then,

1. If I is an ideal of R, then $\phi(I)$ is an ideal of $\mathrm{im}(\phi)$
2. if I' is an ideal of R', then $\phi^{-1}(I')$is an ideal of R.’

Theorem 1.16. *Let R be commutative ring with unity. Then $M \subseteq R$ is a maximal ideal $\iff$ R/M is a field*

We skip this proof as we want to focus on the result.

Remark. 1. $n\mathbb{Z} \subseteq \mathbb{Z}$ is a maximal ideal $\iff$ $n = p$ is a prime.

2. for any field F, $\langle x \rangle \subseteq F[x]$ s a maximal ideal since the evaluation map $F[x] \to F,\ f(x) \mapsto f(0)$ induces an isomorphism $F[x]/\langle x \rangle \cong F$
3. $\langle x - a \rangle \subseteq F[x]$ is a maximal ideal for any $a \in F$ item Generally given $a_1, ..., a_n \in F, \langle x_1 - a_1, ..., x_n - a_n \rangle \subseteq F[x_1, ..., x_n]$ is a maximal ideal.

Definition 1.8. A proper ideal $I \subsetneq R$ of a commutative ring R is called a prime ideal if $ab \in I \implies a \in I$ or $b \in I$

Proposition 1.17. *Let R be a commutative ring with unity. Then a proper ideal $I \subsetneq R$ is a prime $\iff$ R/I is an integral domain.*

Corollary 1.18. *Every maximal ideal in a commutative ring with unity is a prime ideal.*

Remark. 1. In **non-commutative** ring with unity R or in commutative **rng**,

{maximal ideal} $\subset$ {prime ideal}

2. I is prime ideal $\iff$ R/I is integral domain
3. I is maximal ideal $\iff$ R/I is field
4. In integral domain $D, \{0\} \subseteq D$ is a prime ideal
5. in $\mathbb{Z}[x], \langle x \rangle \subseteq \mathbb{Z}[x]$ is prime but not maximal ($\mathbb{Z}$ is not a field(no unity))
6. in $\mathbb{Z}$, we have $\langle n \rangle \subseteq \mathbb{Z}$ is prime $\iff$ $\langle n \rangle \subseteq \mathbb{Z}$ is maximal $\iff$ n is a prime number
7. $\{PID\} \subset$ {Integral domain} $\subset$ {commutative ring with unity}

Proposition 1.19. *Let F be a field. Then for all nonzero $d, f \in F[x]$, $\exists! q, e \in F[x]$ s.t.*

$$f = qd + r$$

with $r = 0$ or $\deg r < \deg d$

Definition 1.9. Let F be a field. $f = \sum c_i a^i \in F[x]$. An element $a \in F$ is a root of f if

$$f(a) = 0$$

in F.

Proposition 1.20. *Let F be field and f be polynomial in $F[x]$, then $a \in F$ is a root of $f \iff (x-a)|f$ in $F[x]$.*

Theorem 1.21. *Let F be a field, f be a nonzero polynomial in $F[x]$.*

1. *if f has degree n, it has at most n roots in F*
2. *if f has degreen and $a_1, ..., a_n \in F$ are distinct roots of f, then*

$$f = c \cdot \prod_{i=1}^{n} (x - a_i)$$

$\exists c \in F^*$

Definition 1.10. An integral domain D is called a principal ideal domain (PID) if every ideal in D is principal

Remark. Any field F is a PID since it has no proper nontrivial ideals

Corollary 1.22. *Let F be a field, $f, g \in F[x]$. Then $\exists!$monic $d \in F[x]$ s.t.*

1. $\langle f, g \rangle = \langle d \rangle$
2. $d|f, d|g$
3. *if $h \in F[x]$ s.t. $h|f, h|g$, then $h|d$*

we say d is the ***gcd(f,g)****.*

Definition 1.11. A nonconstant polynomial $p \in F[x]$ is said to be **irreducible** if there do not exist $f, g \in F[x]$ with $\deg f, \deg g < \deg p$, such that $p = fg$

Corollary 1.23. *Every nonconstant polynomial $f \in F[x]$ may be written as $f = cp_1 \cdots p_n$ where $c \in F^*$ and each p_i is monic irreducible in $F[x]$, and the factorization is unique up to reordering of the factors. (Since $\{PID\} \subset \{UFD\}$)*

Proposition 1.24. *1. $F[x]$ is a PID Let $f(x) \in F[x]$ be nonconstant polynomial. Then TFAE:*

2. $f(x)$ is irreducible over F

3. $\langle f(x) \rangle$ is maximal

4. $\langle f(x) \rangle$ is prime

Theorem 1.25. *Let F be a field, $f(x) \in F[x]$. Then, TFAE:*

1. $F[x]/\langle f(x) \rangle$ is a field

2. $F[x]/\langle f(x) \rangle$ is an integral domain

3. p is irreducible in $F[x]$

Corollary 1.26. *Let F be field. Let $p(x) \in F[x]$ be an irreducible polynomial in $F[x]$, suppose that $p|f \cdot g, f, g \in F[x]$. Then either $p|f$ or $p|g$*

Proof. Trivial since p is irreducible, then we have $\langle p \rangle$ is prime ideal, hence$F/\langle p \rangle$ is integral domain, so by $p|fg \implies \bar{f} \cdot \bar{g} = \bar{0} \in F[x]/\langle p \rangle$, then we must have either $\bar{f} = \bar{0}$ or $\bar{g} = \bar{0} \implies p|f$ or $p|g$ □

1.3 Field Extension

Consider the two field F, F' and the ring homomorphism between then $\phi : F \hookrightarrow F'$. Note that the ring homomorphism between fields must be injective.

Also recall that if we want to enlarge the integral domain D into a field, consider D is commutative, $(0 \neq 1)$, the set $D \times (D \backslash \{0\}) := \{(a, b) : a, b \in D, b \neq 0\}$, defines the equivalence relation $(a, b) \equiv (c, d) \iff ad = bc$, then by defining the addition and multiplication on $D \times (D \backslash \{0\})$ by:

$$(a, b) + (c, d) := (ad + bc, bd)$$

$$(a, b) \cdot (c, d) = (ac, bd)$$

Then the set of equivalence classes $Frac(D)$with the above addition and multiplication forms a field with the addition identity $0 = [(0, 1)]$, multiplication identity $1 = [(1, 1)]$.The unit of the element $[(a, b)]$ of the field $Frac(D)$ is $[(b, a)]$.

Definition 1.12. If we have field F and its ring of polynomial $F[x]$, then denote $F(x) := Frac(F[x]) = \{\frac{f}{g} : f, g \in F[x], g \neq 0\}$ is called the field of **rational functions over F**

Definition 1.13. Consider a Field E, we say that F is a **subfield** of the field E, if F is a subring of E which is also a field.

Remark. (i) we say that E is an **extension** of F

(ii) we say that E/F is a **field extension**. Note that E/F IS NOT the QUOTIENT RING!!!

(iii) Let $\alpha \in E$, consider the evaluation map $\phi_\alpha : F[x] \to E$ by $f \mapsto f(\alpha)$ ($\phi_\alpha(f) = f(\alpha)$), it is a ring homomorpism and also $\phi_\alpha|_F = \mathrm{id}_F$

(iv) The image of ϕ_α is a subring

$$F[\alpha] := \mathrm{im}(\phi_\alpha) = \{f(\alpha) : f \in F[x]\}$$

in E.

(v) Since E is a field, any subring is integral domain $\implies$ $F[\alpha]$ is an integral domain, and also the subfield

$$F(\alpha) := \{\frac{f(\alpha)}{g(\alpha)} : f, g \in F[x], g(\alpha) \neq 0\}$$

in E is precisely the field of fraction of $F[\alpha]$(i.e. $F(\alpha) = Frac(F[\alpha])$)

(vi) **case 1:**$\ker(\phi_\alpha) = \{0\}$, then α is not a root of any nonzero polynomial $f \in F[x]$, then we say $\alpha \in E$ is **transcendental** over F, and ϕ_α gives an isomorphism $F[x] \cong F[\alpha]$

(vii) **case 2:** $\ker(\phi_\alpha) \neq 0$, then α is a root of some nonzero polynomial $f \in F[x]$, In this case we asy $\alpha \in E$ is **algebraic** over F. Since $F[x]$ is a PID, we have $\ker(\phi_\alpha) = \langle p \rangle$ $\exists p \in F[x]$.Then by First Isomorphism Theorem, we have

$$\bar{\phi_\alpha} : F[x]/\langle p \rangle \cong F[\alpha]$$

, Since we know that $F[\alpha]$ is an integral domain, then p is irreducible and $F[x]/\langle p \rangle \cong F[\alpha]$ is hence a field, so at last,

$$F[x]/\langle p \rangle \cong F[\alpha] = F(\alpha)$$

(viii) so that to extent the field F, we construct structure of every new elt $f(\alpha)$ where α is in E.

(ix) if α is not any root of the polynomial in $F[x]$, then $F[x] \cong F(\alpha)$, if there is a polynomial $p \in F[x] s.t. p(\alpha) = 0$, then we have $F[x]/\langle p \rangle \cong F(\alpha)$

(x) in the theorem below, i summerize the concept into short sentence:

(a) clearly if there is polynomial $f \in F[x]$ such that $f(\alpha) = 0$, then it is element in the principal ideal $\langle p \rangle$, so $p|f$

(b) it is isomorphism since they are of same structure by above.

(c) if p is irreducible in F[x] with $p(\alpha) = 0$, then first, $f \in F[x]/\langle p \rangle$ is mod p and second, by division algorithm that if $f = pg+r$, then $f = r$ in $F[x]/\langle p \rangle$, where $\deg r < p$, so any element in $F[x]/\langle p \rangle$ is of form $c_0x_0 + c_1x_1 + ... + c_{n-1}x_{n-1}$, which is also the structure of element of $F(\alpha)$. And since each $c_i \in F$, we have that there are at most $|F|^n$ elements in $F[x]/\langle p \rangle$, and we say that it is the basis of the $F[x]/\langle p \rangle$, where each element in it can be expressed as form of the basis $\langle c_0, c_1, ..., c_{n-1} \rangle$, where $c_j = 0$ if $j > \deg r, r \in F[x]/\langle p \rangle$

(d) the proposition 9 in Finite field is the direct consequences of 3 in Theorem 13.1.1

Theorem 1.27. *Let E/F be a field extension and α be an element of E.*

1. *if α is algebraic over F, then α is a root of an irreducible polynomial $p \in F[x]$ such that $p|f$ $, \forall f \in F[x]$ with $f(\alpha) = 0$*

2. *For p be an irreducible polynomial $F[x]$ of which α is a root. Then the map $\bar{\phi}_\alpha : F[x]/\langle p \rangle \to F(\alpha)$ defined by*

$$\phi(\sum_{i=1}^{n} c_j x^j + \langle p \rangle) = \sum_{j=1}^{n} c_j \alpha^j$$

is a ring isomorphism mapping $x + \langle p \rangle$to α and $a + \langle p \rangle$ to a $\forall a \in F$.

(Here $\sum_{i=1}^{n} c_j x^j + \langle p \rangle \in F[x]/\langle p \rangle$)

3. *Let p be irreducible polynomial in $F[x]$ with the root α. Then each element in $F(\alpha)$ has unique expression of the form*

$$c_0 + ... + c_{n-1}\alpha^{n-1}$$

, $c_i \in F, n = \deg p$

4. *if $\alpha, \beta \in E$ are both roots of an irreducible polynomial $p \in F[x]$,then $\exists \sigma : F(\alpha) \xrightarrow{\cong} F(\beta)$ with $\sigma(\alpha) = \beta$, $\sigma(s) = s$ $, \forall s \in F$*

Theorem 1.28 (Kronecker)**.** *If F is a field and f is nonconstant polynomial in $F[x]$, then $\exists$ field extension E/F such that $f \in F[x] \subset E[x]$ is a product of linear polynomial in $E[x]$.*

In other word, $\exists E/F$ such that:

$$f = c(x - \alpha_1) \cdots (x - \alpha_n)$$

for some $c, \alpha_i \in E$

1.4 Finite Fields

Let F be a finite field, $\mathbb{F}_p$ be finite field with p element. Then we must have char$F = p$ where p is prime. So F is a field extension of $\mathbb{F}_p$. $(F/\mathbb{F}_p)$

Proposition 1.29. *Let F be a finite field, then the number of elements of F is equal to p^n for some prime p and $n \in \mathbb{N}$.*

Proof. Note that F is a vector space over $\mathbb{F}_p$, since the cardinality of F is finite, the dimension n of F over $\mathbb{F}_p$ must necessarily be finite. Hence there exist n basis elements $\alpha_1, ..., \alpha_n$ in F such that each element of F may be expressed uniquely as

$$c_1\alpha_1 + ... + c_n\alpha_n$$

, where$c_i \in \mathbb{F}_p$. Since $\mathbb{F}_p$. has p elements, it follows that F has p^n elements. □

Theorem 1.30. *Given any prime $p, n \in \mathbb{N}$, there exists a finite field F with p^n elements.*

Proposition 1.31. *Let F be a field, f a nonzero irreducible polynomial in $F[x]$, then $F[x]/\langle f\rangle$ is a vector space of dimension* deg f *over F.*

Proof. Let $E = F[x]/\langle f\rangle$, then E is a field extension of F which contains a root α of f, namely $\alpha = \bar{x} := x + \langle f\rangle$. By theorem 1.7, $E = F(\alpha)$ and every element in E may be expressed uniquely in the form:

$$c_0 + ... + c_{n-1}\alpha^{n-1}, \quad c_i \in F, n = \deg f$$

, This shows that E is a vector space of dimension deg f over F. □

Corollary 1.32. *if F is a finite field with $|F|$ element, and f is an irreducible polynomial of degree n in $F[x]$, then the field $F[x]/\langle f\rangle$ has $|F|^n$ elements*

Proof. clearly, each element of the field $F[x]/\langle f\rangle$can be written uniquely as :

$$c_0 + c_1\alpha + ... + c_{n-1}\alpha^{n-1}, \ n = \deg f$$

, since each $c_i \in F$ and there could be total of $|F|$ different possibility of c_i for each i, and there are n such i's. So there could be total of $|F|^n$ elements in the field. □

Chapter 2

Polynomial Over Finite Fields

Throughout this paper, we set $\mathbb{F}$ be a finite field with $|\mathbb{F}| = q = p^f$ where p is the characteristic of $\mathbb{F}$.

And use direct sum as formality for abelian group.

Theorem 2.1 (Unique Factorization of polynomial)**.** *Let $\mathbb{F}$ be a field, every nonconstant polynomial in $\mathbb{F}[T]$ can be written as a product of irreducible. Moreover, the irreducible factorization is unique: if $f(T) = p_1(T)\cdots p_r(T), f(T) = q_1(T)\cdots q_s(T)$, where p_i, q_j are irreducible, then $r = s$, and after relabelling, we got $p_i(T) = c_i q_i(T), \forall i$ where $c_i \in \mathbb{F}^*$.*

Proposition 2.2 (Division Algorithm for polynomial)**.** *Let $f, g \in F[x]$ with $g \neq 0$, then there exist elements $q, r \in F[x]$ such that $f = qg + r$ where r is either 0 or* $\deg r < \deg g$.

Moreover, q and r are uniquely determined by these condition.

Proof. Define $\deg f = n, \deg g = m$, where $f = \alpha T^n + \alpha_1 T^{n-1} + ... + \alpha_n, g = \beta T^n + \beta_1 T^{n-1} + ... + \beta_n$. we do induction. Suppose that on the base case that $\deg f < \deg g$, then define $q \equiv 0, r = f$, then we have $f = qg + r = 0 + f$, where then $\deg r = \deg f <$ $\deg g$.

Suppose that we have $\deg g = m < n = \deg f$, then define

$$f' = f - \alpha\beta^{-1}T^{n-m}g$$

, it is clear that $\deg(-\alpha\beta^{-1}T^{n-m}g) = n$, so we must have $\deg f' = \deg(f + -\alpha\beta^{-1}T^{n-m}g) <$ $\deg f$.Then by induction hypothesis, $\exists! q', r'$ that $f' = q'g + r' \implies f = \alpha\beta^{-1}T^{n-m}g + q'g + r' = (\alpha\beta^{-1}T^{n-m} + q')g + r'$ with $r = 0$ or $\deg r < \deg g$, so let $q := \alpha\beta^{-1}T^{n-m} + q', r := r'$, then we have $f = qg + r$, where $r = 0$ or $\deg r < \deg g$.

For uniqueness part, assume that $\exists q'', r''$ such that $f = q''g + r''$,so $q''g + r'' = qg + r \implies (q'' - q)g = r - r''$, notice that $\deg((q'' - q)g) = \deg g + \deg(q'' - q) = \deg(r - r'')$. However, as $\deg r'', \deg r < \deg g$, so $\deg(r'' - r) \leq \deg g$, which the above equality is only possible if $\deg(q'' - q) = 0$, i.e. $q = q''$, then we have $r = r''$ as the consequences. □

Remark. Immediately, we got 2 facts:

1. $\deg(fg) = \deg(f) + \deg(g)$,$\text{sgn}(fg) = \text{sgn}(f)\text{sgn}(g)$

2. $\deg(f+g) \leq \max(\deg(f), \deg(g))$

Proof. suppose we define $g = \beta_0 T^m + ... + \beta_m$, $f = \alpha_0 T^n + ... + \alpha_n$ where $\beta_0, \alpha_0 \neq 0$. then $fg = \gamma_0 T^{n+m} + ... \gamma_{n+m}$, where $\gamma_k = \sum_{i+j=k} \alpha_i \beta_j$, so $\deg(fg) = n+m = \deg(f)+\deg(g)$,and $\text{sgn}(fg) = \alpha_0\beta_0 = \text{sgn}(f)\text{sgn}(g)$. And WLOG, assume $n \geq m$,$f + g = \alpha_0 T^n + .. + (\alpha_m + \beta_m)T^m + ... + (\alpha_0 + \beta_0) \implies \deg(f+g) = n \leq \max(\deg(g), \deg(f))$, and the equality holds when $n = m$. □

Proposition 2.3. *Suppose $g \in \mathbb{F}[T]$ and $g \neq 0$. Then $\mathbb{F}[x]/\langle g \rangle$ is a finite ring with $q^{\deg(g)}$ elements.*

Proof. since for $f \in \mathbb{F}[T]$, by division algorithm, $\exists! q, r \in \mathbb{F}[T]$ such that $f = qg + r$, as $qg \in \langle g \rangle$, so f has the representation r in $\langle g \rangle$, and the uniqueness is ensured by division algorithm. Since $\deg r < \deg g = m$, so any r can be expressed as

$$r = \gamma_0 T^{m-1} + \gamma_1 T^{m-2} + ... + \gamma_{m-1}$$

. Since any $\gamma_i \in \mathbb{F}$, and $|\mathbb{F}| = q$, so there are in total of $q^m = q^{\deg g}$ different combinations of r. So we conclude that $\mathbb{F}[T]/\langle g \rangle$ has $q^{\deg g}$ elements. □

Remark. Note that r need not to be of degree m-1, i.e. we can have $r = \gamma_{m-2} T + \gamma_{m-1}$, which means that $\gamma_0 = \gamma_1 = ... = \gamma_{m-3} = 0$

Definition 2.1. Let $g \in \mathbb{F}[T]$. If $g \neq 0$, set $|g| = q^{\deg g}$. If $g = 0$, set $|g| = 0$.

Remark. The $|g|$ DOES NOT mean the order of the element g,instead, it is a measure of the size of g. In other word, we are counting how many possible remainders of $f \in \mathbb{F}[T]$ when it is divided by g. For example if we have any $g|h$,meaning that h is in the prime ideal $\langle g \rangle$, then $\exists f \in \mathbb{F}[T]$ s.t. $h = fg$, then $\deg h = \deg g + \deg f$, so $|h| = q^{\deg h} = q^{\deg g + \deg f} = |g| \cdot |f|$, so $|g| \mid |h|$.

Corollary 2.4. *Let $f, g \in \mathbb{F}[T]$, then we have*

1. $|fg| = |f| \cdot |g|$

2. $|f+g| \leq \max(|f|, |g|)$

Proof. $|fg| = q^{\deg(fg)} = q^{\deg f + \deg g} = q^{\deg f} q^{\deg g} = |f| \cdot |g|$, and $|f+g| = q^{\deg(f+g)} = q^{\max(f,g)} = \max(|f|, |g|)$. □

Proposition 2.5. *The group of units in $\mathbb{F}[T]$ is $\mathbb{F}^*$. In particular, it is a finite cyclic group with $q-1$ elements.*

Proof. consider any $f \in \mathbb{F}[T]$, if f is a unit, we need $fg = 1$ for some $g \in \mathbb{F}[T]$. Note that $fg = 1$ is possible $\iff$ f, g is constant since $\deg f + \deg g = \deg(fg) = \deg 1 = 0 \implies \deg f = 0, \deg g = 0$. So the set of unit of $\mathbb{F}[T]$ is $\mathbb{F}^*$. Since any finite subgroup of $\mathbb{F}^*$ must be cyclic, in addition, if the field is finite, $\mathbb{F}^*$ is cyclic finite group. So we have $\mathbb{F}^* < \mathbb{F}^*$ and $|\mathbb{F}^*| = |\mathbb{F}| - 1 = q - 1$. □

Now, since by the unique factorization of polynomials, we can express any polynomial in $\mathbb{F}[T]$ as $f = p_1 \cdots p_t$, where p_i is irreducible. And the uniqueness property can be sharpen into following statement: $\forall f \in \mathbb{F}[T], f \neq 0, f = \alpha P_1^{e_1} \cdots P_t^{e_t}$, where $\alpha \in \mathbb{F}^*$, each $P_i \neq P_j$ is a monic polynomial for $i \neq j$, each e_i is non-negative integer.

To continue our journey, i must introduce an important tool: the Chinese Remainder Theorem.

Proposition 2.6 (Chinese Remainder Theorem for polynomial)**.** *Let* $m_1, ..., m_t \in \mathbb{F}[T]$,*s.t.* $\gcd(m_i, m_j) = 1$ *for* $i \neq j$.*Let* $m = m_1 \cdots m_t$, *and* $\pi : \mathbb{F}[T]/\langle m\rangle \hookrightarrow \mathbb{F}[T]/\langle m_i\rangle$ *be the natural homomorphism map. Then the map*

$$\phi : \mathbb{F}[T]/\langle m\rangle \to \mathbb{F}[T]/\langle m_1\rangle \oplus \cdots \oplus \mathbb{F}[T]/\langle m_t\rangle$$

given by

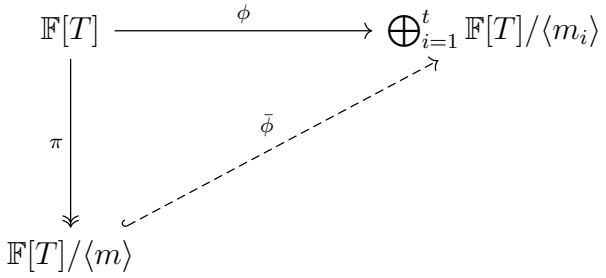
$$\phi(a) = (\phi_1(a), ..., \phi_t(a))$$

is a ring isomorphism.

i.e.

$$\mathbb{F}[T]/\langle m\rangle \cong \bigoplus_{i=1}^{t} \mathbb{F}[T]/\langle m_i\rangle$$

Proof. we want to show the isomorphism by using first isomorphism theorem, hence we need to show the following relations:



first we show that π is a canonical projection:

Consider $\pi : \mathbb{F}[T] \twoheadrightarrow \mathbb{F}[T]/\langle m_i\rangle$ by $f \mapsto f + \langle m_i\rangle$.Clearly, for any $f + \langle m_i\rangle, \exists f \in \mathbb{F}[T]$ such that $\pi_i(f) = f + \langle m_i\rangle$, so it is surjective, which means that $\mathrm{im}\pi = \mathbb{F}[T]/\langle m_i\rangle$.

For the kernel, we claim that $\ker\phi = \langle m_i\rangle$: for $f \in \ker\phi = \{f \in \mathbb{F}[T] | \phi(f) = 0\} = \{\mathbb{F}[T] | f \in \langle m_i\rangle, \forall i \in [1, t]\} \subset \langle m\rangle$ suppose that $f \in \langle m\rangle$, then $f = hm_1 \cdots m_t$ for

some $h \in \mathbb{F}[T]$, which $\phi(h) = (0, ..., 0) = 0 \implies f \in \ker\phi$. So we can conclude that $\ker\phi = \langle m \rangle$.

For the ring homomorphism: consider $a, b \in \mathbb{F}[T]/, \phi(f+g) = (f+g) + \langle m \rangle = (f + \langle m \rangle) + (g + \langle m \rangle) = \phi(f) + \phi(g)$, and $\phi(fg) = (fg) + \langle m \rangle = (f + \langle m \rangle)(g + \langle m \rangle) = \phi(f)\phi(g)$.

So π is a canonical projection.

To show that the im$\phi = \bigoplus_i \mathbb{F}[T]/\langle m_i \rangle$, first consider the map $\phi(a) = (a + \langle m_1 \rangle, ..., a + \langle m_t \rangle)$, let an arbitrary $\phi(a) \in \bigoplus_i \mathbb{F}[T]/\langle m_i \rangle$, then first consider the $M_i := \frac{m}{m_i}$, $\implies$ $\gcd(m_i, M_i) = 1$, then we would have the following two congruent:

$$M_i M_i' \equiv 1 \mod m_i, \quad M_i M_i' \equiv 1 \mod m_j, \quad i \neq j$$

and M_i' is the inverse of M_i.

so we can express a in terms of Mi :

$$a = a_1 M_1 M_1' + ... + a_t M_t M_t'$$

and $a = a_1 M_1 M_1' + ... + a_t M_t M_t' \equiv 0 + ... + a_i \cdot 1 + ... + 0 \mod m_i \implies a \equiv a_i \mod m_i \ \forall i$. So for any arbitrary $(a_1, ..., a_t) = (a + \langle m_1 \rangle, ..., a + \langle m_t \rangle) \in \bigoplus_i \mathbb{F}/\langle m_i \rangle, \exists a \in \mathbb{F}[T]$ s.t. $(a_1, ..., a_t) = \phi(a)$.We then can conclude that im$\phi = \bigoplus_i \mathbb{F}[T]/\langle m_i \rangle$.

Hence by First isomorphism Theorem,

$$\mathbb{F}[T]/\langle m \rangle \cong \bigoplus_{i=1}^{t} \mathbb{F}[T]/\langle m_i \rangle$$

□

Before the next corollary, i introduce a useful tool for the restriction of ring isomorphism to unit group isomorphism:

Proposition 2.7. *Suppose that $\phi : A \to B$ is a ring isomorphism, then the same map on the multiplicative group $(A^*, \times), (B^*, \times)$ is also a group isomorphism.*

Proof. First, to show the homomorphism, by Ring homomorphism, we already got: $\phi(1_A) = 1_B \implies \phi(1_{A^*}) = 1_{B^*}$ as they got the same identity. And $\forall x, y \in A^* \implies x, y \in A$, $\phi(xy) = \phi(x)\phi(y)$. So it is a group homomorphism.

For injectivity,$\forall x, y \in A^*$ if $\phi(x) = \phi(y)$, since $x, y \in A \implies$ by injectivity of the ring map, $x = y$.

Finally, suppose that $\forall y \in B^* \implies y \in B$, so $\exists x \in A$ s.t. $y = \phi(x)$. As $y \in B^*, \exists y^{-1} \in B^* \implies y^{-1} \in B$, by surjectivity of ring again, $\exists w \in A$ s.t. $y^{-1} = \phi(w)$. Hence $\phi(xw) = \phi(x)\phi(w) = yy^{-1} = 1_{B^*} \implies xw = \phi^{-1}(1_{B^*}) = 1_{A^*}$, which means that the $x, w \in A^*$ and also $w = x^{-1}$. □

Corollary 2.8. *The same map ϕ restricted to the unites of $\mathbb{F}[T], (\mathbb{F}[T])^*$, gives rise to a group isomorphism*

$$(\mathbb{F}[T]/\langle m\rangle)^* \cong \prod_{i=1}^{t}(\mathbb{F}[T]/\langle m_i\rangle)^*$$

Method 1. By the above proposition, since $\mathbb{F}[T]/\langle m\rangle \cong \bigoplus_{i=1}^{t}\mathbb{F}[T]/\langle m_i\rangle$ is ring isomorphism,

then immediately, we have $(\mathbb{F}[T]/\langle m\rangle)^* \cong (\prod_{i=1}^{t}\mathbb{F}[T]/\langle m_i\rangle)^*$,

which is just $(\mathbb{F}[T]/\langle m\rangle)^* \cong \prod_{i=1}^{t}(\mathbb{F}[T]/\langle m_i\rangle)^*$ □

Method 2. we have the map $\phi : f + \langle m\rangle \mapsto f + \langle m_i\rangle$, we already got that $\phi(ab) = \phi(a)\phi(b)$ by above ring homomorphism. Suppose that $1 \in (\mathbb{F}[T]\langle m\rangle)^*$ is identity, so $\forall f \in (\mathbb{F}[T]\langle m\rangle)^*, 1 \cdot f = f \cdot 1 = f.$, we have $\phi(1) = (1 \mod m_1, ..., 1 \mod m_t)$, which is identity on the multiplicative group$\prod_{i=1}^{t}(\mathbb{F}[T]/\langle m_i\rangle)^*$

For the injectivity, since $\ker\phi = \{a \in (\mathbb{F}[T]/\langle m\rangle)^* | \phi(a) = 1\} = \{a \equiv 1 \mod m_i \forall i\} = \{a \equiv (1, ..., 1) \mod m\} = \{1\}$.

For surjectivity, since for any unit in $\mathbb{F}/\langle m_i\rangle$ can be lifted to $\mathbb{F}/\langle m\rangle$.

□

Since now that we have already constructed the valuable tool in function field, for $f \in \mathbb{F}[T]$ be non-zero and not a unit, suppose that $f = \alpha P_1^{e_1} \cdots P_t^{e_t}$ is its prime decomposition, then, by above propositions, we have $\gcd(P_i, P_j) = 1 \forall i \neq j$, so

$$(\mathbb{F}[T]/\langle f\rangle)^* \cong (\mathbb{F}[T]/\langle P_1^{e_1}\rangle)^* \times \cdots \times (\mathbb{F}[T]/\langle P_t^{e_t}\rangle)^* := \prod_{i=1}^{t}(\mathbb{F}[T]/\langle P_i^{e_i}\rangle)^*$$

Where this isomorphism reduced the task to that of determining the structure of the groups intead of rings. First, we state a useful proposition that determine the element and group structure of our group:

Proposition 2.9. *Let $P \in \mathbb{F}[T]$ be an irreducible polynomial. Then, $(\mathbb{F}[T]/\langle P\rangle)^*$ is a cyclic group with $|P| - 1$ elements.*

Proof. Since $\mathbb{F}[T]$ is a field and hence a PID,we have $\langle P\rangle$ is a maximal ideal and also a prime ideal(by P is irreducible), then $\mathbb{F}[T]/\langle P\rangle$ is a field using theorem in section 1. As $P \in \mathbb{F}[T]$, by proposition 2.2, it is a finite field with $|P|$ elements and using proposition 2.3 it is a finite cyclic group .By excluding the addition identity, $(\mathbb{F}[T]/\langle P\rangle)^*$ has $|P| - 1$ elements. □

We now consider the situation when $e > 1$, here we consider something that is different at $\mathbb{F}[T]$ from the situation in $\mathbb{Z}$. If p is an odd prime number in $\mathbb{Z}$, then it is a standard result that $(\mathbb{Z}/\langle p^e\rangle)^*$ is cyclic for all positive integer e. To take care of the structure of the field $\mathbb{F}[T]/\langle P_i^{e^i}\rangle$, proposition 2.7 would be useful.

Proposition 2.10. *Let $P \in \mathbb{F}[T]$,be an irreducible polynomial and e a positive integer. The order of $(\mathbb{F}[T]/\langle P^e\rangle)^*$ is $|P|^{e-1}(|P|-1)$. Let $\ker\varphi_e$ be the kernel of the natural homomorphism $\varphi_e : (\mathbb{F}[T]/\langle P^e\rangle)^* \to (\mathbb{F}[T]/\langle P\rangle)^*$. It is a p-group of order $|P|^{e-1}$. As e tends to infinity, the minimal number of generators of $\ker\varphi$ tends to infinity.*

Remark. In this proposition, we want to show that different from number theory, in finite field, is it very hard to have a primitive root for the group $(\mathbb{F}[T]/\langle P^e\rangle)^*$.

Before beginning the proof, I would personally give some facts for insight of this proposition.

Note.
1. If a group G is cyclic, then any subgroup of G must also be cyclic.
2. So, if there exists a non-cyclic subgroup $H < G$, then G must not be cyclic.
3. the primitive root is an element in a group such that it generate the whole group. In other word, it is a generator of the group.

Hence, the idea of proposition 2.7 is to demonstrate the element of generators of kernel of the map is bounded below by some variables, which when e is sufficiently large, it is hard for kernel to be a cyclic group, hence the super group $\mathbb{F}[T]/\langle P^e\rangle$ is forced to be non-cyclic, hence showing non-existence of primitive root for large e.

We will separate the proof into 3 parts:

1. find the order of the group $(\mathbb{F}[T]/\langle P^e\rangle)^*$
2. shows that the kernel is a p-group of order $|P|^{e-1}$
3. shows the number of elements of generators of kernel is bounded below for each $e \in [1,\infty)$
4. In conclusion, for large e, the kernel must not be cyclic.
5. Hence showing the non-existence of primitive roots (generator of $\mathbb{F}[T]/\langle P^e\rangle$).

Before proving the proposition, I would like to introduce a useful tool that will help show the uniqueness of the maximal ideal:

Definition 2.2. For a group G, we said the p is the exponent of the group G if p is the smallest integer p such that $|g|^p = e,\ \forall g \in G$.

Proposition 2.11. *Suppose that R is a Ring. M is maximal ideal of R. Then M is a unique maximal ideal of $R \iff M$ is the set of all non-unit of R.*

Proof. ($\Longrightarrow$) Suppose that M is a unique maximal ideal. Assume that S is the set of non-unit of R. Clearly, for $x \in S$, $x \in \langle x\rangle \subset M \implies x \in M \implies S \subset M$.

($\Longleftarrow$)Suppose that $x \in M$, then we must have x is not a unit else ,$\exists r \in R$s.t. $rx = 1$, which means that $x \in \langle 1\rangle$ and hence $\langle M\rangle = R$. So any non-unit $x \in M, x \in S \implies M \subset S$.So we conclude $M = S$. □

Proof of proposition 2.7. First we need to show that the ring $\mathbb{F}[T]/\langle P^e\rangle$ is a local ring, which means it has a unique maximal ideal. To find the maximal ideal, since $\mathbb{F}[T]$ is a field ,and hence a PID, so any generators of $\mathbb{F}[T]$ is a principle ideal, which is generated by single polynomial in $\mathbb{F}[T]$.Note that P is an irreducible polynomial, so any polynomial that is divisible by P is $P^k, k = 1, 2, ..., e$. So, we do have the relation on the principle ideal:

$$\langle P^e\rangle \subseteq \langle P^{e-1}\rangle \subseteq \cdots \subseteq \langle P\rangle \subseteq \langle 1\rangle = \mathbb{F}[T]$$

. Clearly, for any $r_k P^i + \langle P^e\rangle \in \langle P^i\rangle/\langle P^e\rangle$, $r_k P^i + \langle P^e\rangle \in \langle P^j\rangle/\langle P^e\rangle$ for any j that $i \geq j, k \in \mathbb{F}[T]$. Hence,

$$\langle P\rangle/\langle P^e\rangle \supset \langle P^2\rangle/\langle P^e\rangle \supset \cdots \supset \langle P^{e-1}\rangle/\langle P^e\rangle$$

is the relation of ideals of $\mathbb{F}[T]/\langle P^e\rangle$. Clearly, $\langle P\rangle/\langle P^e\rangle$ is the maximal ideal. To show that this maximal ideal is unique,we show that it is a set of all non-unit in the ring. First, we claim that the set of unit is $S' = \{f + \langle P^e\rangle | \gcd(f, P^e) = 1\}$.By Bezout-identity, since $\gcd(f, P^e) = 1$, so $\exists u, v \in \mathbb{F}[T]$ that $fu + P^e v = 1 \implies [fu] + 0 = [1] \in \mathbb{F}[T]/\langle P^e\rangle$. So,$S' \supset \{f + \langle P^e\rangle | \gcd(P^e, f) = 1\}$. Suppose that f is a unit, then $\exists [g] \in \mathbb{F}[T]/\langle P^e\rangle$,$[fg] = [f][g] = 1 \implies fg + rP^e = 1$, $\exists r \in \mathbb{F}[T]$ hence $\gcd(f, P^e) = 1$ so $\{f + \langle P^e\rangle | \gcd(f, P^e) = 1\} \supset S' \implies S' = \{f + \langle P^e\rangle | \gcd(f, P^e) = 1\}$. So, by setting S be the set of non-unit, then $f \in S \iff \gcd(f, P^e) \neq 1$.Note that the only factor of P^e is $1, P, ..., P^e$, if $\gcd(f, P^e) \neq 1 \implies \gcd(f, P^e) = P^k, \exists k \geq 1 \implies P|f \implies f \in \langle P\rangle$. So $S \subset \langle P\rangle$. The case $\langle P\rangle \subset S$ is trivial.SO, $S = \langle P\rangle$,hence any non-unit $f + \langle P^e\rangle \in \mathbb{F}[T]/\langle P^e\rangle$ is the set $\langle P\rangle/\langle P^e\rangle$, we then conclude that it is a unique maximal ideal by above proposition.

Now we show that the cardinality of $\langle P\rangle/\langle P^e\rangle$ is $|P|^{e-1}$. Consider $f \in \langle P\rangle/\langle P^e\rangle$, by division algorithm, $\exists! u, r \in \mathbb{F}[T]$, that

$$f = uP^e + r, \quad \deg r < \deg P^e$$

so $\deg r \leq \deg P^{e-1}$. By proposition 2.2, $\deg r$ has $|P|^{e-1}$ possible combinations hence $|\langle P\rangle/\langle P^e\rangle| = |P|^{e-1}$. Then,

$$\begin{aligned}(\mathbb{F}[T]/\langle P^e\rangle)^* = (\mathbb{F}[T]/\langle P^e\rangle)\backslash(\langle P\rangle/\langle P^e\rangle) \implies |(\mathbb{F}[T]/\langle P^e\rangle)^*| &= |(\mathbb{F}[T]/\langle P^e\rangle)| - |(\langle P\rangle/\langle P^e\rangle)| \\ &= |P|^e - |P|^{e-1} = |P|^{e-1}(|P| - 1)\end{aligned}$$

For the second part,consider the multiplicative group canonical projection map:

$$\phi_e : (\mathbb{F}[T]/\langle P^e\rangle)^* \twoheadrightarrow (\mathbb{F}[T]/\langle P\rangle)^*$$

$$f + \langle P^e \rangle \mapsto f + \langle P \rangle$$

Clearly it is surjective by construction, and is a group homomorphism. Define the kernel of this map by $\ker \phi_e = \{\bar{f} \in (\mathbb{F}[T]/\langle P^e \rangle)^* | \phi_e(\bar{f}) = 1_{(\mathbb{F}[T]/\langle P \rangle)^*}\} = \{f + \langle P^e \rangle | f \equiv 1 \mod P\} = \{1 + \langle P \rangle + \langle P^e \rangle\}$. Consider $f \in \ker \phi_e \implies f = qP^e + r,\ \exists! q, r,\ \deg r < \deg P^e. \implies |\ker \phi_e| = |P|^{e-1}$.

For the third part, consider the simple case $e = 2$: The kernel $\ker \phi_2 = \{f + \langle P^2 \rangle | f \equiv 1 \mod P\} = \{1 + \langle P \rangle + \langle P^2 \rangle\} = \{1 + bP, b \in \mathbb{F}[T]\}$, consider arbitrary $a \in \ker \phi_2$, then

$$a^p = (1 + bP)^p = 1 + \sum_{k=1}^{p-1} \binom{p}{k} (bP)^k + b^p P^p$$

Since $\text{char}(F) = p$, and $p | \binom{p}{k}$, $\forall p - 1 \geq k \geq 1 \implies \binom{p}{k} = 0$ for $k \in [1, p-1]$, so

$$a^p = (1 + bP)^p = 1 + b^p P^p \overset{p \geq 2}{\equiv} 1 \mod P^2$$

In the other word, any element in the $\ker \phi_2$ must have order l where $p | l$. And since p is a prime, so if $\exists k, |a|^k = e\ , k | p$, then $k = 1$ or p so $k = p$. Hence, the group $\ker \phi_2$ has exponent p. and note that $|\ker \phi_2| = |P|^{2-1} = |P| = q^{\deg P} = (p^f)^{\deg P} = p^{f \deg P}$.

Clearly, $\ker \phi_2$ is finite and abelian as it is inherited from the ring. So by the Fundamental theorem of Finitely Generated Abelian Group,

$$\ker \phi_2 \cong \bigoplus_{i=1}^{f \deg P} \mathbb{Z}/p\mathbb{Z}$$

By induction, consider the general case e, pick s as smallest integer such that $p^s \geq e$, then choose arbitrary $a \in \ker \phi_e$,

$$a^{p^s} = (1 + bP)^{p^s} = 1 + (bP)^{p^s} + \sum_{k=1}^{p^s - 1} \binom{p^s}{k} (bP)^k$$

similar to the above case, for $k \in [1, p^s - 1]$,$p | \binom{p^s}{k}$, $\implies \binom{p^s}{k} = 0, k \in [1, p^s - 1]$, so we left with

$$a^{p^s} = 1 + (bP)^{p^s} \equiv 1 \mod P^e$$

so again we show that any element in $\ker \phi_e$ has that $|a|^{p^s} = 1$, if $|a|^l = 1 \implies l | p^s \implies l = 1, p, ..., p^s$. But the case that $l = 1, 2, ..., p^{s-1}$ is not possible as we assume s to be the smallest s such that $e \leq p^s$, so any $r < s \implies p^r < e, \implies P^r \not\equiv 1 \mod P^e$. So p^s is the exponent of $\ker \phi_e$, and also $|\ker \phi_e| = |P|^{e-1} = p^{(e-1) f \deg P}$. Again, then $\ker \phi_e$ is a p-group and a finite abelian group. So by Fundamental Theorem of Finitely

Generated Abelian Group,

$$\ker \phi_e \cong \bigoplus_{k=1}^{(e-1)f \deg P} \mathbb{Z}/p^s\mathbb{Z} \cong (\mathbb{Z}/p^s\mathbb{Z})^{(e-1)f \deg P}$$

.

Let d be the minimal number of generators of $\ker \phi_e$. (In here, the number of generators means how many elements in the kernel to generate the whole group, but not the generators defined in group theory.) Consider the map

$$\Psi : (\mathbb{Z}/p^s\mathbb{Z})^d \longrightarrow \ker \phi_e$$

To show that it is a surjective map, intuitively, pick an arbitrary element in $(\mathbb{Z}/p^s\mathbb{Z})^d$, consider the d generators in the kernel, to construct those generators, we use "combination" to construct each d in the kernel.

$$\Psi(a_1, ..., a_d) = g_1^{a_1} \cdots g_d^{a_d}$$

, where the $g_1, ..., g_d$ are the elements in kernel that generate the group.

Note that, for all i, a_i is of order p^s in $(\mathbb{Z}/p^s\mathbb{Z})^*$, and p^s is the exponent of the kernel, which means that $g_i^{p^s} = 1,\ \forall i = 1, ..., d$, so shifting input by p^s doesn't change the value of output.

For subjectivity, since any element x_i in the kernel can be written as the combination of $g_1^{a_i^{(1)}} \cdots g_d^{a_i^{(d)}}$, so any element in the co-domain has a pre-image in the domain.(Note here we use multiplication to write out the element of the kernel as in polynomial we use multiplication, although the kernel is an abelian group)

. So we obtained the inequality

$$|(\mathbb{Z}/p^s\mathbb{Z})^d| \geq |\ker \phi_e| \implies p^{ds} \geq p^{(e-1)f \deg P} \implies d \geq \frac{(e-1)f \deg P}{s}$$

Since we do have another inequality $p^s \geq e \implies p^{s-1} < e \implies s < \log_p e + 1 \implies \frac{1}{s} > \frac{1}{\log_p ep}$ so

$$d \geq \frac{(e-1)f \deg P}{\log_p ep} \xrightarrow{d\to\infty} \infty$$

since $\log_p ep$ converges to infinity slower than e.Which shows that when the number e goes to infinity, the minimal generator of the kernel goes to infinity, which is essential in topics about primitive root.

□

Definition 2.3. let $f \in \mathbb{F}[T]$ be a non-zero polynomial. Then $\Phi(f)$ is defined to be the number of elements in the group $(\mathbb{F}[T]/\langle f \rangle)^*$

Remark. We can also define the Euler $\phi-$ function in other way: suppose that $\{r \in \mathbb{F}[T] | \deg r < \deg f\}$ be the set of representatives for $\mathbb{F}[T]/\langle f \rangle$.

Such r represents a unit in $\mathbb{F}[T]/\langle f \rangle \iff \gcd(r, f) = 1$. Thus $\Phi(f)$ can also be the number of non-zero polynomials of degree less than $\deg f$ and relatively prime to f.

Proposition 2.12.

$$\Phi(f) = |f| \prod_{P|f} (1 - \frac{1}{|P|})$$

Proof. Let $f = \alpha P_1^{e_1} P_2^{e_2} \cdots P_t^{e_t}$ be the prime decomposition of f. By below of corollary 2.2, we have $(\mathbb{F}[T]/\langle f \rangle)^* \cong \prod_{k=1}^{t} (\mathbb{F}[T]/\langle P_k^{e_k} \rangle)^*$ and by proposition 2.7, we can compare their cardinality where

$$\begin{aligned}
\Phi(f) = |(\mathbb{F}[T]/\langle f \rangle)^*| &\overset{\text{by} \cong}{=} |\prod_{k=1}^{t} (\mathbb{F}[T]/\langle P_k^{e_k} \rangle)^*| = \prod_{k}^{t} |P_k|^{e_k - 1}(|P_k| - 1) \\
&= \prod_{k}^{t} |q^{\deg \alpha_{(=0)}}| \cdot |P_k|^{e_k} \prod_{k}^{t} (1 - \frac{1}{|P_k|}) \\
&= |f| \prod_{P|f} (1 - \frac{1}{|P|})
\end{aligned}$$

□

Proposition 2.13 (Euler's Little Theorem)**.** *If $f \in \mathbb{F}[T], f \neq 0$, and $a \in \mathbb{F}[T]$ is relatively prime to f, i.e. $\gcd(a, f) = 1$, then*

$$a^{\Phi(f)} \equiv 1 \ , \mod f$$

Proof. By above proposition, we know that $|(\mathbb{F}[T]/\langle f \rangle)^*| = \Phi(f)$, and $\bar{a} = a + \langle f \rangle \in (\mathbb{F}[T]/\langle f \rangle)^*$,since the order of any element in the group must be divisor of the cardinality of the group,

$$\bar{a}^{\Phi(f)} = \bar{a}^{|(\mathbb{F}[T]/\langle f \rangle)^*|} = \bar{1}$$

. Hence writing out,

$$a^{\Phi(f)} \equiv 1 \mod f$$

□

Corollary 2.14 (Fermat's Little Theorem)**.** *Let $P \in \mathbb{F}[T]$ be irreducible and $a \in \mathbb{F}[T]$ be a polynomial not divisible by P. Then,*

$$a^{|P|-1} \equiv 1 \ , \mod P$$

Proof. Since P is irreducible, then $\gcd(a, P) = 1 \iff a \not| P, P \not| a. \iff a \notin \langle P \rangle$. By above proposition,

$$\bar{a}^{|P|-1} \overset{prop.2.6}{=} \bar{a}^{(\mathbb{F}[T]/\langle P \rangle)^*} = \bar{a}^{\Phi(P)} = \bar{1}$$

, which is just

$$a^{|P|-1} \equiv 1 \mod P$$

□

Proposition 2.15 (Lagrange's Theorem)**.** *Let $P \in \mathbb{F}[T]$ be irreducible and of degree d. Suppose X is an indeterminate. Then,*

$$X^{|P|-1} - 1 \equiv \prod_{0 \le \deg f < d} (X - f) \ , \mod P$$

Proof. Recall that if we set, by division algorithm that, for $g \in \mathbb{F}[T]/\langle P \rangle$, $g = qP + f$, then the set of representatives for this quotient ring is $\{f \in \mathbb{F}[T] \mid \deg f < \deg P = d\}$. If we throw away $f = 0$,we get the set of representatives for $(\mathbb{F}[T]/\langle P \rangle)^*$. And $\#\{f \in \mathbb{F}[T] \mid \deg f < d\} = |P| - 1$ by proposition 2.3, 2.6. As the set of root of $X^{|P|-1} - \bar{1}$ is $\{f \in \mathbb{F}[T] \mid \deg f < d\}$, and the set of root of $\prod_{0 \le \deg f < d}(X - \bar{f})$ is $\{\mathbb{F}[T] \mid \deg f < d\}$ i.e. they have the same set of root. We consider

$$g(X) = (X^{\Phi(P)} - \bar{1}) - (\prod_{0 \le \deg f < d} (X - \bar{f}))$$

, notice that first $\deg(g) \le |P| - 2$, and $g(X)$ has $|P| - 1$ roots, which is possible only if $g(X) \equiv 0$. Hence $X^{|P|-1} - \bar{1} = \prod_{0 \le \deg f < d}(X - \bar{f})$, which when writen into modulo case, is

$$X^{|P|-1} - 1 \equiv \prod_{0 \le \deg f < d} (X - f) \mod P$$

□

Corollary 2.16. *Let $d|(|P| - 1)$. The congruence $X^d \equiv 1 \mod P$ has exactly d solutions. Equivalently, the equation $X^d = \bar{1}$ has exactly d solutions in $(\mathbb{F}[T]/\langle P \rangle)^*$*

Proof. Since $d \mid (|P| - 1) \implies (X^d - 1) \mid (X^{|P|-1} - 1)$ and by proposition 2.11, $X^{|P|-1} - \bar{1} = \prod_{0 \le \deg f < d}(X - \bar{f})$, which is written as the product of linear factors.Hence the polynomial $X^d - 1$ must also be written into product of d linear factors, hence $X^d = \bar{1}$ hence $X^d \equiv 1 \mod P$. □

Corollary 2.17 (Wilson's Theorem)**.** *With the same notation,*

$$\prod_{0 \le \deg f < \deg P} f \equiv -1 \ , \mod P$$

Proof. Set $X = 0$ in proposition 2.11, we got

$$-1 \equiv \prod_{0 \le \deg f < d} -f \equiv (-1)^{|P|-1} \prod_{0 \le \deg f < d} f \mod P$$

If $\mathrm{char}(\mathbb{F}) = 2k+1 \implies |P|-1 = 2k \implies (-1)^{2k} \prod_{0 \le \deg f < \deg P} f = \prod_{0 \le \deg < d}(-1) \prod_{0 \le \deg f < d} f = \prod_{0 \le \deg f < d} f$ and the result follows.

If $\mathrm{char}(\mathbb{F}) = 2$,then $-1 = 1 \implies \prod_{0 \le \deg f < \deg P} f \equiv \prod_{0 \le \deg f \deg P} -1 \overset{\text{bychar}(\mathbb{F})=2}{=} 1 \mod P$, so result also follows. □

Remark. It is interesting that the Left hand side for Wilson's Theorem for polynomial only depends on the degree of P instead of P itself.

As a final topic of this chapter, i would give some of the theory of d-th power residues. Which is important in discussing the quadratic reciprocity and general reciprocity laws for $\mathbb{F}[T]$.

Definition 2.4. If $f \in \mathbb{F}[T], a \in \mathbb{F}[T], \gcd(a, f) = 1$, we said that a is a d-th power residue modulo f if the equation $x^d \equiv a \mod f$ is solvable in $\mathbb{F}[T]$. Equivalently, $\bar{a}$ is a d-th power in $(\mathbb{F}[T]/\langle f \rangle)^*$.

Corollary 2.18. *Suppose that $f = \alpha P_1^{e_1} \dots P_t^{e_t}$ is the prime decomposition of f, then a is a d-th power residue modulo $f \iff a$ is a d-th power residue modulo $P_i^{e_i}$, $\forall i = 1, ..., t$.*

Proof. Since by the corollary of Chinese Remainder Theorem, which is $\mathbb{F}[T]/\langle f \rangle \cong \prod_i \mathbb{F}[T]/\langle P_i^{e_i} \rangle$, consider the map $\phi : x \mapsto (x \mod P_1^{e_1}, ..., x \mod P_t^{e_t})$,then $\phi(x^d) = (x^d \mod P_1^{e_1}, ..., x^d \mod P_t^{e_t})$. So $x^d \equiv a \mod f \iff x^d \equiv a \mod P_i^{e_i}$, $\forall i = 1, ..., t$ □

The above corollary successfully reduces the problem into the case where the modulus is a prime power.

Proposition 2.19 (Euler's criterion)**.** *Let P be irreducible and $a \in \mathbb{F}[T]$ not divisible by P. Assume $d|(|P| - 1)$. The congruence $X^d \equiv a \mod P^e$ is solvable if and only if*

$$a^{\frac{|P|-1}{d}} \equiv 1 \ , \mod P$$

. There are $\frac{\Phi(P^e)}{d}$ d-th power residues modulo P^e.

Proof. Assume to begin with e=1. If $b^d \equiv a \mod P$, then b is also not divisible by P. $a^{\frac{|P|-1}{d}} \equiv b^{|P|-1} \equiv 1 \mod P$ by Fermat's little theorem.This shows the condition is necessary. To show it is sufficient, recall that for any d-th root of unity of $\mathbb{F}[T]/\langle P \rangle$, $X^d = 1 + \langle P \rangle$ by Corollary 2.4 are in the field $\mathbb{F}[T]/\langle P \rangle$.Consider the homomorphism $\psi : (\mathbb{F}[T]/\langle P \rangle)^* \longrightarrow (\mathbb{F}[T]/\langle P \rangle)^*$ by $x \mapsto x^d$. Then $\ker(\psi) = \{X \in (\mathbb{F}[T]/\langle P \rangle)^* \mid X^d \equiv$

$1 \mod P\} = \{X^d \equiv 1 \mod P\} \implies \#\ker(\psi) \overset{\text{Corollary 2.4}}{=} d$ and $\text{im}(\psi) = \{X^d \mid X \in (\mathbb{F}[T]/\langle P\rangle)^*\}$, which is the d-th powers set. Thus, by first isomorphism theorem,

$$\frac{(\mathbb{F}[T]/\langle P\rangle)^*}{\ker(\psi)} \cong \text{im}(\psi)$$

So $|\text{im}(\psi)| = \frac{|P|-1}{d}$, which implies that there are precisely $\frac{|P|-1}{d}$ d-th powers in $(\mathbb{F}[T]/\langle P\rangle)^*$ and they all satisfies $X^{\frac{|P|-1}{d}} \equiv 1 \mod P$, so they are all precisely the root of this equation. This proves the assertions in case e=1.

For the general case, consider the natural homomorphism from Proposition 2.7,

$$\phi_e : (\mathbb{F}[T]/\langle P^e\rangle)^* \longrightarrow (\mathbb{F}[T]/\langle P\rangle)^*$$

, we have shown that it is a surjective map and that the kernel $\ker\phi_e = \{1+\langle P\rangle+\langle P^e\rangle\}$ is a p-group. Since $|(\mathbb{F}[T]/\langle P\rangle)^*| = |P| - 1$,and $\gcd(|P| - 1, p) = 1$,and by first isomorphism theorem again, as

$$\frac{(\mathbb{F}[T]/\langle P^e\rangle)^*}{\ker\phi_e} \cong (\mathbb{F}[T]/\langle P\rangle)^*$$

, so

$$(\mathbb{F}[T]/\langle P^e\rangle)^* \cong \ker\phi_e \times (\mathbb{F}[T]/\langle P\rangle)^*$$

Since $\gcd(d, p) = 1$,raising to the d-th power in an abelian p-group is an automorphism. Thus, $a \in \mathbb{F}[T]$ is a d-th power modulo $P^e \iff$ it is a d-th power modulo P. The latter has been shown to hold iff $a^{\frac{|P|-1}{d}} \equiv 1 \mod P$. Now, consider the homomorphism from $(\mathbb{F}[T]/\langle P^e\rangle)^* \to (\mathbb{F}[T]/\langle P^e\rangle)^*, a \mapsto a^d$, it is clear that the kernel of this map has d elements and the image is the subgroup of d-th powers. It follows that the latter group has order $\frac{\Phi(P^e)}{d}$. This concludes the proof. □

Chapter 3

Prime, Arithmetic Functions, and Zeta Function

From this chapter onward, we will use A to denote $\mathbb{F}[T]$ for convenience.

3.1 Zeta Function

Definition 3.1. The zeta function of A, denoted by $\zeta_A(s)$, is defined by the infinite series

$$\zeta_A(s) = \sum_{f \in A, f \text{ monic}} \frac{1}{|f|^s}$$

Remark. Note that if $\deg f = d$, then $|f| = q^d$, besides, if f is monic, say $f = T^d + \alpha_{d-1}T^{d-1} + \cdots + \alpha_1 T + \alpha_0$, then there would be totally q^d such f. since there are total of q^d such possible combination for coefficient for f. Hence,

$$\zeta_A(s) = \sum_{d=0}^{\infty}\Big(\sum_{\substack{\deg f = d \\ f \text{ monic}}} \frac{1}{|f|^s}\Big) = \sum_{d=0}^{\infty} \sum_{\substack{\deg f = d \\ f \text{ monic}}} q^{-ds} = \sum_{d=0}^{\infty} q^d \cdot q^{-ds} = \sum_{d=0}^{\infty} (q^{1-s})^d = \frac{1}{1-q^{1-s}}$$

Hence we get our new presentation for $\zeta_A(s)$:

$$\zeta_A(s) = \sum_{\substack{f \in A \\ f \text{ monic}}} \frac{1}{|f|^s} = \frac{1}{1-q^{1-s}}$$

Remark. Clearly, the above series converges absolutely for any $\Re(s) > 1$

However, we are not satisfied with only $\Re(s) > 1$, we want more!

To consider the analytic continuation as a meromorphic function on the whole complex plane with only 1 pole being the simple pole at $s = 1$.

First of all, i would give some definitions for the analysis.

Definition 3.2. a set $\Omega \subseteq \mathbb{C}$ is said to be a **region** if it is a open connected set. In other words, we cannot find two open non-empty set V, U with $V \cap U = \emptyset$, such that $\Omega = V \cup U$.

Definition 3.3. A function f on an open set Ω is **meromorphic** if there exists a sequence of points $\{z_0, z_1, ...\}$ that has no limit points in Ω such that

1. the function f is holomorphic in $\Omega - \{z_0, z_1, ..., \}$(pole must be isolated singularities,else if limit exist, we cannot find an open set that only contains the pole, so cannot find residue)

2. f has poles at the points $\{z_0, z_1, ...\}$(so we can write f as fraction)

Theorem 3.1 (Identity Theorem)**.** *Suppose that f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points$\{w_k\}_{k=0}^{\infty}$, with a limit point in Ω. Then f is identically 0.*

Note. So in human language, every open neighbourhood around the limit point L, must contain at least 1 point in the sequence except the limit pooint itself $\to \infty$ zeros near limit $\to$ f=0(since if a hoolomorphic function is not zero, then its zeros must be isolated(need proof)).

Proof. In other word, if the zeros of a holomorphic function f in the connected open set Ω accumulate in Ω, then $f = 0$.

Suppose that $z_0 \in \Omega$ is a limit point for the sequence $\{w_k\}_{k=1}^{\infty}$ and that $f(w_k) = 0$. First, we show that f is identically zero in a small disc containing z_0. For that, choose a disc D centered at z_0 and contained in Ω, and consider the power series expansion of f in that disc

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

If f is not identically zero, then $\exists m$, a smallest integer such that $a_m \neq 0$. But then we can write

$$f(z) = a_m (z - z_0)^m (1 + g(z - z_0))$$

, where $g(z - z_0)$ converges to 0 as $z \to z_0$. Take $z = w_k \neq z_0$ for a sequence of points converging to z_0, we get a contradiction since $a_m (w_k - z_0)^m \neq 0$ and $1 + g(w_k - z_0) \neq 0$, but $f(w_k) = 0$.

We conclude that the proof using the fact that Ω is connected. Let U denote the interior of the set of points where $f(z) = 0$. Then U is open by definition and non-empty by the argument just given. The set U is also closed since if $z_n \in U$ and $z_n \to z$, then $f(z) = 0$ by continuity, and f vanishes in a neighborhood of z by the argument above. Hence $z \in U$. Now if we let V denote the complement of U in Ω, we conclude that U and V are both open, disjoint, and

$$\Omega = U \cup V$$

. Since Ω is connected we conclude that either U or V is empty.(Here we use one of the two equivalent definition of connectedness). Since $z_0 \in U$, we find that $U = \Omega$ and the proof is complete. □

Corollary 3.2. *Suppose f and g are holomorphic in a region Ω and $f(z) = g(z)$ for all z in some non-empty open subset of Ω(or generally for z in some sequence of distinct points with limit point in Ω). Then $f(z) = g(z)$ throughout Ω.*

Definition 3.4 (analytic continuation)**.** Suppose we are given a pair of functions f, F analytic in regions Ω, Ω', respectively, with $\Omega \subset \Omega'$. If $f = F$ on the smaller set Ω, then we say that F **is an analytic continuation of** f into the region Ω'. And by above corollary, the analytic continuation F of f is unique.

Theorem 3.3 (Analytic Continuation of Zeta function)**.** *The zeta function for polynomial*

$$\zeta_A(s) = \frac{1}{1 - q^{1-s}}$$

is defined as a meromorphic function on the whole complex plane with a simple pole at $s = 1$.

Proof. Note that since we can write the $\zeta_A(s)$ as a rational function, but not in series. In particular, note that $\zeta_A(s) = \frac{1}{q^{1-s}}$ is a well-defined complex function for any $s \in \mathbb{C}$(except at $s = 1$). So if we define the analytic continuation of $\zeta_A(s)$ to be $F(s) = \frac{1}{1-q^{1-s}}$ on $\mathbb{C}$, we see that $\zeta_A(s) = F(s)$ on $\Re(s) > 1$,so $F(s) = \frac{1}{1-q^{1-s}}$ is the unique analytic continuation on $\mathbb{C}$ except the point $s = 1$. So we conclude that $\zeta_A(s) = F(s)$ is a holomorphic function on $\mathbb{C}\backslash\{1\}$ with the simple pole at $s = 1$. To find the residue of zeta function, consider $1 - q^{1-s} = 0 \iff q^{1-s} = 1 \iff (1 - s)\ln(q) = \ln 1 = 2\pi i n, n \in \mathbb{Z}$, so $s_n = 1 - \frac{2\pi i n}{\ln q}$. Since it is a simple pole, and for any $n \in \mathbb{Z}$, the term $q^{-\frac{2\pi i n}{\ln q}}$ will collapse to $e^{2\pi i n} = 1$,consider the Taylor expansion of $q^{1-s} = e^{(1-s)\ln q} = 1 + (1 - s)\ln q + \frac{[(1-s)\ln q]^2}{2} + ... \implies$

$1 - q^{1-s} = 1 - 1 - (1 - s)\ln q - O((1 - s)^2) - ... = (s - 1)\ln q - O((1 - s)^2) - ...$, so

$$F(s) = \frac{1}{(s-1)\ln q} \cdot \frac{1}{1 - \frac{O((1-s)^2)}{(s-1)\ln q}} = \frac{1}{(s-1)\ln q} \cdot \frac{1}{1 + O((1-s))}$$

$$\mathrm{Res}_{s=1} F(s) = \lim_{s\to 1}(s - 1)F(s) = \frac{1}{\ln q} \lim \frac{1}{1 + O(1 - s)} = \frac{1}{\ln q}$$

□

Definition 3.5. $\xi_A(s) = q^{-s}(1 - q^{-s})^{-1}\zeta_A(s)$ be a functional equation.

Proposition 3.4. $\xi_A(1 - s) = \xi_A(s)$ *for* $\Re(s) > 1$.

Proof. $\xi_A(1-s) = q^{s-1}(1-q^{s-1})^{-1}\zeta_A(1-s) = \frac{q^{s-1}}{1-q^{s-1}}\frac{1}{1-q^s} = \frac{q^s}{q-q^s}\frac{1}{1-q^s} = \frac{q^{-2s}q^s}{q^{-s}(q-q^s)}\frac{1}{q^{-s}(1-q^s)} = \frac{q^{-s}}{q^{1-s}-1}\frac{1}{q^{-s}-1}$

$= \frac{1}{q^s(1-q^{-s})}\frac{1}{1-q^{1-s}} = q^{-s}(1-q^{-s})^{-1}\zeta_A(s) = \xi_A(s)$ □

Remark. The use of above functional equation didn't show its value here beacause of the rationalality of zeta function in finite field. However, the zeta function in function field is non trivial, it cannot be represented by a simple rational function: $\zeta_X(s) = \frac{P(q^{-s})}{(1-q^{-s})(1-q^{1-s})}$, which requires the help of functional equation to discuss its property in higher genus $g \geq 1$.where the functional equation shows the symmetry at the line $\Re(s) = \frac{1}{2}$, which is the primary tool for development in proving the Riemann Hypothesis in function field.

Proposition 3.5 (Euler Product Formula)**.** *For all* $\Re(s) > 1$*,*

$$\zeta_A(s) = \prod_{\substack{P \text{ irreducible} \\ P \text{ monic}}} (1 - \frac{1}{|P|^s})^{-1}$$

Proof. Note that left hand side, by geometric series,

$$(1 - |P|^{-s})^{-1} = 1 + |P|^{-s} + |P|^{-2s} +$$

So

$$\prod_{\substack{P \text{ irreducible} \\ P \text{ monic}}} (1 - |P|^{-s})^{-1} = \prod_{P} (1 + |P|^{-s} + |P|^{-2s} + ...)$$

Then each term in the product can be expressed as $\frac{1}{|P_1|^{se_1}|P_2|^{se_2}\cdots|P_t|^{se_t}}$ Note that the field $\mathbb{F}[T]$ is also UFD, so any monic f can be expressed as one of the above combination. so

$$\prod_{\substack{P \text{ irreducible} \\ P \text{ monic}}} (1 - |P|^{-s})^{-1} = \prod_{P} (1 + |P|^{-s} + |P|^{-2s} + ...) = \sum_{\substack{f \in A \\ f \text{ monic}}} \frac{1}{|f|^s} = \zeta_A(s)$$

□

Remark. From above, we can immediate deduce a fact regarding the polynomials for $\mathbb{F}[T]$.

The number of Irreducible polynomials in $\mathbb{F}[T]$ is infinite.

I present 2 methods:

1. Note that suppose on contrary that there are finitely many irreducible polynomials in $\mathbb{F}[T]$, then the right hand side of the equation of the euler formula $\zeta_A(s) = \prod_P (1 - \frac{1}{|P|^s})^{-1}$ will be defined at $s = 1 : \zeta_A(s) = \prod_P (1 - |P|^{-s})^{-1} = \frac{1}{(1-|P_1|^{-s})(1-|P_2|^{-s})\cdots(1-|P_t|^{-s})}$, which is a defined value, however on left hand side,

at $s = 1$, there is a pole so the zeta function will not be defined(blow up) at $s = 1$. Hence we must have infinitely many irreducible monic polynomials in $\mathbb{F}[T]$

2. Suppose S is a finite set of all irreducible polynomial in $\mathbb{F}[T]$: $S = \{P_1, P_2, ..., P_t\}$. Consider the following $P = P_1P_2 \cdots P_t + 1$, clearly, $P = P_1P_2 \cdots P_t + 1 \equiv 1 \mod P_i$ for any $i = 1, ..., t$. Thus, P is a polynomial of positive degree but not divisible by any element in S. Note that P cannot be an element inside S since if it is the case, we would have $P \equiv 0 \mod P_j$ for some j. So we showed that we can always construct a new irreducible polynomial from S if S is finite hence contradict that S contains all the irreducible polynomials.

Example 3.1. Consider $\mathbb{F}_2[T]$ i.e. we have the field has only 2 element $q = 2$. Clearly, for any $f \in \mathbb{F}_2[T]$, $|f| = 2^{\deg f}$. We consider all the monic polynomial of $\mathbb{F}_2[T]$: Since the total combination of all degree 1 polynomial :

$$T + 1, T + 0$$

,the norm is 2.

And total number of degree 2 polynomial:

$$T^2 + 0, T^2 + 1, T^2 + T + 0, T^2 + T + 1$$

, and the norm is 4.

Clearly, for degree 2 polynomial, only $T^2 + T + 1$ is irreducible.

Others: $T^2 = T \cdot T, T^2 + T = T(T^2 + 1), T^2 + 1 = (T + 1^2) = T^2 + 2T + 1 = T^2 + 1$

So writing out in Euler Product

$$\prod_P (1 - |P|^{-s})^{-1} = (1 - 2^{-s})^{-1}(1 - 2^{-s})^{-1}(1 - 4^{-s})^{-1} \cdots =$$

And in $\zeta_{\mathbb{F}_2[T]}(s) = \frac{1}{1-q^{1-s}} = \frac{1}{1-2^{1-s}}$.

3.2 Prime Number Theorem

Definition 3.6. we said that $f(x_n)$ is asymptotic to $g(x_n)$ if $\frac{f(x_n)}{g(x_n)} \to 1$ when $n \to \infty$

Theorem 3.6 (Prime Number Theorem)**.** *Suppose that $x \in \mathbb{R}$, $\pi(x)$ be the number of positive prime numbers less than or equals to x. Then,*

$$\pi(x) \sim \frac{x}{\log(x)}$$

We will show that the classical prime number theorem is also true in finite field.

Definition 3.7. Define a_d to be the number of monic irreducibles of degree d.

Definition 3.8 (radius of convergence)**.** For any formal series $\sum c_n z^n$, $\exists R$, called radius of convergence, such that

1. if $|z| < R$, then the series converges absolutely.
2. if $|z| > R$, the series diverges.
3. if $|z| = R$, diverges/ converges.

Proposition 3.7 (Weierstrass's M-test)**.** *Let $\{f_n\}$ be sequence of function defined on a set $\Omega \subset \mathbb{C}$, taking value on $\mathbb{C}$.Assume,*

1. *$\exists \{M_n\}_{n=1}^{\infty}$, a sequence of positive number such that $|f_n(x)| \leq M_n$, $\forall n \geq 1, \forall x \in \Omega$.*
2. *$\sum_{n=1}^{\infty} M_n$ converges.*

Then, the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly and absolutely.

Proposition 3.8. *For any $n \geq 1$,*

$$q^n = \sum_{d|n} d a_d$$

where a_d is the number of monic irreducibles of degree d, and q is the cardinality of $\mathbb{F}$.

Proof. using the Euler product form for zeta function,

$$\zeta_A(s) = \prod_{\substack{P \text{ monic} \\ P \text{ irreducible}}} (1 - \frac{1}{|P|^s})^{-1} = \prod_{d=0}^{\infty} \prod_{\deg P = d} (1 - q^{-s \deg P})^{-1} = \prod_{d=0}^{\infty} (1 - q^{-ds})^{-a_d}$$

Recall that $\zeta_A(s) = \frac{1}{1-q^{1-s}}$, let $u = q^{-s}$,

note that when $\Re(s) > 1, |u| = |q^{-s}| = |q^{\Re(-s)}| \cdot |e^{\Im(-s)\ln q i}| \overset{\text{unit circle for i}}{=} |q^{\Re(-s)}| < 1$, we obtained the identity

$$\frac{1}{1-qu} = \prod_{d=0}^{\infty} (1 - u^d)^{-a_d}$$

We define $Z(u) = \prod_d^{\infty} (1 - u^d)^{-a_d}$,

$$\ln Z(u) = \ln(\prod_d (1 - u^d)^{-a_d}) = \sum_d (-a_d \ln(1 - u^d))$$

By Taylor expansion of $\ln(1 - u^d)$,

$$\sum_d (-a_d(-\sum_{k=1}^{\infty} \frac{u^{dk}}{k})) = \sum_{d=1}^{\infty} a_d \sum_{k=1}^{\infty} \frac{u^{dk}}{k}$$

Taking the derivative respect to u, we need to apply the Weierstrass M-test: Consider

$$f_{d,k}(u) = a_d \frac{u^{dk}}{k}$$

, since the number $a_d \leq q^d$ as there could only have at most q^d degree d polynomial, and also the radius of convergence of our zeta function series is $R = \frac{1}{q}$,so we pick a r such that $|u| \leq r < \frac{1}{q} = R < 1$(and also we have $rq < 1$). So

$$|f_{d,k}| \leq q^d \frac{r^{dk}}{k} \leq q^d r^{dk} =: M_{d,k}$$

. And since all the term of $M_{d,k}$ are positive, we can reorder the summation:

$$\begin{aligned}
\sum_{d=1}^{\infty}\sum_{k=1}^{\infty} M_{d,k} &= \sum_{d=1}^{\infty}\sum_{k=1}^{\infty} q^d r^{dk} \\
&= \sum_{d=1}^{\infty} q^d \frac{r^d}{1-r^d} \\
&\leq \sum_{d=1}^{\infty} (qr)^d = \frac{qr}{1-qr} < \infty
\end{aligned}$$

So, by Weierstrass M-test, the series $Z(u)$ converges uniformly and absolutly on the disk $|u| \leq r < \frac{1}{q}$, and we can put the derivative inside:

$$\frac{d}{du} \ln Z(u) = \sum_{d=1}^{\infty}\sum_{k=1}^{\infty} a_d \frac{d}{du}\frac{u^{dk}}{k} = \sum_{d=1}^{\infty}\sum_{k=1}^{\infty} d a_d u^{dk-1}$$

$$\implies u\frac{d}{du} \ln Z(u) = \sum_{d=1}^{\infty}\sum_{k=1}^{\infty} d a_d u^{dk}$$

Taking $n = dk \implies k = \frac{n}{d}$ and $d|n$,

$$\sum_{d=1}^{\infty}\sum_{k=1}^{\infty} d a_d u^{dk} = \sum_{n=1}^{\infty}(\sum_{d|n} d a_d) u^n$$

On the left hand side, since $Z(u) = \frac{1}{1-qu} \implies \ln Z(u) = -\ln(1-qu)$. So

$$\frac{d}{du} \ln Z(u) = \frac{q}{1-qu} \implies u\frac{d}{du} \ln Z(u) = \frac{qu}{1-qu} = \sum_{d=1}^{\infty} (qu)^d = \sum_{d=0}^{\infty} q^d u^d$$

So,

$$\sum_{d|n} d a_d = q^n$$

□

Definition 3.9. the mobius function $\mu(n)$ is defined by

$$\mu(n) = \begin{cases} \mu(n) = 1 & \text{if } n = 1 \\ \mu(n) = (-1)^k & \text{if } n \text{ is the product of } k \text{ distinct prime} \\ \mu(n) = 0 & \text{if } n \text{ has a squared prime factor} \end{cases}$$

Theorem 3.9 (Mobius inversion formula)**.** *if g, f are both arithmetic functions satisfying*

$$g(n) = \sum_{d|n} f(d), \ \forall n \geq 1$$

Then,

$$f(n) = \sum_{d|n} \mu(d) g(\frac{n}{d}), \ \forall n \geq 1$$

Corollary 3.10. *for any $n \geq 1$,*

$$a_n = \frac{1}{n} \sum_{d|n} \mu(d) q^{\frac{n}{d}}$$

Proof. By using the Mobius Inversion formula, we immediately get

$$a_n = \frac{1}{n} \sum_{d|n} \mu(d) q^{\frac{n}{d}}$$

□

Example 3.2. since we showed that there are only 1 monic Irreducible polynomials of degree 2, we will verify it with our formula here. Since $n = 2$, we show that $a_2 = 1$. by

$$\frac{1}{2} \sum_{d|2} \mu(d) q^{\frac{2}{d}} = \frac{1}{2}(\mu(1)q^2 + \mu(2)q^1) = \frac{1}{2}(1 \cdot 2^2 + (-1) \cdot 2) = 1$$

What follows, $n = 3$, $a_3 = 1/3(q^3 + \mu(3)q) = 1/3(8 - 2) = 2$, which if we count the degree 3 monic irreducible, $T^3 + T^2 + 1, T^3 + T + 1$.(Note that since in $\mathbb{F}_2$, $1 = -1$, so $T^3+T^2+T+1 = T^3-T^2+T+1 = T^3-T^2-T+1 = T^3-T^2+T-1 = T^3+T^2+T-1 = ...$, all have common factor $T + 1 = T - 1$).

Theorem 3.11 (Prime Number Theorem for polynomials)**.** *Let a_n denote the number of monic irreducible polynomials in $A = \mathbb{F}[T]$ of degree n. Then,*

$$a_n = \frac{q^n}{n} + O(\frac{q^{\frac{n}{2}}}{n})$$

Proof. Fix n. From corollary 3.2, we see that the highest power of q we see occurs when $d = 1$, i.e. occurs at q^n,where we got $na_n = \mu(1)q^n + \sum_{\substack{d|n \\ d>1}} \mu(d)q^{\frac{n}{d}}$.The next is $q^{\frac{n}{2}}$ in case $2|n$, where $na_n - \mu(1)q^n - \mu(2)q^{\frac{n}{2}} = \sum_{\substack{d|n \\ d>2}} \mu(d)q^{\frac{n}{d}}$ And all other terms have the form $\pm q^m,\ m \leq \frac{n}{3}$.

Besides, for the mobius function $\mu(d) = 1$ if $d = 1$, $|\mu(d)| \leq 1$ if $d \geq 1$.

So we obtained that

$$|na_n - q^n| = |\sum_{\substack{d|n \\ \frac{n}{d}<\frac{n}{3}}} \mu(d)q^{\frac{n}{d}} + \mu(2)q^{\frac{n}{2}}| \leq \sum_{\frac{n}{d}=1}^{\lfloor\frac{n}{3}\rfloor} |\mu(d)q^{\frac{n}{d}}| + q^{\frac{n}{2}} \leq \sum_{k=1}^{\lfloor\frac{n}{3}\rfloor} q^k + q^{\frac{n}{2}}$$

$$= \frac{q(q^{\lfloor\frac{n}{3}\rfloor-1} - 1)}{q-1} + q^{\frac{n}{2}} \leq q \cdot q^{\lfloor\frac{n}{3}\rfloor-1} + q^{\frac{n}{2}} \leq q^{\frac{n}{3}} + q^{\frac{n}{2}}$$

$$|a_n - \frac{q^n}{n}| \leq \frac{q^{\frac{n}{2}}}{n} + \frac{q^{\frac{n}{2}}}{n} = O(\frac{q^{\frac{n}{2}}}{n})$$

□

Remark. By substituting $x = q^n$ into the Prime Polynomial Theorem, we obtain the asymptotic formula:

$$a_n = \frac{x}{\log_q(x)} + O\left(\frac{\sqrt{x}}{\log_q(x)}\right)$$

This beautifully mirrors the classical Prime Number Theorem, where $\pi(x)$ counts the number of primes whose size (absolute value) is at most x. To understand this analogy perfectly, we must examine what "size" means in the function field $\mathbb{F}_q[T]$.

In the classical integers $\mathbb{Z}$, the absolute value $|n|$ of any non-zero integer n is exactly equal to the cardinality of the finite quotient ring $\mathbb{Z}/n\mathbb{Z}$. For example, if $n = 10$, the quotient ring $\mathbb{Z}/10\mathbb{Z}$ contains exactly $|10| = 10$ elements. Thus, the "size" of an integer is algebraically defined by the number of residue classes modulo n.

We apply this exact same algebraic definition to the polynomial ring $A = \mathbb{F}_q[T]$. For any non-zero polynomial $f \in A$, we define its norm (or absolute value) $|f|$ to be the cardinality of the quotient ring $A/\langle f\rangle$:

$$|f| := |\mathbb{F}_q[T]/\langle f\rangle|$$

Because the quotient ring $A/\langle f\rangle$ forms an $\mathbb{F}_q$-vector space of dimension exactly equal to $\deg f$, the total number of elements in this space is $q^{\deg f}$. Therefore, the algebraic norm naturally evaluates to $|f| = q^{\deg f}$.

Under this substitution, if a polynomial has degree n, its size is exactly $x = q^n$. Therefore, a_n acts as the exact analogue of $\pi(x)$, counting the number of irreducible, monic polynomials of size exactly x.

Having established the Prime Polynomial Theorem, it is natural to ask how else the Zeta function can be utilized. In analytic number theory, Zeta functions and Dirichlet series act as powerful generating functions. By studying the formal power series associated with $\zeta_A(s)$, we can solve complex counting problems and compute exact averages for arithmetic functions—for example, determining the exact probability that a random monic polynomial of degree n is square-free.

3.3 Number Theoretic Function

After the PNT, it is natural to ask for other uses for the zeta function, since in complex analysis, we usually use zeta function for counting problem. For example, what is the number of square-free monics of degree n?

Definition 3.10. Let b_n be the number of square-free monics of degree n

Definition 3.11. suppose f is monic polynomial in $\mathbb{F}[T]$, then we define the function

$$\delta(f) = \begin{cases} 1 & \text{if } f \text{ is square-free} \\ 0 & \text{otherwise} \end{cases}$$

Proposition 3.12. *Let b_n be the number of square-free monics in $\mathbb{F}[T] = A$ of degree n. Then $b_1 = q$, and for $n > 1$, $b_n = q^n(1 - q^{-1})$*

Proof. Consider the following,

$$\prod_{\substack{P \text{ monic} \\ P \text{ irreducible}}} (1 + |P|^{-s}) = \sum_{f \in A} \frac{\delta(f)}{|f|^s}$$

,if we let $u = q^{-s}$, On right hand side, Since

$$\prod_{\substack{\text{P monic} \\ \text{P irreducible}}} (1 + |P|^{-s}) = 1 + \sum_{e_1 \le 1, e_2 \le 1, \ldots} \left(\frac{1}{|P_1|^{e_1}|P_2|^{e_2}|P_3|^{e_3} \ldots}\right)^s = \sum_{f \in A} \frac{\delta(f)}{|f|^s}$$

$$= \sum_{n=0}^{\infty} \sum_{\substack{\deg f = n \\ f \text{ monic}}} \delta(f) q^{-ns} = \sum_{n=0}^{\infty} b_n u^n$$

.

On Left hand side, consider the identity $\frac{1-w^2}{1-w}$, and substitute $|P|^{-s} = w$,

$$\prod_P (1 + |P|^{-s}) = \prod \left(\frac{1 - w^2}{1 - w}\right) = \frac{\prod (1 - |P|^{-s})^{-1}}{\prod (1 - |P|^{-2s})^{-1}} = \frac{\zeta_A(s)}{\zeta_A(2s)}$$

$$= \frac{1-q^{1-2s}}{1-q^{1-s}} = \frac{1-qu^2}{1-qu}$$

So, combining we have

$$\frac{1-qu^2}{1-qu} = \sum_{n=0}^{\infty} b_n u^n$$

Using geometric series expansion, $\frac{1-qu^2}{1-qu} = (1-qu^2)\sum_{n=0}^{\infty} q^n u^n = \sum_{n=0}^{\infty} q^n u^n - \sum_{k=0}^{\infty} q^{k+1}u^{k+2} \overset{k+2=j}{=} \sum_{n=0}^{\infty} q^n u^n - q^{-1}\sum_{j=2}^{\infty} q^j u^j = 1 + qu + \sum_{n=2}^{\infty}(1-q^{-1})q^n u^n$

So,

$$\begin{cases} qu = b_1 u \\ \sum_{n=2}^{\infty}(1-q^{-1})q^n u^n = \sum_{n=2}^{\infty} b_n u^n \end{cases}$$

Comparing the coefficient on both side,

$$b_n = \begin{cases} q^n(1-q^{-1}) & n > 1 \\ q & n = 1 \end{cases}$$

□

Remark. So what is the use of proposition 3.5:

First, i give a result in integers, and lift it to the finite field cases. If i let B_n be the number of positive, square-free integers less than or equal to n ($x \neq p_1 \cdots p_i^2 \cdots p_n$), then $\frac{B_n}{n} \longrightarrow \frac{6}{\pi^2}$ as $n \longrightarrow \infty$. In less precise language, since B_n/n is the counting of number of positive- square-free integer lower than or equals to n per n, we are talking about the probability that a square-free positive number is $6/\pi^2$. Similarly, we see that the probability that a monic polynomial of degree n is square-free is precisely b_n/q^n, and equals $(1-q^{-1})$ for $n > 1$. So in entire $\mathbb{F}[T] = A$, the probability that the monic polynomial in A is square-free is $1-q^{-1}$.

An interesting fact: $1/\zeta_{\mathbb{Z}}(2) = \frac{6}{\pi^2}$, $1/\zeta_{\mathbb{F}[T]}(2) = 1-q^{-1}$.

For the k-th case would be similar.

Define $\delta_k(f) := \begin{cases} 1 & \text{if f is k-th free} \\ 0 & \text{otherwise} \end{cases}$ and γ_n is the number of k-th free moncis. Consider the following

$$\prod_{\substack{\text{P monic} \\ \text{P irreducible}}} (1 + |P|^{-s} + \cdots + |P|^{-(k-1)s}) = 1 + \sum_{e_1<k, e_2<k \ldots} \left(\frac{1}{|P_1^{e_1}||P_2^{e_2}|\cdots}\right) = \sum_{f \in A} \frac{\delta_k(f)}{|f|^s}$$

$$\sum_{n=0}^{\infty} \sum_{\substack{\deg f = n \\ \text{f monic}}} \delta_k(f) q^{-ns} = \sum_{n=0}^{\infty} \gamma_n u^n$$

So we see that the step is identical, so the result follows:

$$\prod(1+|P|^{-s}+\cdots+|P|^{-(k-1)s})=\frac{\zeta_A(s)}{\zeta_A(ks)}=\frac{1-q^{1-ks}}{1-q^{1-s}}=\frac{1-qu^k}{1-qu}=\sum_{n=0}^{\infty}\gamma_n u^n A$$

$$\frac{1-qu^k}{1-qu}=(1-qu^k)(1+qu+(qu)^2+\cdots)=(1+qu+(qu)^2+\cdots)+(-qu^k-q^2u^{k+1}-\cdots)$$

$$\gamma_n=\begin{cases}q^n(1-q^{1-k}) & n\geq k\\ q^n & n<k\end{cases}$$

So we know that the probability that a monic polynomial of degreee n is k-th free is precisely $\gamma_n/q^n=(1-q^{1-k})$ if $n\geq k$, q^n if $n<k$. Our next goal is to introduce analogues of some well-known number-theoretic functions and to discuss their property if they are instead in field.

Definition 3.12. The Mobius function for polynomial is stated as

$$\mu(f)=\begin{cases}0 & \text{if } f \text{ is not square-free}\\ (-1)^k & \text{if } f \text{ is a constant times a product of } t \text{ distinct monic irreducibles}\end{cases}$$

Definition 3.13. suppose $f\in\mathbb{F}[T]$, then $d(f)$ is defined to be a number of of monic divisors of f.

And we define $\sigma(f)=\sum_{g|f}|g|$, the sum of all monic divisor of f.

Before showing that 3 useful formula for number-theortic functions, we should show the multiplicativity for them first:

Definition 3.14. A complex-valued function λ on $\mathbb{F}[T]\backslash\{0\}$ is said to be multiplicative if

$$\lambda(fg)=\lambda(f)\lambda(g)$$

whenever f,g are relatively prime.

Remark. if we assume $\lambda=1$ on $\mathbb{F}^*$, if $f=\alpha P_1^{e_1}\cdots P_t^{e_t}$ be the prime decomposition of f, if λ is multiplicative,

$$\lambda(f)=\lambda(P_1^{e_1})\cdots\lambda(P_t^{e_t})$$

Proposition 3.13. *For $f\in\mathbb{F}[T]$,the function $\sigma(f),d(f)$ is multiplicative.*

Proof. Suppose that we have f,g are relatively prime in $\mathbb{F}[T]$, meaning that they have no common irreducible factors. Consider f,g prime factorization be

$$f=\alpha P_1^{e_1}\cdots P_t^{e_t}, g=\beta Q_1^{e_1}\cdots Q_r^{e_r}$$

first, consider $d(f)$ be the number of monic divisors of f,$d(g)$ be the number of monic divisors of g. Clearly $d(fg) = d(\alpha\beta P_1^{e_1}\cdots P_t^{e_t}Q_1^{e_1}\cdots Q_r^{e_r}) = d(P_1^{e_1})\cdots d(P_t^{e_t})d(Q_1^{e_1})\cdots d(Q_r^{e_r}) = d(f)d(g)$

For the $\sigma(fg) = \sigma(f)\sigma(g)$, since if we want a $d|fg$, then we can factor $d = d_1d_2$ where $d_1|f, d_2|g$ where $\gcd(d_1, d_2) = 1$. So $|d_1d_2| = q^{\deg(d_1d_2)} = q^{\deg d_1+\deg d_2} = q^{\deg d_1}q^{\deg d_2} = |d_1|\cdot|d_2|$.

$\sigma(fg) = \sum_{d|(fg)}|d| = \sum_{d_1|f}\sum_{d_2|g}|d_1\cdot d_2| = \sum_{d_1|f}\sum_{d_2|g}|d_1|\cdot|d_2| = (\sum_{d_1|f}|d_1|)(\sum_{d_2|g}|d_2|) = \sigma(f)\sigma(g)$. □

Proposition 3.14. *Let the prime decomposition of $f = \alpha P_1^{e_1}\cdots P_t^{e_t}$. Then,*

$$\Phi(f) = |f|\prod_{P|f}(1-|P|^{-1})$$

$$d(f) = \prod_{j=1}^{t}(1+e_j)$$

$$\sigma(f) = \prod_{j=1}^{t}\frac{|P_j|^{e_j+1}-1}{|P_j|-1}$$

Proof. Before showing the 3 formulas, we might need to show the $\Phi(f)$ is given at last chapter. For $d(P^e)$, if P is a monic irreducible, the divisors of P^e is simply $1, P, P^2, ..., P^e$. So $d(P^e) = e+1$. Hence by multiplicativity of d, $d(f) = \prod_{j=0}^{t}d(P_j^{e_j}) = \prod_{j=0}^{t}(1+e_j)$. And for $\sigma(P^e) = \sum_{g|P^e}|g| = 1+|P|+|P|^2+\cdots|P|^e = \frac{|P|^{e+1}-1}{|P|-1}$. Also by multiplicativity for σ,$\sigma(f) = \sigma(P_1)\ldots\sigma(P_t^{e_t}) = \prod_{j=1}^{t}\frac{|P_j|^{e_j+1}-1}{|P_j|-1}$. □

Example 3.3. we use $\mathbb{F}_2[T]$,consider $f = T^{16}+T = T(T+1)(T^2+T+1)(T^4+T+1)(T^4+T^3+1)(T^4+T^3+T^2+T+1)$.

First we show that the it is actually the prime decomposition:$T(T+1)(T^2+T+1)(T^4+T+1)(T^4+T^3+1)(T^4+T^3+T^2+T+1) = (T^2+T)(T^2+T+1)(T^4+T+1)(T^4+T^3+1)(T^4+T^3+T^2+T+1) = (T^4+T)(T^4+T+1)(T^4+T^3+1)(T^4+T^3+T^2+T+1) = (T^8+T^4+T^2+1)(T^8+T^7+T^6+T^5+T^4+T^7+T^6+T^5+T^4+T^3+T^4+T^3+T^2+T+1) = (T^8+T^4+T^2+T)(T^8+T^4+T^2+T+1) = T^{16}+T+2(T^{12}+T^{10}+T^9+T^8+T^6+T^5+T^4+T^3+T^2) = T^{16}+T$.

Now, $|T| = |T+1| = 2, |T^2+T+1| = 4, |T^4+T^3+1| = |T^4+T+1| = |T^4+T^3+T^2+T+1| = 16$

$$\Phi(f) = |f|\prod_{P|f}(1-|P|^{-1}) = 2^{16}(1-\frac{1}{2})(1-\frac{1}{2})(1-\frac{1}{4})(1-\frac{1}{16})^3 = \frac{3\cdot 15^3}{2^2 2^2 2^{12}} = 3\cdot 15^3$$

$$d(f) = \prod_{j=1}^{t}(1 + e_j) = 2^6$$

since the monic factor can form the combination of divisors by "include" or "exclude", so there are totally 2^6.

$$\sigma(f) = \prod_{j=1}^{t} \frac{|P_j|^{e_j+1} - 1}{|P_j| - 1} = (2+1)(2+1)(4+1)(16+1)^3 = 9 \cdot 5 \cdot 17^3$$

As demonstrated in this example, computing arithmetic functions like $\Phi(f)$, $d(f)$, and $\sigma(f)$ pointwise relies entirely on knowing the exact prime factorization of f. While this is computable for individual polynomials, the distribution of prime factors becomes highly chaotic as the degree increases.

More importantly, if our goal is counting—specifically, understanding the *average* behavior of these functions over *all* monic polynomials of a fixed degree n—summing these factorization-dependent formulas one by one is analytically intractable. This structural limitation motivates the introduction of Dirichlet series. By leveraging Dirichlet series and the Zeta function as generating functions, we can completely bypass the need for individual prime factorizations. Instead, these powerful analytical tools will allow us to elegantly extract the exact average values of these arithmetic functions across all polynomials of degree n, depending only on n and the cardinality of the base field q.

3.4 Average Function an d Dirichlet Series

Definition 3.15. Let $h(x)$ be a complex valued function $h : \mathbb{Z}^+ \to \mathbb{C}$. We define

$$\alpha := \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} h(k)$$

Given that the limit exists

Remark.

1. Then we can say that α is the average value (expected value) of the function h
2. if we define $h(n) = \begin{cases} 1 & \text{if } n \text{ is square free} \\ 0 & \text{otherwise} \end{cases}$,then $h(n)$ is the Mobius function and the average of this function becomes $\frac{b_n}{q^n} \implies \alpha_h = \lim_n \sum h(k) = \lim \frac{b_n}{q^n} = \frac{6}{\pi^2}$

Since the sum $\sum_{k=1}^{n} h(k)$ sometimes grows too fast for the average value to exist. Often, we can show the growth is dominated by a simple function of n.

Example 3.4. Euler ϕ-function. we can show that

$$\sum_{k=1}^{n} \phi(k) = \frac{3}{\pi^2} n^2 + O(n \log n)$$

Proof. since $\phi(n) = \sum_{d|n} \mu(d) \frac{n}{d}$ □

Now on, we will shift our focus from counting problem to the characteristics and properties of average value in the context of polynomial.

Definition 3.16. Let $h(x)$ be the function on the set of monic polynomials. For $n > 0$, we define

$$\mathrm{Ave}_n(h) = \frac{1}{q^n} \sum_{\substack{f \text{ monic} \\ \deg(f)=n}} h(f)$$

Remark. Note that this is clearly the average function of the function $h(f)$, on the set of minoc polynomials of degree n **along** in $\mathbb{F}_q[T]$ like the one we defined above in integer ring $\mathbb{Z}$:$\frac{1}{n}\sum_{k=1}^{n} h(k)$, since $\frac{1}{q^n}$ is the number of degree n monic polynomial in the finite field. Hence $\lim_{n\to\infty} \mathrm{Ave}_n(h)$ would be the average value of h provided the limit exists.

In summery:

Feature	$\mathbb{Z}$	$\mathbb{F}_q[T]$
Function h	$h : \mathbb{Z}^+ \to \mathbb{C}$	$h : \mathbb{F}_q[T] \to \mathbb{C}$
Partial Ave.	$\mathrm{Ave}_n = \frac{1}{n}\sum_{k=1}^{n} h(k)$	$\mathrm{Ave}_n = \frac{1}{q^n}\sum_{\substack{\deg(f)=n \\ h \text{ monic}}} h(f)$
Asymptotic	$\lim_{n\to\infty} \mathrm{Ave}_n(h)$	$\lim_{n\to\infty} \mathrm{Ave}_n(h)$

Proposition 3.15. *Suppose the limit* $\lim_n \mathrm{Ave}_n(h)$ *exists, then the following are equal:*

$$\lim_{n\to\infty} \frac{1}{1+q+q^2+\cdots+q^n} \sum_{\substack{\deg(f)\le n \\ h \ monic}} h(f) = \lim_{n\to\infty} \frac{1}{q^n} \sum_{\substack{\deg(f)=n \\ h \ monic}} h(f) =: \alpha$$

Proof. using basic analysis is enough. Consider the limit

$$S_{h,k} := \sum_{\deg f = k, \text{monic}} h(f) \ , \ B_{n,h} := \frac{1}{1+q+\cdots+q^n} \sum_{\deg f \le n, \text{monic}} h(f)$$

Clearly,

$$\mathrm{Ave}_k(h) = \frac{S_{k,h}}{q^k}, \sum_{\deg f \le n, \text{monic}} h(f) = \sum_{k=0}^{n} S_{k,h}$$

Hence,

$$B_{n,h} = \frac{\sum_{k=0}^{n} S_{k,h}}{\sum_{k=0}^{n} q^k} = \frac{\sum_{k=0}^{n} q^k \mathrm{Ave}_k(h)}{\sum_{k=0}^{n} q^k}$$

Note that $\lim_n \mathrm{Ave}_n(h) = \alpha$, by definition of limit

$$\text{Fix } \varepsilon > 0, \exists N_1 \in \mathbb{N} \text{ s.t. } \forall k \geq N_1, |\mathrm{Ave_k(h)} - \alpha| < \frac{\varepsilon}{2}$$

So,

$$B_{n,h} - \alpha = \frac{\sum_{k=0}^{n} q^k \mathrm{Ave}_k(h)}{\sum_{k=0}^{n} q^k} - \alpha = \frac{\sum_{k=0}^{n} q^k (\mathrm{Ave}_k(h) - \alpha)}{\sum_{k=0}^{n} q^k}$$

$$\begin{aligned} |B_{n,h} - \alpha| \leq & \frac{\sum_{k=0}^{N-1} q^k (|\mathrm{Ave}_k(h) - \alpha|)}{\sum_{k=0}^{n} q^k} + \frac{\sum_{k=N}^{n} q^k (|\mathrm{Ave}_k(h) - \alpha)|}{\sum_{k=0}^{n} q^k} \\ < & \frac{\sum_{k=0}^{N-1} q^k (|\mathrm{Ave}_k(h) - \alpha|)}{\sum_{k=0}^{n} q^k} + \frac{\varepsilon}{2} \frac{\sum_{k=N}^{n} q^k}{\sum_{k=0}^{n} q^k} \end{aligned}$$

we define the global bound $|\mathrm{Ave}_k(h) - \alpha| \leq M$, $\forall k \geq 1$, then, since $\sum_{k=0}^{n} q^k = \frac{1-q^{n+1}}{1-q}, \sum_{k=0}^{N-1} = \frac{1-q^N}{1-q} \implies \frac{\sum_{k=0}^{N-1} q^k}{\sum_{k=0}^{n} q^k} = \frac{1-q^N}{1-q^{n+1}} =: \frac{C_N}{1-q^{n+1}}$

Define another $N_2 \in \mathbb{N}$ such that $k \geq N_2, \frac{C_n}{1-q^{n+1}} < \frac{\varepsilon}{2M}$

Hence, for $k \geq N > \max\{N_1, N_2\}$,

$$|B_{n,h} - \alpha| < \frac{\sum_{k=0}^{N-1} q^k (|\mathrm{Ave}_k(h) - \alpha|)}{\sum_{k=0}^{n} q^k} + \varepsilon \frac{\sum_{k=N}^{n} q^k}{\sum_{k=0}^{n} q^k} < M \frac{C_N}{1 - q^{n+1}} + \frac{\varepsilon}{2} \frac{\sum_{k=N}^{n} q^k}{\sum_{k=0}^{n} q^k} < M \frac{\varepsilon}{2M} + \frac{\varepsilon}{2} = \varepsilon$$

□

Note that this limit does not always exists, but it is a good indicator on how the function $h(f)$ grows, i.e. the average rate of growth of $h(f)$.

Definition 3.17.

$$H(n) = \sum_{\substack{f \text{ monic} \\ \deg f = n}} h(f)$$

is defined to be the sum of $h(f)$ over all monic polynomials of degree n.

Definition 3.18 (Dirichlet series)**.** we define the associated Dirichlet series to be

$$D_h(s) = \sum_{f \text{ monic}} \frac{h(f)}{|f|^s} = \sum_{n=0}^{\infty} \sum_{\deg f = n, \text{monic}} \frac{h(f)}{q^{ns}} = \sum_{n=0}^{\infty} \frac{H(n)}{q^{ns}} = \sum_{n=0}^{\infty} H(n) u^n$$

where $s \in \mathbb{C}$

Corollary 3.16. *Note that above converges with conditions: if $|h(f)| = O(|f|^\beta)$, then $D_h(s)$ converges for $\Re(s) > 1 + \beta$.*

Proof. note that consider $a_f = |h(f)|, b_f = |f|^s$, since $|h(f)| = O(|f|^\beta)$, we would have □

Note the right hand side of Dirichlet Series is simply $\sum H(n)u^n$, hence it becomes the power series in u whose coefficient are the averages $H(n)$.Note the Dirichlet Series is very useful when we are encountering the 'counting the sum problem' like sum of all monics of degree n or others.

Definition 3.19. Define $D(n)$ to be the sum of $d(f)$ over all monics of degree n.

Proposition 3.17. *if d is the monic divisor of f, then*

$$D_d(s) = \zeta_A(s)^2 = (1 - qu)^{-2}$$

. Consequently,

$$D(n) = (n + 1)q^n$$

Proof. since by definition of zeta function,

$$\zeta_A(s)^2 = (\sum_h \frac{1}{|h|^s})(\sum_g \frac{1}{|g|^s}) = \sum_f (\sum_{\substack{h,g \\ hg=f}} 1)\frac{1}{|f|^s} = \sum_f \frac{1_{\{h,g:hg=f\}}}{|f|^s} = \sum_f \frac{d(f)}{|f|^s} = D_d(s)$$

Also,by another definition of Dirichlet series,

$$D_d(s) = \sum \frac{D(n)}{q^{ns}} = \sum_{n=0}^{\infty} D(n)u^n = \zeta_A(s)^2 = (1 - qu)^{-2} = \sum_{n=0}^{\infty}(n + 1)q^n u^n$$

, by comparing the coefficient, we got $D(n) = (n + 1)q^n$ □

Remark.

1. Note that $\mathrm{Ave}_n(d) = n + 1$, by $\lim \mathrm{Ave}_n(d) \to \infty$,the average value of $d(f)$ is the way we defined DNE.

2. On average, the number of monic divisor of f : $d(f) \sim O(n)$, grows with the degree of f.

3. If we set $x = q^n$, then we get $D(n) = x\log(x) + x$, which is a closed analogous result for the integer$\sum_{k=1}^{n} d(k) = x\log(x)+(2\gamma-1)x+O(\sqrt{x})$ where $\gamma \sim .577216$ is Euler constant.

4. Note that the formula in polynomial has no error term which is due to the simple nature of zeta function $\zeta_A(s)$

3.5 Multiplicative function and corresponding Dirichlet Series

Definition 3.20. Suppose that $h(f)$ is multiplicative. Then we have

$$D_h(s) = \prod_P (\sum_{k=0}^{\infty} \frac{h(P^k)}{|P|^{ks}})$$

Remark. 1. Recall that if $h(f)$ is multiplicative, then $h(fg) = h(f)h(g)$ if $\gcd(f,g) = 1$. Then let $f = \alpha P_1^{e_1} \cdots P_t^{e_t}$, then by

$$\prod_p (\sum_{k=0}^{\infty} \frac{h(P^k)}{|P|^{ks}}) = \prod_P (1 + \frac{h(P)}{|P|^s} + \frac{h(P^2)}{|P|^{2s}} + \cdots) = \sum_{e_1,e_2,\ldots} \frac{h(P_1)^{e_1} \cdots h(P_t)^{e_t}}{|P_1|^{e_1} \cdots |P_t|^{e_t}} = \sum_{f \text{ monic}} \frac{h(f)}{|f|^s} = D_h(s)$$

2. Note that this expression is more powerful as we can use this to determine the Dirichlet series for $\Phi(f), \sigma(f)$, hence find their total sum formula.

3. Consider the function $\mu(f)$, recall that $\mu(f)$ is the mobius function where is $(-1)^k$ if f is a constant times a product of t distinct monic irreducible, otherwise is 0. Then $\sum_{k=0}^{\infty} \frac{\mu(P^k)}{|P|^{ks}} = \sum_{k=0}^{\infty} \frac{-1}{|P|^{ks}} = 1 - |P|^{-s}$, hence $D_\mu(s) = \prod_P (1 - |P|^{-s}) = \zeta_A(s)^{-1}$

Definition 3.21. Let λ, ρ be two complex valued function on the monic polynomials. The Dirichlet product of the two complex valued function are defined as follows:

$$(\lambda * \rho)(f) = \sum_{\substack{h,g \\ hg=f}} \lambda(h)\rho(g)$$

Remark. Note that the Dirichlet product is closely related to multiplication of Dirichlet series.

Proposition 3.18.

$$D_\lambda(s) D_\rho(s) = D_{\lambda * \rho}(s)$$

Proof.

$$D_\lambda(s) D_\rho(s) = (\sum_h \frac{\lambda(h)}{|h|^s})(\sum_g \frac{\rho(g)}{|g|^s}) = \sum_f (\sum_{\substack{h,g \\ hg=f}} \lambda(h)\rho(g)) \frac{1}{|f|^s} = \sum_f ((\lambda * \rho)(f)) \frac{1}{|f|^s} = D_{\lambda * \rho}(s)$$

□

Remark. Note that by proceeding to calculate the average value of $\Phi(f)$, we know that $\Phi(f) = |f| \prod_{P|f} (1 - |P|^{-1})$.

Corollary 3.19. *Define* $\lambda(f) = |f|$, *then*

$$D_\Phi(s) = \zeta_A(s)^{-1}\zeta_A(s-1)$$

Proof. since $\prod_P(1-|P|^{-1}) = \prod_{j=1}^k(1-|P_j|^{-1}) = \sum_{e_1,\dots,e_k\in\{0,1\}}(-1)^{e_1+\cdots+e_k}|P_1|^{-e_1}\cdots|P_k|^{-e_k}$ $= \sum_{J\subseteq\{1,..,k\}}(-1)^J\prod_{i\in J}|P_i|^{-1}$. Since Mobius function $\mu(g) = (-1)^m$ if g is a product of m distinct primes, $\mu(g) = 0$ if g is not square -free .So choosing a subset of distinct prime factors P_i is equiv to choosing a square-free divisor g of f.Hence $\prod_{P|f}(1-|P|^{-1}) = \sum_{g|f}\mu(g)|g|^{-1} \implies \Phi(f) = |f|\prod_{P|f}(1-|P|^{-1}) = \sum_{g|f}\mu(g)|f/g| = \sum_{\substack{f/g,g\\ f/g\cdot g=f}}\lambda(f/g)\mu(g) := (\mu*\lambda)(f)$. Hence by Proposition 3.10, $D_\Phi(s) = D_{\mu*\lambda}(s) = D_\mu(s)D_\lambda(s) = \zeta_A(s)^{-1}\zeta_A(s-1)$ □

Proposition 3.20.

$$\sum_{\substack{\deg f=n\\ f\ monic}}\Phi(f) = q^{2n}(1-q^{-1})$$

Proof. Define $A(n) = \sum_{\substack{\deg f=n\\ f\text{ monic}}}\Phi(f)$,since $\sum_{n=0}^\infty A(n)u^n = D_\Phi(s) = \sum_{\substack{\deg f=n\\ f\text{ monic}}}\Phi(f) = \zeta_A(s)^{-1}\zeta_A(s-1) \overset{u=q^{-s}}{=} \frac{1-qu}{1-q^2u}$,Hence,expanding $(1-q^2u)^{-1} = \sum_{n=0}^\infty(q^2u)^n$,$\sum_{n=0}^\infty A(n)u^n = \sum_{n=0}^\infty(1-qu)(q^{2n}u^n)$, so coefficient of $u^n : q^{2n}-q^{2n-1}$, so $A(n) = q^{2n}-q^{2n-1}$,The result follows. □

Remark. To do the similar analysis for the function $\sigma(f)$, we need some extra tools to shorten the proof.

Definition 3.22. Let $\mathbf{1}(f) := \begin{cases}1 & \text{if } f \text{ is monic}\\ 0 & \text{otherwise}\end{cases}$

Note. For any complex valued function λ on monics, we would have $(\mathbf{1}*\lambda)(f) = \sum_{g|f}\lambda(g)$: by definition, $(\mathbf{1}*\lambda)(f) = \sum_{\substack{h,g\\ hg=f}}\mathbf{1}(h)\lambda(g) = \sum_{g|f}\lambda(g)$. In particular, if $\lambda(f) = |f|$, since we already have f monic, and since h monic, we must have g be monic, then by definition on $\sigma(f) = \sum_{g|f}|g|$, the sum of all monic divisor of f, then $(\mathbf{1}*\lambda)(f) = \sigma(f)$.

Corollary 3.21. *Define* $\lambda(f) = |f|$, *then*

$$D_\sigma(s) = \zeta_A(s)\zeta_A(s-1)$$

Proof. By above note, $D_\sigma(s) = D_{(\mathbf{1}*\lambda)}(s) \overset{Prop.3.10}{=} D_{\mathbf{1}}(s)D_\lambda(s) = \zeta_A(s)\zeta_A(s-1)$ □

Proposition 3.22.

$$\sum_{\substack{\deg f=n\\ f\ monic}}\sigma(f) = q^{2n}\cdot\frac{1-q^{-n-1}}{1-q^{-1}}$$

Proof. Define $S(n) := \sum_{\substack{\deg f=n \\ f \text{ monic}}} \sigma(f)$, again , $\sum_{n=0}^{\infty} S(n)u^n = D_\sigma(s) = \zeta_A(s)\zeta_A(s-1) = (1-qu)^{-1}(1-q^2u)^{-1} = \sum_{k=0}^{\infty}(qu)^k \sum_{l=0}^{\infty}(q^2u)^l$, then comparing the coefficient at n, we would obtain $S(n) = \sum_{k+l=n} q^k q^{2l}$. Also, $k+l=n \iff k = n-l$, then $S(n) = \sum_{l=0}^{n} q^{n-l}q^{2l} = \sum_{l=0}^{n} q^n q^l = q^n \frac{q^{n+1}-1}{q-1} = q^{2n}\frac{1-q^{-n-1}}{1-q^{-1}}$. □

Remark. The method of obtaining average value results via the zeta function has now been amply demonstrated. Note, having this average value function, it is far more convenient for us to see the trend of the three function Φ, d, σ: the Euler-Phi function, number of monic divisors, sum of all monic divisor when they reach infinity: i.e. the $d(f), \sigma(f), d(f) \sim O(n)$.

3.6 Extra function: $\omega(f)$

Definition 3.23. The $\omega(f)$ is defined to be the number of distinct monic prime factors of f.

Remark. If $f = P_1^{e_1} \cdots P_r^{e_r}$, then $\omega(f) = r$.Also, we can write $\omega(f) = \sum_{P|f} 1$.Below we want to investigate the average value of all monic polynomial of degree n:Avg$(n) = \frac{1}{q^n} \sum_{\deg f = n} \omega(f)$.Before it, let's define a useful lemma first.

Lemma 3.23.

$$\sum_{d=1}^{n} \frac{1}{d} = \log n + \gamma + O(\frac{1}{n})$$

Proof. Define $a_n = H_n - \ln(n)$ where H_n is n-th partial sum of harmonic series.Since $\frac{1}{k} = \sum_{k-1}^{k} \frac{1}{k} dx$.Since $x < k \implies \frac{1}{k} > \frac{1}{x}$.Hence, $H(n) = \sum_{k=1}^{n} \frac{1}{k} = \sum \int_{k-1}^{k} \frac{1}{k} dx > \int_1^n \frac{1}{x} dx = \ln x \implies a_n = H_n - \ln(n) > 0$, which shows it is bounded. Next, to show the sequence is decreasing, consider $a_{n+1} - a_n = H_{n+1} - H_n + \ln(\frac{n}{n+1})$.Since $H_{n+1} - H_n = \frac{1}{n+1}, a_{n+1} - a_n = \frac{1}{n+1} - \ln(1 + \frac{1}{n})$.Since by Taylor Expansion, $\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k!}$,then $a_{n+1} - a_n = \frac{1}{n+1} - (\frac{1}{n} - \frac{2}{n^2} + \frac{1}{3n^3} - \cdots)$,note that $\frac{1}{n+1} - \frac{1}{n} < 0, \frac{1}{2n^2} - \frac{1}{3n^3} < 0, \cdots$.We conclude that $a_{n+1} - a_n < 0$.So it is monotone decreasing and so $\lim_{n\to\infty} a_n = \gamma = \lim_{n\to\infty}(H_n - \ln n)$.Now to show that the error term of $H(n) - \ln n$ and its limit γ is decreasing at a rate proportional to $\frac{1}{n}$,consider $a_n - \gamma = \sum_{k=n}^{\infty}(a_k - a_{k+1})$,since $a_k - a_{k+1} = \ln(1 + \frac{1}{k}) - \frac{1}{k+1}$ and that $\ln(1 + \frac{1}{k}) = \frac{1}{k} - \frac{1}{2k^2} + O(\frac{1}{k^3}), \frac{1}{k+1} = \frac{1}{k}(\frac{1}{1+\frac{1}{k}}) = \frac{1}{k} \sum_j (-\frac{1}{k})^j = \frac{1}{k} - \frac{1}{k^2} + O(\frac{1}{k^3})$.Then, $a_k - a_{k+1} = \frac{1}{2k^2} + O(\frac{1}{k^3})$, where $a_n - \gamma = \sum_{k=n}^{\infty}(\frac{1}{2k^2} + O(\frac{1}{k^3}))$.Now , consider that $\frac{1}{(p-1)n^{p-1}} = \int_{n-1}^{\infty} \frac{1}{x^p} dx < \sum_{k=n}^{\infty} < \int_n^{\infty} \frac{1}{x^p} dx = \frac{1}{(p-1)(n-1)^{p-1}} = \frac{1}{n^{p-1}(1-\frac{1}{n})^{p-1}(p-1)} \leq \frac{2^{p-1}}{p-1} \cdot \frac{1}{n^{p-1}}$ for $n \geq 2$.Then $\sum_{k=n}^{\infty} \frac{1}{k^p} = O(\frac{1}{n^{p-1}})$.So $a_n - \gamma = \sum_{k=n}^{\infty} \frac{1}{2k^2} + \sum O(\frac{1}{k^3}) = \frac{1}{2n} + O(\frac{1}{n^2}) = O(\frac{1}{n}) \implies H_n - \ln n - \gamma = O(\frac{1}{n})$. □

Remark. Note that the γ is the Euler constant where $\gamma \approx 0.5772$.

Proposition 3.24.

$$\sum_{\substack{f \text{ monic} \\ \deg f = n}} \omega(f) = \sum_{d=1}^{n} a_d q^{n-q}$$

Proof. Since we have that $\omega(f) = \sum_{P|f} 1$, consider that

$$\sum_{\deg f = n} \omega(f) = \sum_{\deg f = n} \sum_{P|f} 1 = \sum_{\deg P \leq n} \sum_{\substack{\deg f = n \\ P|f}} 1$$

Now, if $P|f$, then there exists some monic polynomial $f = P \cdot g$, and since $n = \deg f = \deg P + \deg g, \deg g = n - \deg P$.Hence the number of choice of g is exactly $q^{n - \deg P}$.

So $\sum_{\deg f=n}\omega(f)=\sum_{\deg P\le n}q^{n-\deg P}$.Then,

$$\deg_{\deg f=n}\omega(f)=\sum_{d=1}^{n}\sum_{\deg P=d}q^{n-d}=\sum_{d=1}^{n}a_dq^{n-d}$$

since we are counting the number of monic irreducible polynomial. □

Proposition 3.25.

$$\mathrm{Ave}_n(\omega)=\frac{1}{q^n}\sum_{\substack{f\ monic\\ \deg f=n}}\omega(f)=\sum_{d=1}^{n}\frac{a_d}{q^d}\approx\log n+\gamma$$

Proof. Since directly by definition,

we know that $\mathrm{Ave}_n(\omega)=\frac{1}{q^n}\sum_{\substack{f\text{ monic}\\ \deg f=n}}\omega(f)=\frac{1}{q^n}\sum_{d=1}^{n}a_dq^{n-d}=\sum_{d=1}^{n}\frac{a_d}{q^d}$.

Recall that PNT:$a_d=\frac{q^d}{d}+O(\frac{q^{\frac{d}{2}}}{d})$,then $\mathrm{Ave}_n(\omega)=\sum_{d=1}^{n}\frac{1}{q^d}(\frac{q^d}{d}+O(\frac{q^{\frac{d}{2}}}{d}))=\sum_{d=1}^{n}\frac{1}{d}+\sum_{d=1}^{n}O(\frac{1}{q^{\frac{d}{2}}d})$,by proposition above, $\sum_{d=1}^{n}O(\frac{1}{q^{\frac{d}{2}}d})+\sum_{d=1}^{n}\frac{1}{d}=\sum_{d=1}^{n}O(\frac{1}{q^{\frac{d}{2}}d})+\log n+\gamma+O(\frac{1}{n})$.To deal with the big O function of $\frac{1}{q^{\frac{d}{2}}d}$,first, define $E(d)=O(\frac{q^{\frac{d}{2}}}{d})$.Since when $d\ge 1,q\ge 2$, the denominator of the term $\sum_{d=1}^{n}O(\frac{1}{dq^{\frac{d}{2}}})$ grows with rate $q^{\frac{d}{2}}$,exponentially,which makes the term converges abs. So $\sum_{d=1}^{\infty}\frac{E(d)}{q^d}=B$ for some constant B.Now consider that $\sum_{d=1}^{\infty}\frac{E(d)}{q^d}=(\sum_{d=1}^{\infty}-\sum_{d=n+1}^{\infty})\frac{E(d)}{q^d}=B-\sum_{d=n+1}^{\infty}O(\frac{1}{dq^{\frac{d}{2}}})$,since when $d\ge n+1,d>n\implies\frac{1}{d}<\frac{1}{n}$.Since $E(d)=O(\frac{q^{\frac{d}{2}}}{d})\implies|E(d)|\le K\frac{q^{\frac{d}{2}}}{d}$ and

$$\Big|\sum_{d=n+1}^{\infty}\frac{E(d)}{q^d}\Big|\le\sum_{d=n+1}^{\infty}\frac{|E(d)|}{q^d}\le\sum_{d=n+1}^{\infty}\frac{K}{dq^{\frac{d}{2}}}\le\frac{K}{n}\sum_{d=n+1}^{\infty}q^{-\frac{d}{2}}$$

Since $n\cdot q^{\frac{n}{2}}$ grows faster than n,so the error $O(\frac{1}{n\cdot q^{\frac{n}{2}}})$ is much smaller than $O(\frac{1}{n})\implies O(\frac{1}{n\cdot q^{\frac{n}{2}}})=O(\frac{1}{n})$ and that Evaluating the geometric series on the right gives $\sum_{d=n+1}^{\infty}q^{-d/2}=O(q^{-n/2})$. Therefore, we obtain:

$$\left|\sum_{d=n+1}^{\infty}\frac{E(d)}{q^d}\right|\le\frac{K}{n}\cdot O\left(q^{-n/2}\right)=O\left(\frac{1}{n\cdot q^{n/2}}\right).\tag{3.1}$$

Since $n\cdot q^{n/2}$ grows strictly faster than n as $n\to\infty$, this remainder decays much faster than $O\left(\frac{1}{n}\right)$. By the absorption property of Big-O notation, we can write $O\left(\frac{1}{n\cdot q^{n/2}}\right)=O\left(\frac{1}{n}\right)$. Thus, the tail sum satisfies:

$$\sum_{d=n+1}^{\infty}\frac{E(d)}{q^d}=O\left(\frac{1}{n}\right).\tag{3.2}$$

. To find the behavior of the finite remainder sum, we express it as the difference between the complete infinite series and its tail remainder:

$$\sum_{d=1}^{n} \frac{E(d)}{q^d} = \sum_{d=1}^{\infty} \frac{E(d)}{q^d} - \sum_{d=n+1}^{\infty} \frac{E(d)}{q^d} = B - \sum_{d=n+1}^{\infty} \frac{E(d)}{q^d}. \tag{3.3}$$

By substituting the tail bound $\sum_{d=n+1}^{\infty} \frac{E(d)}{q^d} = O\left(\frac{1}{n}\right)$ derived above, we obtain:

$$\sum_{d=1}^{n} \frac{E(d)}{q^d} = B + O\left(\frac{1}{n}\right). \tag{3.4}$$

Which finalizes the calculation for the average value:

$$\mathrm{Ave}_n(\omega) = \left(\log n + \gamma + O\left(\frac{1}{n}\right)\right) + \left(B + O\left(\frac{1}{n}\right)\right) = \log n + C + O\left(\frac{1}{n}\right), \tag{3.5}$$

where $C = \gamma + B$ is a unified constant.

□

Chapter 4

Reciprocal Law

Suppose we are in $\mathbb{F}[T] =: A$, where $\mathbb{F}$ is finite field of cardinality q. It is natural to ask if given a polynomial a, is it a perfect d-th power of some polynomial $x \in A$modulo P in A?(i.e. is $x^d \equiv a \mod P$ solvable?). In this chapter we will investigate it with a powerful tool.

Recall in Chapter 2, we have a proposition that is related to solvability of polynomial:

Note (Proposition 2.12). Let P be irreducible and $a \in A, P \not| a$. Assume that $d|(|P|-1)$. The congruence $X^d \equiv a \mod P$ is solvable $\iff a^{\frac{|P|-1}{d}} \equiv 1 \mod P$ And there are $\Phi(P)/d$ d-th power residue modulo P.

Note that $\Phi(P)/d = (|P|-1)/d$ and algebraically proposition 2.12 is $a^{\frac{|P|-1}{d}} \equiv 1 \mod P$ is the set of kernel of the map

$$\begin{aligned} \phi_P : (\mathbb{F}[T]/\langle P\rangle)^* &\longrightarrow \mathbb{F}^* \\ a &\mapsto a^{\frac{|P|-1}{d}} \mod P \end{aligned}$$

where $\ker\phi = \{a \in (\mathbb{F}[T]/\langle P\rangle)^* | a^{\frac{|P|-1}{d}} \equiv 1 \mod P\}$,which is exactly the set of all polynomial a such that it is solvable mod P (exists a polynomial in A where it is perfect d-th power of a).

Note that $\mathrm{im}\phi = \{a^{\frac{|P|-1}{d}} \in \mathbb{F}^* | a \in (\mathbb{F}[T]/\langle P\rangle)^*\}$, which is the set of d-th root unity since we have $y \in \mathrm{im}\phi, y^d = (a^{\frac{|P|-1}{d}})^d \equiv 1 \mod P$.

In other word,

Corollary 4.1 (The Roots of Unity Bijection)**.** *Let $K = \mathbb{F}_q$ and $L = \mathbb{F}_q[T]/\langle P\rangle$. Define the sets of d-th roots of unity in both fields:*

$$S_K := \{x \in K \mid x^d = 1\} \quad \textit{and} \quad S_L := \{x \in L \mid x^d = 1\}$$

The natural embedding of the constants into the quotient field restricts to a strict bijection between S_K and S_L. Consequently, any element in S_L can be written uniquely as a constant from S_K.

Proof. Consider the natural inclusion map (an injective ring homomorphism) that embeds the base field into the quotient field:

$$\begin{aligned} i : K &\hookrightarrow L \\ c &\mapsto c \pmod{P} \end{aligned}$$

Because $i(c)^d = i(c^d)$, if we restrict the domain of this map to just the roots of unity S_K, the image will entirely land inside S_L. Let's call this restricted map ι:

$$\iota : S_K \longrightarrow S_L$$

We want to show that ι is a bijection (both injective and surjective).

1. Injectivity: Because the original map i is injective (two different constants cannot be congruent modulo a polynomial P of degree ≥ 1), the restricted map ι is also trivially injective.

2. Surjectivity (The Counting Argument): Recall that K^* is a cyclic group of order $q-1$. Because $d \mid q-1$, K^* contains exactly one subgroup of order d, meaning $|S_K| = d$.

Now consider L. Because it is a field, the polynomial $X^d - 1 = 0$ can have at most d distinct roots in L. Therefore, $|S_L| \leq d$.

Because ι is an injective map from a set of size d to a set of size at most d, it forces $|S_L|$ to be exactly d, and it forces the map to be surjective. Thus, ι is a perfect bijection.

Conclusion: Let $y \equiv a^{\frac{|P|-1}{d}} \pmod{P}$. We know $y^d \equiv 1 \pmod{P}$, which means $y \in S_L$. Because $\iota : S_K \to S_L$ is a bijection, there exists a **unique** constant $\alpha \in S_K$ (where $S_K \subset \mathbb{F}_q^*$) such that $\iota(\alpha) = y$.

Written in equivalence notation, this means:

$$a^{\frac{|P|-1}{d}} \equiv \alpha \pmod{P}$$

We define the d-th power residue symbol $\left(\frac{a}{P}\right)_d$ to be this unique constant α. □

Remark. Hence, the bijection ι helps us transition algebraically from the complicated polynomial evaluation $a^{\frac{|P|-1}{d}} \pmod{P}$ in $(\mathbb{F}_q[T]/\langle P\rangle)^*$ to a unique simple constant α in the base field $\mathbb{F}_q^*$. We formally define our residue symbol map as $\phi(a) = \alpha$.

To check the solvability of $x^d \equiv a \pmod{P}$, we simply evaluate $\phi(a)$ and see if the output constant α is exactly 1. If $\alpha = 1$, then our input polynomial a lies in the kernel:

$$\ker \phi = \left\{ a \in (\mathbb{F}_q[T]/\langle P\rangle)^* \;\middle|\; a^{\frac{|P|-1}{d}} \equiv 1 \mod P \right\}$$

which represents the exact set of solvable polynomials (perfect d-th powers). If a is not

solvable, $\alpha \neq 1$; however, by the surjectivity of the map onto the roots of unity, the constant α is guaranteed to be one of the other d-th roots of unity in $\mathbb{F}_q^*$.

Definition 4.1. If $P \nmid a$, let $\left(\frac{a}{P}\right)_d$ be the unique element of $\mathbb{F}^*$ such that

$$a^{\frac{|P|-1}{d}} \equiv \left(\frac{a}{P}\right)_d \mod P$$

If $P|a$, define $\left(\frac{a}{P}\right)_d = 0$. The symbol $\left(\frac{a}{P}\right)_d$ is called the d-th power residue symbol.

Remark. When $d = 2$, this symbol is just like the Legendre symbol of Number Theory.The situation is a bit more flexible in $\mathbb{F}_q$ since $(\mathbb{F}_q)^*$ is cyclic of order $q-1$, where $\mathbb{Z}^*$ is just $\{\pm 1\}$.Note the value of residue symbol is in the finite field $\mathbb{F}$

Proposition 4.2. *The d-th power residue symbol has the following properties:*

1. $\left(\frac{a}{P}\right)_d = \left(\frac{b}{P}\right)_d$ *if* $a \equiv b \mod P$
2. $\left(\frac{ab}{P}\right)_d = \left(\frac{a}{P}\right)_d \left(\frac{b}{P}\right)_d$
3. $\left(\frac{a}{P}\right)_d = 1 \iff x^d \equiv a \mod P$ *is solvable*
4. *Let* $\zeta \in \mathbb{F}_q$ *be an element of order dividing* d*. There exists an* $a \in \mathbb{F}_q$ *such that* $\left(\frac{a}{P}\right)_d = \zeta$

Proof.

1. The first assertion follows immediately since by definition, $a \equiv b \mod P \implies a^{\frac{|P|-1}{d}} \equiv b^{\frac{|P|-1}{d}} \mod P \implies \left(\frac{a}{P}\right)_d = \left(\frac{b}{P}\right)_d$
2. $\left(\frac{ab}{P}\right)_d = (ab)^{\frac{|P|-1}{d}} \mod P \equiv a^{\frac{|P|-1}{d}} b^{\frac{|P|-1}{d}} \mod P = \left(\frac{a}{P}\right)_d \left(\frac{b}{P}\right)_d$
3. $\left(\frac{a}{P}\right)_d \equiv a^{\frac{|P|-1}{d}} \mod P \iff a \in \ker\phi \iff x^d \equiv a \mod P$ is solvable by Proposition 2.12
4. Note that by above remark, we already know that the image of the map ϕ is the set of d-th root unity , hence each $\phi(a) \in \text{im}\phi$, hence ζ with order dividing d,$\zeta \in \text{im}\phi$, $\exists a \in \mathbb{F}$ such that $\phi(a) = \zeta$.

□

Example 4.1. Suppose that the base field is $\mathbb{F}_2$, then consider $P(T) = T^2 + T + 1$ in $\mathbb{F}_2[T]$,clearly $\mathbb{F}_2[T]/\langle P\rangle$ has $|P| = 4$ elements and the remainder of degree less than 2 is $K := \mathbb{F}_2[T]/\langle P\rangle = \{0, 1, T, T+1\}$.We have $|P| - 1 = 3$, by definition, if $P \nmid A$, $\left(\frac{A}{P}\right)_3 \equiv A(T)^{\frac{4-1}{3}} \equiv A(T) \mod P$, we have the 3rd root of unity in this field are equal to the quotient group$(\mathbb{F}_2[T]/\langle P\rangle)^* := \{1, T, T+1\}$.

For example of $A(T) = T + 1$, then

$$\left(\frac{T+1}{P}\right)_3 \equiv T + 1 \mod P$$

if $A(T) = T^3$, then

$$\left(\frac{T^3}{P}\right)_3 \equiv T^3 \equiv T^{|P|-1} \equiv 1 \mod P$$

if $A(T) = T$, then

$$\left(\frac{T}{P}\right)_3 \equiv T \mod P$$

So from above we can see that the image set is the roots of unity $\{1, T, T+1\}$, by our calculation,

1. $A(T) = T^3$ maps to 1
2. $A(T) = T$ maps to T
3. $(AT) = T + 1$ maps to $T + 1$

Proposition 4.3. *Let $\alpha \in \mathbb{F}$. Then,*

$$\left(\frac{\alpha}{P}\right)_d = \alpha^{\frac{q-1}{d}\deg P}$$

Proof. Let $\delta = \deg P$. Then,

$$\frac{|P|-1}{d} = \frac{q^\delta - 1}{d} = (1 + q + \cdots + q^{\delta-1})\frac{q-1}{d}$$

Since $\forall \alpha \in \mathbb{F}, \alpha^q = \alpha \implies \alpha^{1+q+\cdots q^{\delta-1}} = \alpha^{1+1+\cdots+1} = \alpha^\delta$ Hence

$$\left(\frac{\alpha}{P}\right)_d = \alpha^{\frac{|P|-1}{d}} = \alpha^{(1+q+\cdots+q^{\delta-1})\frac{q-1}{d}} = \alpha^\delta \alpha^{\frac{q-1}{d}} = \alpha^{\frac{q-1}{d}\deg P}$$

□

Notice that if $d|\deg P$, then every constant is automatically a d'th power residue modulo P.

Before stating the theorem, i state useful lemma

Lemma 4.4. *Suppose that $\mathbb{F}$ is a finite field with cardinality q,P is irreducible in $\mathbb{F}[T]$, $\mathbb{F}'$ be an algebraic closure of $\mathbb{F}$ where $\alpha \in \mathbb{F}'$.If α is a root of a polynomial $P \in A = \mathbb{F}'[T]$ where $\deg P = \delta$. Then*

1. *$P(T) = (T - \alpha)(T - \alpha^q) \cdots (T - \alpha^{q^{\delta-1}})$*
2. *If $P(T) \in \mathbb{F}'[T] \implies P(T) \equiv P(\alpha) \mod (T - \alpha)$*
3. *If $P(T) \in \mathbb{F}[T] \implies P(T)^q = P(T^q)$*

Proof. 1. Note that α is a root of the polynomial $P(T)$, so $P(\alpha) = 0$. Now since by statement 3,$P(T)^q = P(T^q)$,$P(\alpha^q) = P(\alpha)^q = 0^q = 0$, hence we also have α^q a root of the polynomial $P(T)$. Similarly by statement 3, we also have $\alpha, \alpha^q, ..., \alpha^{q^{\delta-1}}$ are all the roots of the polynomial P, and they are unique: suppose not, then WLOG assume that $i < j < \delta$, such that $\alpha^{q^i} = \alpha^{q^j} \implies a^{q^{j-i}} = \alpha$. Since P is irreducible of degree δ over the field $\mathbb{F}_q$(as $\alpha \notin \mathbb{F}$ but in $\mathbb{F}'$), it must have exactly δ distinct roots in $\mathbb{F}'$. .Thus the polynomial can be expressed as $P(T) = (T-\alpha)(T-\alpha^q)\cdots(T-\alpha^{q^{\delta-1}})$.

2. By division algorithm , $P(T) = (T-\alpha)Q(T)+R(T)$, since $P(\alpha) = (\alpha-\alpha)Q(\alpha)+R \implies P(T) = (T-\alpha)Q(T)+P(\alpha) \implies P(T) \equiv P(\alpha) \mod P$

3. Consider $P(T) = \sum_{j=0}^{\delta} c_j T^j$,by multinomial theorem ,
$P(T)^q = (\sum_{j=0}^{\delta} c_j T^j)^q = \sum_{l_0+...+l_\delta=q} \binom{q}{l_0\, l_1 \cdots l_\delta} (c_0T^0)^{l_0}\cdots(c_\delta T^\delta)^{l_\delta}$
$= \sum_{j=0}^{\delta}(c_jT^j)^{l_j} + \sum_{0<l_0,...,l_\delta<q} \binom{q}{l_0\, l_1 \cdots l_\delta} (c_0T^0)^{l_0}\cdots(c_\delta T^\delta)^{l_\delta} = \sum_{j=0}^{\delta}(c_jT^j)^{l_j} = \sum_{j=0}^{\delta} c_j(T^j)^q$Since $q|\binom{q}{l_0\, l_1 \cdots l_\delta}$ for $0 < l_0, ..., l_\delta < q, l_0 + ... + l_\delta = q$ and also $c_j^q = c_j$ Hence we have $P(T)^q = (\sum_{j=0}^{\delta} c_j T^j)^q = \sum_{j=0}^{\delta} c_j (T^j)^q = P(T^q)$

□

Theorem 4.5 (The d-th power reciprocity law)**.** *Let P, Q be monic irreducible polynomials of degrees δ, ν respectively. Then,*

$$\left(\frac{Q}{P}\right)_d = (-1)^{\frac{q-1}{d}\delta\nu}\left(\frac{P}{Q}\right)_d$$

Proof. Define $(\frac{a}{P}) := (\frac{a}{P})_{q-1}$, then $\left(\frac{a}{P}\right)_d = a^{\frac{|P|-1}{d}} = (a^{\frac{|P|-1}{q-1}})^{\frac{q-1}{d}} = \left(\frac{a}{P}\right)^{\frac{q-1}{d}}$.The theorem would follow in full generality if we could show

$$\left(\frac{Q}{P}\right) = (-1)^{\delta\nu}\left(\frac{P}{Q}\right)$$

Since the general result would follow by raising both sides to the $\frac{q-1}{d}$ power
$:\left(\frac{Q}{P}\right)_d = \left(\frac{Q}{P}\right)^{\frac{q-1}{d}} = (-1)^{\delta\nu\frac{q-1}{d}}\left(\frac{P}{Q}\right)^{\frac{q-1}{d}} = (-1)^{\delta\nu\frac{q-1}{d}}\left(\frac{P}{Q}\right)_d$

Let α be a root of P, β be a root of Q. Let $\mathbb{F}'$ be a finite field which contains $\mathbb{F}, \alpha, \beta$. Then by above lemma, we got

$$P(T) = (T-\alpha)(T-\alpha^q)\cdots(T-\alpha^{q^{\delta-1}})$$

$$Q(T) = (T-\beta)(T-\beta^q)\cdots(T-\beta^{q^{\nu-1}})$$

Now, take congruences in the ring $A' = \mathbb{F}'[T]$, if $f(T) \in A'$, we have $f(T) \equiv f(\alpha)$

mod $(T-\alpha)$.Then,

$$Q(T)^{1+q+\cdots+q^{\delta-1}} \equiv Q(T)Q(T^q)\cdots Q(T^{q^{\delta-1}}) \equiv Q(\alpha)Q(\alpha^q)\cdots Q(\alpha^{q^{\delta-1}}) \mod (T-\alpha)$$

By symmetry, this congruence holds modulo $(T-\alpha^{q^i})$ for all i and it follows that it holds modulo P:

$$Q(T)^{1+q+\cdots+q^{\delta-1}} \equiv Q(\alpha)Q(\alpha^q)\cdots Q(\alpha^{q^{\delta-1}}) \mod (T-\alpha^{q^i}),\ \forall i = 0,1,2,...,\delta-1$$

Then combining with $Q(T) = (T-\beta)(T-\beta^q)\cdots(T-\beta^{q^{\nu-1}})$,

$$\left(\frac{Q}{P}\right) \equiv Q^{\frac{q^\delta-1}{q-1}} \equiv Q^{1+q+q^2+\cdots+q^{\delta-1}} \equiv Q(\alpha)\cdots Q(\alpha^{q^{\delta-1}}) \equiv \prod_{i=0}^{\delta-1}\prod_{j=0}^{\nu-1}(\alpha^{q^i}-\beta^{q^j}) \mod P$$

Similarly,

$$P(T)^{1+q+\cdots q^{\delta-1}} \equiv P(\beta)P(\beta^q)\cdots P(\beta^{q^{\nu-1}}) \mod (T-\beta^{q^j}),\ \forall j = 0,1,2,...,\nu-1$$

,combining with $P(T) = (T-\alpha)\cdots(T-\alpha^{q^{\delta-1}})$,

$$\left(\frac{P}{Q}\right) \equiv P^{\frac{q^\delta-1}{q-1}} \equiv P^{1+q+\cdots+q^{\delta-1}} \equiv P(\beta)P(\beta^q)\cdots P(\beta^{q^{\nu-1}}) \equiv \prod_{j=0}^{\nu-1}\prod_{i=0}^{\delta-1}(\beta^{q^j}-\alpha^{q^i}) \mod Q$$

Since both side of above congruence are in $\mathbb{F}'$ so they must be equal.Thus,

$$\left(\frac{Q}{P}\right) = \prod_{i=0}^{\delta-1}\prod_{j=0}^{\nu-1}(\alpha^{q^i}-\beta^{q^j}) = (-1)^{\delta\nu}\prod_{j=0}^{\nu-1}\prod_{i=0}^{\delta-1}(\beta^{q^j}-\alpha^{q^i}) = (-1)^{\delta\nu}\left(\frac{P}{Q}\right)$$

Finally raising both side with $\frac{q-1}{d}$ power and we are done. □

Since we are done dealing with the irreducible case, it is natural to ask about the general case where an arbitrary non-zero element $b \in \mathbb{F}$.

Definition 4.2. Let $b \in A := \mathbb{F}, b \neq 0$, and $b = \beta Q_1^{f_1}Q_2^{f_2}\cdots Q_s^{f_s}$ be a prime decomposition of b. If $a \in A$, define

$$\left(\frac{a}{b}\right)_d = \prod_{j=1}^{s}\left(\frac{a}{Q_j}\right)_d^{f_j}$$

Remark. Note that this definition ignores $\beta = \mathrm{sgn}(b)$ and so the symbol only depends on the principal ideal $\langle b\rangle$ generated by b. The basic properties of this extended symbol are easily derived from those of t he d-th power residue symbol.

Proposition 4.6. *The symbol* $\left(\frac{a}{b}\right)_d$ *has the following properties:*

1. *If* $a_1 \equiv a_2 \mod b$, *then* $\left(\frac{a_1}{b}\right)_d = \left(\frac{a_2}{b}\right)_d$
2. $\left(\frac{a_1 a_2}{b}\right)_d = \left(\frac{a_1}{b}\right)_d \left(\frac{a_2}{b}\right)_d$
3. $\left(\frac{a}{b_1 b_2}\right)_d = \left(\frac{a}{b_2}\right)_d \left(\frac{a}{b_2}\right)_d$
4. $\left(\frac{a}{b}\right)_d \neq 0 \iff \gcd(a,b) = 1$
5. *If* $x^d \equiv a \mod b$ *is solvable, then* $\left(\frac{a}{b}\right)_d = 1$, *provided that* $\gcd(a,b) = 1$

Proof. 1. if $a_1 \equiv a_2 \mod b \implies a_1 \equiv a_2 \mod Q_i \; \forall i = 1, 2, ..., s$, then $\left(\frac{a_1}{Q_i}\right)_d = \left(\frac{a_2}{Q_i}\right)_d \implies \left(\frac{a_1}{Q_i}\right)_d^{f_i} = \left(\frac{a_2}{Q_i}\right)_d^{f_i}, \forall i \implies \prod_{i=1}^s \left(\frac{a_1}{Q_i}\right)_d^{f_i} = \prod_{i=1}^s \left(\frac{a_2}{Q_i}\right)_d^{f_i} \implies \left(\frac{a_1}{b}\right)_d = \left(\frac{a_2}{b}\right)_d$

2. $\left(\frac{a_1 a_2}{b}\right)_d = \prod_{i=1}^s \left(\frac{a_1 a_2}{Q_i}\right)_d^{f_i} = \prod_{i=1}^s (\left(\frac{a_2}{Q_i}\right)_d \left(\frac{a_1}{Q_i}\right)_d)^{f_i} = \prod_{i=1}^s \left(\frac{a_1}{Q_i}\right)_d^{f_i} \prod_{j=1}^s \left(\frac{a_2}{Q_j}\right)_d^{f_j} = \left(\frac{a_1}{b}\right)_d \left(\frac{a_2}{b}\right)_d$

3. Define $b_1 = \beta_1 Q_1^{f_1} \cdots Q_s^{f_s}, b_2 = \beta_2 P_1^{g_1} \cdots P_t^{g_t} \implies b_1 b_2 = \beta_1 \beta_2 Q_1^{f_1} \cdots Q_s^{f_s} P_1^{g_1} \cdots P_t^{g_t}$
   So$\left(\frac{a}{b_1 b_2}\right)_d := \prod_j \left(\frac{a}{Q_j}\right)_d^{f_j} \prod_i \left(\frac{a}{P_i}\right)_d^{g_i} = \left(\frac{a}{b_1}\right)_d \left(\frac{a}{b_2}\right)_d$

4. $\left(\frac{a}{b}\right)_d \neq 0 \iff \prod_j \left(\frac{a}{Q_j}\right)_d^{f_j} \neq 0 \iff \left(\frac{a}{Q_j}\right)_d \neq 0 \; , \; \forall j \iff Q_j \not| a \; , \; \forall j \iff \gcd(a,b) = 1$

5. Suppose $\exists c^d \equiv a \mod b$, by properties 1,2, $\left(\frac{a}{b}\right)_d = \left(\frac{c^d}{b}\right)_d = \left(\frac{c}{b}\right)_d^d = 1$

□

Remark. Note that for property 5 in proposition 4.3, the converse is not generally true. For example if Q is a monic irreducible not dividing $a, b = Q^d$, then by property 3, we have $\left(\frac{a}{b}\right)_d = \left(\frac{a}{Q^d}\right)_d = \left(\frac{a}{Q}\right)_d^d = 1$. However, not every element of $(\mathbb{F}[T]/\langle b\rangle)^* = (\mathbb{F}[T]/\langle Q^d\rangle)^*$ is a d-th power.In fact, the group of d-th powers has index d as we saw in proposition 2.12.

Also, the property 4 in Proposition 4.1 is not generally true for general symbol: As a mapping from $(\mathbb{F}[T]/\langle b\rangle)^* \to \mathbb{F}^*$, the symbol $\left(\frac{a}{Q^d}\right)_d$ only takes on the value 1 and no other element of order dividing d.

We now show the general reciprocity law which works for arbitrary elements of A.

Definition 4.3. For $f \in A := \mathbb{F}[T], f \neq 0$,define $\mathrm{sgn}_d(f)$ to be the leading coefficient of f raised to the $\frac{q-1}{d}$power. i.e. $\mathrm{sgn}_d(f) = \beta^{\frac{q-1}{d}}$ if β is the leading coefficient of f.

Theorem 4.7 (The general reciprocity law)**.** *Let* $a, b \in A := \mathbb{F}[T]$ *be relatively prime, non-zero elements. Then,*

$$\left(\frac{a}{b}\right)_d \left(\frac{b}{a}\right)_d^{-1} = (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a}$$

Proof. When a, b are monic irreducibles, this reduces to Theorem 4.1. Define $b = \alpha Q_1^{f_1} \cdots Q_s^{f_s}, a = \beta P_1^{g_1} \cdots P_t^{g_t}$.Where each $\deg Q_j = \nu_j, \deg P_i = \delta_i$. By definition,

$$\begin{aligned}
\left(\frac{a}{b}\right)_d &= \prod_{j=1}^{s} \left(\frac{a}{Q_j}\right)_d^{f_j} \\
&= \prod_{j=1}^{s} \left(\frac{\beta P_1^{g_1} \cdots P_t^{g_t}}{Q_j}\right)_d^{f_j} \\
&= \prod_{j=1}^{s} \left(\frac{\beta}{Q_j}\right)_d^{f_j} \prod_{i=1}^{t} \left(\frac{P_i}{Q_j}\right)_d^{f_j g_i} \\
&= \prod_{j=1}^{s} \beta^{\frac{q-1}{d} \deg Q_j} \prod_{i=1}^{t} (-1)^{\frac{q-1}{d} \delta_i \nu_j f_j g_i} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (\beta^{\frac{q-1}{d}})^{\sum_{j=1}^{s} \deg Q_j} \prod_{j=1}^{s} (-1)^{\frac{q-1}{d} \sum_{i=1}^{t} (\delta_i g_i) \nu_j f_j} \prod_{i=1}^{t} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (\beta^{\frac{q-1}{d}})^{\deg b} (-1)^{\frac{q-1}{d} \sum_{i=1}^{t} (\deg P_i) g_i \sum_{j=1}^{s} (\deg Q_j) f_j} \prod_{j=1}^{s} \prod_{i=1}^{t} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \prod_{j=1}^{s} \prod_{i=1}^{t} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a} (\alpha^{\frac{q-1}{d}})^{\deg a} \prod_{j=1}^{s} \prod_{i=1}^{t} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a} \prod_{i=1}^{t} \alpha^{\frac{q-1}{d} \sum_{i=1}^{t} \deg P_i} \prod_{j=1}^{s} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a} \prod_{i=1}^{t} \left(\frac{\alpha}{P_i}\right)_d^{g_i} \prod_{j=1}^{s} \left(\frac{Q_j}{P_i}\right)_d^{f_j g_i} \\
&= (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a} \prod_{i=1}^{t} \left(\frac{\alpha Q_1^{f_1} \cdots Q_s^{f_s}}{P_i}\right)_d^{g_i} \\
&= (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a} \left(\frac{b}{a}\right)_d
\end{aligned}$$

Hence we obtained the general reciprocity law

$$\left(\frac{a}{b}\right)_d \left(\frac{b}{a}\right)_d^{-1} = (-1)^{\frac{q-1}{d} \deg a \deg b} \mathrm{sgn}_d(a)^{\deg b} \mathrm{sgn}_d(b)^{-\deg a}$$

□

Example 4.2. Choose the base field as $\mathbb{F}_7, d = 3$,

$$a(T) = 3T^8 + 2T^2 + 5 \quad , \quad b(T) = 5T^3 + 2$$

So, to find $\left(\frac{3T^8+2T^2+5}{5T^3+2}\right)_3$

First, we need to find $3T^8+2T^2+5 \mod (5T^3+2)$: Since $5T^3 \equiv -2 \mod (5T^3+2)$, by $-2 \equiv 5 \mod 7$, we have $5T^3 \equiv 5 \mod (5T^3+2) \implies T^3 \equiv 1 \mod (5T^3+2)$ if we multiply both side by 3.

Then, $a(T) = 3(T^3)^3T^2 + 2T^2 + 5 \equiv 5T^2 + 5 \mod (5T^3 + 2)$. By proposition 4.3 ,since $3T^8 + 2T^2 + 5 \equiv 5T^2 + 5 \mod (5T^3 + 2) \implies \left(\frac{3T^8+2T^2+5}{5T^3+2}\right)_3 = \left(\frac{5T^2+5}{5T^3+2}\right)_3$. Now since $\deg(5T^2 + 5) = 2 \implies \operatorname{sgn}_3(5T^2 + 5) = 5^{\frac{7-1}{3}} = 5^2 = 25 \equiv 4 \mod 7$, similarly, $\deg(5T^3 + 2) \implies \operatorname{sgn}_3(5T^3 + 2) = 5^2 \equiv 4 \mod 7$. So,

$$\left(\frac{3T^8 + 2T^2 + 5}{5T^3 + 2}\right)_3 = \left(\frac{5T^2 + 5}{5T^3 + 2}\right)_3 = (-1)^{\frac{6}{3}\cdot 2\cdot 3}(4)^3(4)^{-2}\left(\frac{5T^3 + 2}{5T^2 + 5}\right)_3 = 4\left(\frac{5T^3 + 2}{5T^2 + 5}\right)_3$$

Now $5T^2+5 \equiv 0 \mod 7 \implies T^2 \equiv -1 \equiv 6 \mod (5T^2+5)$. So $5T^3+2 = 5T(T^2)+2 \equiv 30T + 2 \equiv 2T + 2 \mod (5T^2 + 5)$ Hence we have

$$4\left(\frac{5T^3 + 2}{5T^2 + 5}\right)_3 = 4\left(\frac{2T + 2}{5T^2 + 5}\right)_3 = 4\left(\frac{2}{5T^2 + 5}\right)_3\left(\frac{T + 1}{5T^2 + 5}\right)_3$$

For the case of $\left(\frac{2}{5T^2+5}\right)_3$, by proposition 4.2, $\left(\frac{2}{5T^2+5}\right)_3 = 2^{\frac{q-1}{d}\deg(5T^2+2)} = 2^4 \equiv 2 \mod 7$

Hence we left with

$$4\left(\frac{5T^3 + 2}{5T^2 + 5}\right)_3 = 4\left(\frac{2}{5T^2 + 5}\right)_3\left(\frac{T + 1}{5T^2 + 5}\right)_3 = 8\left(\frac{T + 1}{5T^2 + 5}\right)_3$$

For the case of $\left(\frac{T+1}{5T^2+5}\right)_3$,by the reciprocal law again, $T + 1$ has $\operatorname{sgn}_3(T + 1) = 1$, $5T^2 + 5$ has $\operatorname{sgn}_3(5T^2 + 5) = 4$ bby above. So

$$\left(\frac{T + 1}{5T^2 + 5}\right)_3 = (-1)^{2\cdot 1\cdot 2}(1)^2(4)^{-1}\left(\frac{5T^2 + 5}{T + 1}\right)_3 = 4^{-1}\left(\frac{5T^2 + 5}{T + 1}\right)_3 = 2\left(\frac{5T^2 + 5}{T + 1}\right)_3$$

as in $\mathbb{F}_7$, the inverse of 4 is $2 : (4 \cdot 8 \equiv 1)$.

Finally, by $T \equiv -1 \mod 7$, $5T^2 + 5 \equiv 5(-1)^2 + 5 = 10 \equiv 3 \mod 7$,hence left with $\left(\frac{5T^2+5}{T+1}\right)_3 = \left(\frac{3}{T+1}\right)_3 = 3^2 \equiv 2 \mod 7$, Hence,

$$\left(\frac{3T^8 + 2T^2 + 5}{5T^3 + 2}\right)_3 = 4\left(\frac{5T^3 + 2}{5T^2 + 5}\right)_3 = 8 \cdot 2 \cdot 9 \equiv 1 \cdot 2 \cdot 2 \equiv 4$$

The reciprocity law can be thought of a golden law from Gauss, but its importance

lies in the fact that it relates two natural questions in an intrinsic way.

Given a polynomial m of positive degree, what are the d-th powers modulo m? Since we know $(\mathbb{F}[T]/\langle m\rangle)^*$ is finite, we can just write down the elements of $(\mathbb{F}[T]/\langle m\rangle)^*$, then raise to the d-th power and make a list of the result. The answer will be a list of cosets/ residue classes mod m. Practically, we might not do it this way as the calculation might be messy and tedious.

One can appeal to the structure of $(\mathbb{F}[T]/\langle m\rangle)^*$ to find shortcuts:Recall that we are assuming $d|(q-1)$. Under this assumption, the answer is $\frac{\Phi(m)}{d^{\lambda(m)}}$, where $\lambda(m)$ is the number of distinct monic prime divisors of m. This follows from Proposition 2.12 and also Chinese Remainder Theorem.

Chapter 5

Lifting Exponential

5.1 Lte in $\mathbb{Z}$

The Lifting The Exponent Lemma (LTE) is a powerful tool for solving Diophantine equations, particularly exponential ones involving prime numbers. Frequently featured in Math Olympiads and advanced research, LTE provides a method to determine the highest power of a prime p (usually $p \geq 3$) that divides $a^n \pm b^n$. This guide outlines LTE within the ring of integers, then extends the concept to function fields of characteristic 0 and finite fields of characteristic p.

Definition 5.1. For $a, b \in \mathbb{Z}$, we say a divisible by b and write $b|a \iff$ there exists some integer q such that $a = qb$

we define $v_p(x)$ to be the greatest power in which a prime p divides x. i.e. if $v_p(x) = \alpha$, then $p^\alpha | x$ and $p^{\alpha+1} \nmid x$, we write it as $p^\alpha || x$.

Proposition 5.1. *$v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x+y) \geq \min\{v_p(x) + v_p(y)\}$*

Proof. since let $v_p(x) = \alpha, v_p(y) = \beta$, then $p^\alpha || x, p^\beta || y \implies p^{\alpha+\beta} | xy$. Clearly if we add 1 to the exponent, we will get contradiction if factorizing out into x and y alone, hence $p^{\alpha+\beta} || xy$. For the $v_p(x+y)$case, consider $x = mp^\alpha, y = np^\beta$ where $\gcd(p, m) = \gcd(n, p) = 1$, WLOG assume that $\alpha \leq \beta$, then we have $v_p(x+y) = v_P(mp^\alpha + np^\beta) = v_p(p^\alpha(m + np^{\beta-\alpha})) = v_p(x) + v_p(m + np^{\beta-\alpha}) \geq v_p(x)$, similary we got $v_p(x+y) \geq v_p(y)$ so $v_p(x+y) \geq \min\{v_p(x) + v_p(y)\}$ □

Note. convention define $v_p(0) = \infty, \forall p$ prime.

Example 5.1. Note that the greatest power of 2 that divides 96 is 2^2 since $2^2 = 4|12|96$ but $2^3 = 8 \nmid 96$. Hence $2^2 || 96, v_2(96) = 2$

Example 5.2. Note that is p, q are two distinct prime, then $v_p(p^\alpha q^\beta) = \alpha \implies p^\alpha || p^\alpha q^\beta$

Lemma 5.2. *Let x, y be integers and let n be a positive integer. Given an arbitrary prime p, such that $\gcd(n, p) = 1$, $p|x - y$ and $p \nmid y, p \nmid x$. We have*

$$v_p(x^n - y^n) = v_p(x - y)$$

In other word, the largest integer α s.t. $p^\alpha || x^n - y^n$ is same as that of $p^\alpha || x - y$

Proof. since if we given that $p|x-y \implies p^\alpha = x-y$, if $p \nmid x^{n-1} + x^{n-2}y + ... + y^{n-1}$, then by above example we must have $p^\alpha || x^n - y^n$. So it suffices to show $p \nmid x^{n-1} + x^{n-2}y + ... + y^{n-1}$. Since $p|x-y \implies x - y \equiv 0 \mod p \implies x \equiv y \mod p \implies x^{n-1} + x^{n-2}y + ... + y^{n-1} \equiv x^{n-1} + x^{n-2}x + ... + x^{n-1} \equiv nx^{n-1} \overset{*}{\not\equiv} 0 \mod p$ since $p \nmid x$. □

Remark (*). Note that at the very last line of above proof, $nx^{n-1} \not\equiv 0 \mod p$ only when we have $\gcd(n,p) = 1 \implies n \not\equiv 0 \mod p$ and that $p \nmid x \implies p \nmid x^k, \forall k \implies x^{n-1} \not\equiv 0 \mod p$, hence together we have $nx^{n-1} \not\equiv 0 \mod p$. Generally if we drop the $\gcd(n,p) = 1$, the above equality does not hold without any further restriction.

Lemma 5.3. *Let x, y be integers and let n be an **odd** positive integer. Given an arbitrary prime p such that $\gcd(p,n) = 1$, $p|x+y$, and $p \nmid x, p \nmid y,$, then*

$$v_p(x^n + y^n) = v_p(x+y)$$

Proof. Since x, y can be negative, using lemma 1, we got

$$v_p(x^n - (-y)^n) = v_p(x - (-y)) \implies v_p(x^n + y^n) = v_p(x+y)$$

as n is negative integer. □

Theorem 5.4 (First Form of Lte)**.** *Let x, y be integers, n be a positive integer, and let p be an **odd** prime such that $p|x-y$ and $p \nmid x, p \nmid y$. Then we have*

$$v_p(x^n - y^n) = v_p(x-y) + v_p(n)$$

Proof. We use induction on $v_p(n)$.

First we show that

$$v_p(x^p - y^p) = v_p(x-y) + 1$$

we need to show

$$\begin{cases} p \mid x^{p-1} + x^{p-2}y + ... + y^{p-1} & (2) \\ p^2 \nmid x^{p-1} + x^{p-2}y + ... + y^{p-1} & (3) \end{cases}$$

in (2), we have

$$x^{p-1} + x^{p-2}y + ... + y^{p-1} \equiv x^{p-1} + x^{p-2}x + ... + x^{p-1} \equiv px^{p-1} \equiv 0 \mod p$$

Let $y = x + kp$ for some integer k. For $t \in [1, p)$, we have

$$\begin{aligned} y^t x^{p-1-t} &\equiv (x+kp)^t x^{p-1-t} \\ &\equiv x^{p-1-t}(\sum_{j=0}^{t} \binom{t}{j} x^{t-j}(kp)^j) \\ &\equiv x^{p-1-t}(x^t + tx^{t-1}(kp) + 0 + ... + 0) \\ &\equiv x^{p-1} + x^{p-2}tkp \mod p^2 \end{aligned}$$

Which means that

$$y^t x^{p-1-t} \equiv x^{p-1} + tkpx^{p-2} \mod p^2, \quad t = 1, 2, ..., p-1$$

Using this, then

$$\begin{aligned} & x^{p-1} + x^{p-2}y + ... + y^{p-1} \\ &\equiv x^{p-1} + (x^{p-1} + kpx^{p-2}) + (x^{p-1} + ktpx^{p-2}) + \cdots + (x^{p-1} + k(p-1)px^{p-2}) \\ &\equiv px^{p-1} + (1 + 2 + \cdots + p - 1)kpx^{p-2} \\ &\equiv px^{p-1} + \frac{p(p-1)}{2}kpx^{p-2} \quad \text{Here } \frac{p-1}{2} \text{ is integer only when p is odd prime.} \\ &\equiv px^{p-1} + p^2\frac{p-1}{2}kx^{p-2} \\ &\equiv px^{p-1} \not\equiv 0 \mod p^2 \end{aligned}$$

Hence (3) established and by

$$\begin{cases} v_p(x^{p-1} + \cdots y^{p-1}) = 1 & \text{using (2),(3)} \\ v_p(x^n - y^n) = v_p((x-y)(x^{p-1} + \cdots y^{p-1})) = v_p(x-y) + v_p(x^{p-1} + \cdots y^{p-1}) \end{cases}$$

we have

$$v_p(x^p - y^p) = v_p(x-y) + 1$$

To extend to the first form, suppose that $n = p^\alpha b$ where $\gcd(p, b) = 1$, meaning

that we have $v_p(n) = \alpha$. Then

$$\begin{aligned}
v_p(x^n - y^n) &= v_p(x^{p^\alpha b} - y^{p^\alpha b}) \\
&= v_p((x^{p^\alpha})^b - (y^{p^\alpha})^b) \\
&\overset{\text{Lemma 3.1}}{=} v_p(x^{p^\alpha} - y^{p^\alpha}) = v_p((x^{p^{\alpha-1}})^p - (y^{p^{\alpha-1}})^p) \\
&\overset{\text{case 1}}{=} v_p(x^{p^{\alpha-1}} - y^{p^{\alpha-1}}) + 1 = v_p((x^{p^{\alpha-2}})^p - (y^{p^{\alpha-2}})^p) + 1 \\
&= v_p(x^{p^{\alpha-2}} - y^{p^{\alpha-2}}) + 2 \\
&\vdots \\
&= v_p(x^p - y^p) + \alpha - 1 \\
&\overset{\text{case 1}}{=} v_p(x - y) + \alpha \\
&= v_p(x - y) + v_p(n)
\end{aligned}$$

□

Theorem 5.5 (Second Form of Lte)**.** *Let x, y be integers, n be an **odd** positive integer, p be an **odd** prime such that $p|x + y$, and $p \nmid x, p \nmid y$. Then we have*

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n)$$

Proof. we, again, follow the proof of lemma 3.2, since x, y can be negative integer, then,

$$v_p(x^n + y^n) = v_p(x^n - (-y^n)) = v_p(x - (-y)) + v_p(n) = v_p(x + y) + v_p(n)$$

, note that we must need n to be odd for the negative sign to be pulled out from the bracket. □

Remark. Note that the above first and second form both requires p to be an odd prime because we require $\frac{p-1}{2}$ to be an integer in the proof. Hence for the special case of p=2, we need other conditions.

Theorem 5.6 (Lte for p=2)**.** *Let x, y be two **odd** integers such that $4|x - y$. Then*

$$v_2(x^n - y^n) = v_2(x - y) + v_2(n)$$

Proof. To apply Lemma 3.1, clearly, $\gcd(n, 2) = 1$, $2|4|x - y$, $2 \nmid x, 2 \nmid y$. By lemma 3.1 we have $v_p(x^n - y^n) = v_p(x - y)$. Define $n = 2^{n'}\beta$ for odd β.So now it suffices to show that

$$v_2(x^{2^{n'}} - y^{2^{n'}}) = v_2(x - y) + n'$$

Note that,

$$x^{2^{n'}} - y^{2^{n'}} = (x^{2^{n'-1}} + y^{2^{n'-1}})(x^{2^{n'-2}} + y^{2^{n'-2}}) \cdots (x^2 + y^2)(x + y)(x - y)$$

As $x \equiv y \equiv \pm 1 \mod 4 \implies x^{2^k} \equiv y^{2^k} \equiv 1 \mod 4 \implies x^{2^k} + y^{2^k} \equiv 2 \mod 4, \ \forall k \in [1, \infty)$. Also, as x, y are both odd, $x + y \equiv 2 \mod 4$ and $4|x - y \implies x^{2^{n'}} - y^{2^{n'}} = (x^{2^{n'-1}} + y^{2^{n'-1}})(x^{2^{n'-2}} + y^{2^{n'-2}}) \cdots (x^{2^1} + y^{2^1})(x^{2^0} + y^{2^0})(x - y) \equiv 2^{n'}(x - y) \mod 4$. Then each $2|x^{2^k} - y^{2^k}, 4 \nmid x^{2^k} - y^{2^k} \implies v_2(x^{2^k} - y^{2^k}) = 1, \ \forall k = 0, 1, ..., n' - 1$

Then using

$$v_2(x^n - y^n) = v_2((x^{2^{n'}})^\beta - (y^{2^{n'}})^\beta) = v_2(x^{2^{n'}} - y^{2^{n'}})$$

$$= v_2(x^{2^{n'-1}} + y^{2^{n'-1}}) \cdots (x^{2^0} + y^{2^0})) + v_2(x - y) \equiv n' + v_2(x - y)$$

□

Theorem 5.7. *Ley x, y be **odd** integers and let n be an **even** positive integer. Then*

$$v_2(x^n - y^n) = v_2(x - y) + v_2(x + y)v_2(n) - 1$$

Proof. Since we know that for odd k, $k^2 = 4k' + 1$ for some integer k'. Hence for odd x, y, we have $4|x^2 - y^2$. Following the proof of Theorem 3.1, consider m as odd integer and k be positive integer such that $n = m \cdot 2^k$ then $v_2(n) = k$,

$$\begin{aligned} v_2(x^n - y^n) &= v_2(x^{m \cdot 2^k} - y^{m \cdot 2^k}) = v_2(x^{\cdot 2^k} - y^{\cdot 2^k}) \\ &= v_2((x^2)^{\cdot 2^{k-1}} - (y^2)^{\cdot 2^{k-1}}) = v_2(x^{\cdot 2^{k-1}} - y^{\cdot 2^{k-1}}) + v_2(2) \\ &= v_2(x^{\cdot 2^{k-2}} - y^{\cdot 2^{k-2}}) + 2 \cdot v_2(2) \\ &\vdots \\ &= v_2(x^2 - y^2) + (k - 1) \cdot v_2(2) = v_2(x^2 - y^2) + k - 1 \\ &= v_2(x - y) + v_2(x + y) + k - 1 \\ &= v_2(x - y) + v_2(x + y) + v_2(n) - 1 \end{aligned}$$

□

Remark. Let p be prime and $x, y \in \mathbb{Z}$ where $p \nmid y, p \nmid x$ Then,

(i) For $n \in \mathbb{Z}^+$

(a) if $p \neq 2, p|x - y$, then

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n)$$

(b) if $p = 2, 4|x - y$, then

$$v_2(x^n - y^n) = v_2(x - y) + v_2(n)$$

(c) if $p = 2, n$ is **even**, and $2|x - y$, then

$$v_2(x^n + y^n) = v_2(x - y) + v_2(x + y) + v_2(n)$$

(ii) For $n \in \mathbb{Z}^+$, **odd**, if $p|x + y$, then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n)$$

(iii) For $n \in \mathbb{Z}^+, \gcd(n, p) = 1, p|x - y$, then

$$v_p(x^n - y^n) = v_p(x - y)$$

(iv) For $n \in \mathbb{Z}^+$, **odd**, $\gcd(n, p) = 1, p|x + y$, then

$$v_p(x^n + y^n) = v_p(x + y)$$

5.2 Problems in $\mathbb{Z}$

We follow some problems in Amir Hossein Parvardi Lte first, then raise some more problems.

Problem 5.2.1. (Russia 1996) Find all positive integers n for which there exist positive integers x, y and k such that $\gcd(x, y) = 1, k > 1$ and $3^n = x^k + y^k$

Proof. Note that if k is even, then x^k, y^k will be perfect square and we have $3|a^2+b^2 \iff a^2 + b^2 \equiv 0 \mod 3 \iff a^2, b^2 \equiv 0 \mod 3 \implies 3|a, b \implies \gcd(a, b) = 3 \neq 1$. Hence k is odd. Suppose $\exists$, prime $p|x + y$, by $\gcd(x, y) = 1$, p is odd. Immediately, using Theorem 2,

$$v_p(x^k + y^k) = v_p(x + y) + v_p(k)$$

Since $p|x + y \implies v_p(x + y) \geq 1 > 0 \implies v_p(3^n) = v_p(x^k + y^k) + v_p(k) > 0 \implies p|3^n \implies p = 3$ So by $3|x + y$,we have $x + y = 3^m$, $\exists m \geq 1$. Then, $n = v_3(3^n) = v_3(3^m) + v_3(k) = v_3(k) + m$

1. $m > 1$:

   By induction, it is clear that $3^a \geq 2 + a, \forall a \geq 1$, and so,using $v_3(k) = a \implies 3^a|k \implies a + 2 \leq k \implies v_3(k) = a \leq k - 2$. Take $M = \max(x, y)$,since $x + y = 3^m \geq 9, m > 1$ then $M \geq 5$

   $$x^k+y^k \geq M^k = M^2 \cdot M^{k-2} \geq \frac{(x+y)^2}{4} \cdot M^{v_3(k)} \geq \frac{3^m}{4} 3^m \cdot 5^{v_3(k)} > 3^m \cdot 3^{v_3(k)} = 3^{m+v_3(k)} = 3^n$$

   Contradiction.

2. $m = 1$:

   Then $n = v_3(k) + 1 \implies v_3(k) = n - 1$. Note that $m = 1$ means $x + y = 3$ so $x = 1, y = 2$ or $(x = 2, y = 1)$. We then have $3^{1+v_3(k)} = 3^n = 1^k + 2^k = 1 + 2^k$. Note that $3^{v_3(k)}|k \implies 3^{v_(k)} \leq k$

   $$1 + 2^k = 3^{v_3(k)+1} = 3 \cdot 3^{v_(k)} \leq 3k \implies 1 + 2^k \leq 3^k$$

   Only $k = 3$ could satisfies the above **E**quality.

Hence the answer is $(x, y, k, n) = (1, 2, 3, 2) = (2, 1, 3, 2)$ .So $n = 2$. □

Problem 5.2.2. (Balkan 1993) Let p be prime number and $m > 1$ be a positive integer. Show that if for some positive integers $x > 1, y > 1$ such that

$$\frac{x^p + y^p}{2} = (\frac{x + y}{2})^m,$$

then $m = p$.

Proof. Note that $\frac{x^p+y^P}{2} \geq (\frac{x+y}{2})^p$ for all positive integers p.Since $\frac{x^p+y^p}{2} = (\frac{x+y}{2})^m$ we must have $m \geq p$.Consider $d = \gcd(x, y) \implies gcd(x', y') = 1$ if we take $x = dx', y = dy'$. So we got

$$(x' + y')^m \cdot d^{m-p} = (x'^p + y'^p)2^{m-1}$$

define q as prime divisor of $x' + y'$, $v = v_q(x' + y')$

1. if p is odd: then $v_q(x'^p+y'^p) = v_q(x'^p+y'^p)+v_q(2^{m-1}) = v_q(d^{m-p})+v_q((x'+y')^m) = m$ as $q \not| d, v = v_q(x' + y')$, so $v_q(x'^p + y'^p) = v + v_q(p)$ and $v_q(d^{m-p}(x' + y')^m) \geq mv$(since q may also be a factor of d). Thus $m \leq 2, p \leq 2$, giving an immediate contradiction.

2. if $p = 2$, note that for $x + y \geq 4$, we would have $\frac{x^2+y^2}{2} < 2(\frac{x+y}{2})^2 \leq (\frac{x+y}{2})^3$, hence $m = 2$.

3. immediate that the remaining case $(x, y) = (1, 2), (2, 1)$ are impossible.

□

5.3 Lte in $K[t]$ char 0

Definition 5.2. we say $P \in K[t]$ is prime if it is an irreducible in $K[t]$ For two polynomials $f, g \in K[t]$, we say g is divisible by f and write $f|g \iff \exists r \in K[t]$ such that $g = rf$

Definition 5.3. Suppose $f, g \in K[t]$, not both zero, we say $d(t) \in K[t]$ is the greatest common divisor of f, g, denote as $d = \gcd(f, g)$ if

(i) $d|f, d|g$

(ii) $\forall h \in K[t]$ such that $h|f, h|g$, then $h|d$

(iii) d is **monic**(to ensure gcd is unique)

Definition 5.4. we define $v_P(f)$ to be the greatest power in which a prime $P(t)$ divides f i.e. $P|f$. In particular, $v_P(f) = \alpha$ then $P^\alpha|f$ but $P^{\alpha+1} \nmid f$. We again write $P^\alpha||f \iff v_P(f) = \alpha$.

Remark. define $v_\infty(f) = -\deg(f)$ and $v_P(0) = \infty$ for all prime P

Corollary 5.8. *1.* $v_P(fg) = v_P(f) + v_P(g)$

2. $v_P(f + g) \geq \min\{v_P(f), v_P(g)\}$

3. for any integer n, $v_P(n) = 0$

Proof. suppose that $v_P(f) = \alpha, v_P(g) = \beta \implies P^{\alpha+\beta}|fg$, note that $P^{\alpha+\beta}||fg$,else if $P^{\alpha+\beta+1}||fg \implies P^{\alpha+1}|f, P^\beta|g$ or $P^\alpha|f, P^{\beta+1}|g$, both contradict the definition on $v_P(f) = \alpha, v_P(g) = \beta$. For infinite case, note that $v_\infty(f) = -\deg f, v_\infty(g) = -\deg g \implies v_\infty(fg) = -\deg(fg) = -\deg f - \deg g = v_\infty(f) + v_\infty(g)$ Similar proof for $v_P(f+g) \geq \min\{v_P(f), v_P(g)\}$ case. Note that for any integer n, $\deg n = 0 \implies P \nmid n$ for any prime. So $v_P(n) = 0$ for any prime P and any integer n. □

Note that differ from $\mathbb{Z}$, $v_P(n) = 0, \forall n \in \mathbb{Z}, \forall$ prime $P \in K[t]$. So, we directly have:

Theorem 5.9 (Lte in function field char 0)**.** *Let* $f, g \in K[t], n \in \mathbb{Z}^+$ *, let* P *be prime such that* $P|f - g$ *and* $P \nmid f, P \nmid g$*. Then*

$$v_P(f^n - g^n) = v_P(f - g)$$

Proof. Consider

$$f^n - g^n = (f - g)(f^{n-1} + f^{n-2}g + \cdots fg^{n-2} + g^{n-1})$$

we show $P \not| (f^{n-1} + f^{n-2}g + \cdots fg^{n-2} + g^{n-1})$ By $P|f - g \implies f \equiv g \mod P$ and $P \not| f \implies P \not| f^k,\ \forall k = 1, 2, .. \implies f^k \not\equiv 0 \mod P$ and also using $P \not| n,\ \forall n \in \mathbb{Z} \implies n \not\equiv 0 \mod P$ Since

$$\begin{aligned} & f^{n-1} + f^{n-2}g + \cdots fg^{n-2} + g^{n-1} \\ & \quad \equiv f^{n-1} + f^{n-2}f + \cdots ff^{n-2} + f^{n-1} \\ & \quad \equiv nf^{n-1} \not\equiv 0 \mod P \end{aligned}$$

$\implies P \not| f^{n-1} + f^{n-2}g + \cdots fg^{n-2} + g^{n-1} \implies v_P(f^{n-1} + f^{n-2}g + \cdots fg^{n-2} + g^{n-1}) = 0$
Then, $v_P(f^n - g^n) = v_P(f - g) + v_P(f^{n-1} + \cdots + g^{n-1}) = v_P(f - g)$ □

Note that there is no restriction comparing to integer version of Lte for odd and even part for the $v_P(n)$ since $v_P(n) = 0$ directly.

5.4 Lte in $K[t]$ char p

Using the same notation as the characteristic 0 function field, with the same definition on characteristic field 0 on previous page, we obtained the following theorem:

Theorem 5.10 (Lte in function field char p)**.** *Let $f, g \in K[t]$, $n \in \mathbb{Z}^+$, let P be a prime such that $P|f-g, P \not| f, P \not| g$. Then*

$$v_P(f^n - g^n) = p^{v_p(n)} \cdot v_P(f-g)$$

Proof. **Case 1**$(n = p)$:

By division algorithm, let $f = g + lP^\alpha$, then

$$f^p = (g + lP^\alpha)^p = g^p + \sum_{j=1^p} \binom{p}{j} g^j (lP)^{p-j} = g^p + 0 + l^p P^{\alpha p}$$

$$f^p - g^p = l^p P^{\alpha p}$$

$$v_P(f^p - g^p) = v_P(l^p P^{\alpha p}) = v_P(P^{\alpha p}) + v_P(l^p) = p\alpha + 0 = p \cdot v_P(f-g)$$

Case 2$(n = p^k)$:

By induction suppose that $v_P(f^{p^k} - g^{p^k}) = p^k \cdot v_P(f-g)$

Then,

$$\begin{aligned} v_P(f^{p^{k+1}} - g^{p^{k+1}}) &= v_P((f^{p^k})^p - (g^{p^k})^p) \\ &\overset{*}{=} p \cdot v_P(f^{p^k} - g^{p^k}) \\ &= p^{k+1} \cdot v_P(f-g) \end{aligned}$$

$*$: since $P|f^{p^k} - g^{p^k}$ by hypothesis, and $P \not| f, g \implies P \not| f^{p^k}, P \not| g^{p^k}$, so by case 1 we got $p \cdot v_P(f^{p^k} - g^{p^k})$

Case 3$(n = m \cdot p^k), \gcd(m,n) = 1, \gcd(m,p) = 1$:

$$\begin{aligned} v_P(f^n - g^n) &\overset{**}{=} v_P((f^m)^{p^k} - (g^m)^{p^k}) \\ &= p^k \cdot v_P(f^m - g^m) \\ &= p^k \cdot (v_P(f-g) + v_P(f^{m-1} + \cdots + g^{m-1})) \end{aligned}$$

Since that $P \not| f, g \implies P \not| g^{m-1-k} f^k,\ \forall k = 0, ..., m-1 \implies v_P(f^{m-1} + \cdots g^{m-1}) = 0$

and also that for $**$:since $P \not| f^m, g^m$, it follows from case 2. Hence we conclude that

$$v_P(f^n - g^n) = p^k \cdot v_P(f - g)$$

where $k = v_p(n)$(small p here). $\square$

Chapter 6

Polynomial Fermat Last Theorem and ABC Theorem

While Fermat's Last Theorem remains difficult in the integer ring $\mathbb{Z}$ due to the lack of a continuous structure, its analogue in function fields is significantly more accessible. The key advantage is the availability of the derivative, which provides a rigorous mechanism to detect and bound repeated roots, ultimately leading to the Mason-Stothers Theorem

Let $\mathbb{F}$ be a field such that $\text{char}(\mathbb{F}) = 0$,$f \in \mathbb{F}[T]$,we write $P \in \mathbb{F}[T]$ as monic irreducible and $P|f$.

6.1 algebraic closed

Definition 6.1. we define $N_0(f)$ as the number of distinct zeros of f

Remark. We write $\gcd(f, g)$ as (f, g) for convenience purposes.

Example 6.1. $f(T) = T(T-1)^2(T+2)^3$, then f has 3 distinct zero so $N_0(f) = 3$

Lemma 6.1. *If $f \in K[T]$, then f has repeated roots $\iff$ f, f' have a common root. In other word, suppose P is irreducible s.t. $P|f$, then $v_P(f') = v_p(f) - 1$*

Proof. Since we are in algebraic closed field, consider α as a root of f, of multiplicity k, then $f(T) = (T-\alpha)^k Q(T)$ where $Q(\alpha) \neq 0$, then $f' = k(T-\alpha)^{k-1}Q(T) + (T-\alpha)^k Q'(T) = (T-\alpha)^{k-1}(kQ(T)+Q'(T)(T-\alpha))$.If $k = 1$, $f'(\alpha) = kQ(\alpha) \neq 0 \iff f$has no repeated root $\iff$ f, f' has no common root. For $k > 1$, similar.

Note that if P irreducible such that $P|f$, then $f(T) = P^{v_P(f)}(T)Q(T)$, where $P \nmid Q$. Then $f' = v_P(f)P^{v_P(f)-1}Q + Q'P^{v_P(f)} \implies v_P(f') = v_P(f) - 1$ □

Lemma 6.2. *Suppose that $f \in K[T]$,then* $\deg(f, f') = \deg(f) - N_0(f)$

Proof. Suppose that f has distinct roots $\alpha_0, \alpha_1, ..., \alpha_t$ with multiplicity $e_0, ..., e_t$ and $\deg f = n$.Then $n = e_0 + ... + e_t$. Note that f has repeated roots, then by Lemma 6.1, since each multiplicity of f' dropped by 1, then f' has same distinct roots but with

multiplicity $e_0 - 1, ..., e_t - 1$, hence

$$(f, f') = \prod_{j=0}^{t} (x - \alpha_j)^{e_j - 1}$$

Then $\deg(f, f') = \sum_{j=0}^{t}(e_j - 1) = \sum_{j=0}^{t} e_j - t = n - t = \deg f - N_0(f)$ □

Remark. The Mason-Stothers Theorem is also regarded as the ABC-Theorem

Theorem 6.3 (Mason-Stothers Theorem)**.** *If $f, g, h \in K[T]$ are pairwise relatively prime, and not all constant, such that $f + g = h$, then*

$$\max\{\deg(f), \deg(g), \deg(h)\} \leq N_0(fgh) - 1$$

Proof. first note that

$$f'g - fg' = f'h - f'f - fh' + ff' = f'h - fh'$$

For the case that if at least two of the polynomials f, g are constants, it forces another polynomial to be constant also. So, Without loss of generality, assume f, g are non-constant polynomials. Now, not that we have $f'g - g'f \neq 0$, else,$f'g = g'f \neq 0$, since $(f, g) = 1$, using Bezout identity, $\exists k_1, k_2 \in K[T]$ such that $fk_1 + gk_2 = 1 \implies fg'k_1 + gg'k_2 = g' \implies gf'k_1 + gg'k_2 = g' \implies g(f'k_1 + g'k_2) = g' \implies g|g'$, contradiction.

Now, observe that $(f, f')|(f'g - fg')$and $(g, g')|(f'g - g'f)$: let $(f', f) = l, f = lk_1, f' = lk_2 \implies f'g - g'f = l(k_2g - g'k_1) \implies (f, f')|(f'g - g'f)$, similar case for (g, g').Also $(h, h')|(f'h - h'f)$, the proof is of similar case as above.Note that f, g, h are pairwise relatively prime, and since we have $(g, g'), (f, f')|(f'g - g'f)$ and $(h, h')|(f'h - h'f) = (f'g - g'f) \implies (f, f')(g, g')(h, h')|(f'g - fg')$

$\implies \deg(f, f') + \deg(g, g') + \deg(h, h') \leq \deg(f'g - fg')$. Since $\deg(f'g - fg') = \min\{\deg(f'g), \deg(fg')\} \leq \deg(fg) - 1 = \deg f + \deg g - 1$ Hence we have

$$\deg(f, f') + \deg(g, g') + \deg(h, h') \leq \deg f + \deg g - 1$$

Now by Lemma 6.2, we have

$$\deg(f, f') = \deg f - N_0(f)$$
$$\deg(g, g') = \deg g - N_0(g)$$
$$\deg(h, h') = \deg h - N_0(h)$$

Hence

$$\deg f - N_0(f) + \deg g - N_0(g) + \deg h - N_0(h) \leq \deg f + \deg g - 1$$

$$\deg(h) \leq N_0(f) + N_0(g) + N_0(h) - 1 = N_0(fgh) - 1$$

since f, g, h are relatively prime polynomial so number of distinct zeros of fgh are same.

By symmetry, we obtained

$$\deg(f) \leq N_0(fgh) - 1$$

$$\deg(g) \leq N_0(fgh) - 1$$

Hence

$$\max\{\deg f, \deg g, \deg h\} \leq N_0(fgh) - 1$$

□

Theorem 6.4 (Polynomial Fermat's Last Theorem)**.** *Let $X, Y, Z \in K[T]$. If X, Y, Z are pairwise relatively prime, and not all constant, then $\forall n \geq 3$,*

$$X^n + Y^n \neq Z^n$$

Proof. Suppose $\exists n \geq 3$ such that $X^n + Y^n = Z^n$, let $M = \max\{\deg X, \deg Y, \deg Z\}$, then

$$\begin{aligned}
3M &= \max\{\deg X^3, \deg Y^3, \deg Z^3\} \\
&\leq \max\{\deg X^n, \deg Y^n, \deg Z^n\} \\
&\leq N_0(X^n Y^n Z^n) - 1 \\
&= N_0(XYZ) - 1 \text{ (Since } X^n, Y^n, Z^n \text{ pairwise relatively prime)} \\
&= N_0(X) + N_0(Y) + N_0(Z) - 1 \\
&\leq \deg X + \deg Y + \deg Z - 1 \text{ (by } N_0(f) \leq N_0(f) + \deg(f, f') = \deg f) \\
&\leq 3M - 1 \text{ Contradiction.}
\end{aligned}$$

□

Remark. Since in algebraic closed field, every irreducible polynomial is linear degree 1.Hence any monic irreducible is of form $(T - \alpha)$, however, in general arbitrary field, we can write any polynomial into product of prime factors, which we need minor adjustment on lemma.

6.2 Generalization to Arbitrary Fields of Characteristic 0

With the same notation and definition.

Let K be a field, where $\text{char}(K) = 0$, $f, g \in K[t]$. Define $(f, g) = \gcd(f, g)$,,

Definition 6.2. We define 2 new function

$$\omega(f) = \sum_{\substack{P|f \\ P \text{ monic irreducible}}} 1$$

,

$$s(f) = \sum_{\substack{P|f \\ P \text{ monic irreducible}}} \deg P$$

Lemma 6.5. *Let $f \in K[t]$, suppose that $P \in K[t]$ is irreducible, and $P|f$,then*

$$v_P(f') = v_P(f) - 1$$

Proof. Suppose that $P|f$,then $\exists Q(t) \in K[t], (Q, P) = 1$ such that $f = Q(t)P(t)^{v_P(f)}$. Then $f'(t) = Q'(t)P(t)^{v_P(f)} + v_P(f)P'(t)P(t)^{v_P(f)-1}Q(t) = P^{v_P(f)-1}(Q'P + v_P(f)P'Q)$. Note that since it is trivial that $P \not| P'$, else $P' = RP^{v_{P'}(P)} \implies \deg P' > \deg P$, contradiction. So$P \not| P'Qv_P(f) \implies P \not|(Q'P + v_P(f)P'Q)$, hence $v_P(f') = v_P(f) - 1$. □

Lemma 6.6. *Let $f \in K[t]$, then*

$$\deg((f, f')) = \deg(f) - s(f)$$

Proof. Define the prime decomposition of f to be $f = \alpha P_1^{e_1} \cdots P_\omega^{e_\omega}$, where $P_1, ..., P_\omega$ are monic distinct irreducible polynomials. By Lemma above, since $P_j|f \implies v_{P_j}(f') = v_{P_j}(f) - 1 = e_j - 1$ for all j. Hence, $(f, f') = P_1^{e_1-1} \cdots P_\omega^{e_\omega-1}$.It follows that $\deg((f, f')) = \sum_{j=1}^{\omega}(e_j - 1)\deg(P_j) = \deg f - s(f)$

□

Theorem 6.7 (Mason-Stother Theorem)**.** *Let $f, g, h \in K[t]$, if they are relatively prime, not all constant, and $f + g = h$. Then*

$$\max\{\deg f, \deg g, \deg h\} \leq s(fgh) - 1$$

Proof. Note the proof is almost identical to the one above, but with minor changes.

First note that

$$f'g - fg' = f'h - f'f - fh' + ff' = f'h - fh'$$

For the case that if at least two of the polynomials f, g are constants, it forces another polynomial to be constant also. So, Without loss of generality, assume f, g are non-constant polynomials. Now, not that we have $f'g - g'f \neq 0$, else,$f'g = g'f \neq 0$, since $(f, g) = 1$, using Bezout identity, $\exists k_1, k_2 \in K[T]$ such that $fk_1 + gk_2 = 1 \implies fg'k_1 + gg'k_2 = g' \implies gf'k_1 + gg'k_2 = g' \implies g(f'k_1 + g'k_2) = g' \implies g|g'$, contradiction.

Now, observe that $(f, f')|(f'g - fg')$and $(g, g')|(f'g - g'f)$: let $(f', f) = l, f = lk_1, f' = lk_2 \implies f'g - g'f = l(k_2g - g'k_1) \implies (f, f')|(f'g - g'f)$, similar case for (g, g').Also $(h, h')|(f'h - h'f)$, the proof is of similar case as above.Note that f, g, h are pairwise relatively prime, and since we have $(g, g'), (f, f')|(f'g - g'f)$ and $(h, h')|(f'h - h'f) = (f'g - g'f) \implies (f, f')(g, g')(h, h')|(f'g - fg')$

$\implies \deg(f, f') + \deg(g, g') + \deg(h, h') \leq \deg(f'g - fg')$. Since $\deg(f'g - fg') = \min\{\deg(f'g), \deg(fg')\} \leq \deg(fg) - 1 = \deg f + \deg g - 1$ Hence we have

$$\deg(f, f') + \deg(g, g') + \deg(h, h') \leq \deg f + \deg g - 1$$

By Lemma 4, $\deg(h) = \deg(h, h') + s(h) \leq \deg f + \deg g - 1 - \deg(f, f') - \deg(g, g') + s(h) = \deg f - \deg(f, f') + \deg g - \deg(g, g') + s(h) - 1 = s(f) + s(g) + s(h) - 1 = s(fgh) - 1$. By symmetry, we also have

$$\deg(f) \leq s(fgh) - 1 \ , \ \deg(g) \leq s(fgh) - 1$$

Hence $\max\{\deg f, \deg g, \deg h\} \leq s(fgh) - 1$ □

Theorem 6.8 (Polynomial Fermat's Last Theorem)**.** *Let $X, Y, Z \in K[T]$. If X, Y, Z are pairwise relatively prime, and not all constant, then $\forall n \geq 3$,*

$$X^n + Y^n \neq Z^n$$

Remark. The proof makes no difference to the previous one in the algebraic closed case.

6.3 Arbitrary Field of Characteristic p

Lemma 6.9. *Define the Wronskian identity $W := A'B - B'A$ where $A, B \in K[t]$. Then $W \neq 0$*

Proof. WLOG, we assume that A', B', C' are not all zero. Since $A + B = C$ gives $A' + B' = C'$, define $B = C - A, B' = C' - A'$. Then the Wronskian becomes

$$W = A'B - B'A = A'(C - A) - (C' - A')A = A'C - A'A - C'A - A'A = A'C - C'A$$

Similarly, if we substitute $A = C - B$, we got

$$W = C'B - B'C$$

Hence, we have 3 equivalent expression of Wronskian:

$$W = A'B - B'A = A'C - C'A = C'B - B'C$$

Now, suppose that $W = 0$. Then, $A'B - B'A = 0 \implies A'B = AB'$. Since considering $(\frac{A}{B})' = \frac{A'B - AB'}{B^2} = 0$.Note that in characteristic $p > 0$, a derivative vanishes $\iff$ the rational function is a p-th power(*See remark in page 67*):$\frac{A}{B} = \frac{f^p}{g^p}$ for coprime f, g.Since $\gcd(A, B) = 1$, we must have $A = c \cdot f^p, B = c \cdot g^p$ where $c \in K^*$.By $C = A + B$, we also have C is a p-th power, meaning that A', B', C' are all zero, contradiction. Hence we conclude that $W \neq 0$. □

Lemma 6.10. *Suppose that the prime factorization of $ABC = P_1^{e_1} \cdots P_t^{e_t}$, then*

$$\prod_{i=1}^{t} P_i^{e_i - 1} \,\Big|\, W$$

Proof. Since $ABC = \prod_{i=1}^{t} P_i^{e_i} \implies \operatorname{rad}(ABC) = \prod_{i=1}^{t} P_i$.Note that $\gcd(A, B, C) = 1$, so any chosen irreducible factor P_i, must divides only 1 from the 3 polynomial.

Hence, $\begin{cases} \text{if } P_i^{e_i} \,|\, A \implies P_i^{e_i - 1} \,|\, A' \implies P_i^{e_i - 1} \,|\, (A'B - AB') \\ \text{if } P_i^{e_i} \,|\, B \implies P_i^{e_i - 1} \,|\, B' \implies P_i^{e_i - 1} \,|\, (A'B - AB') \\ \text{if } P_i^{e_i} \,|\, C \implies P_i^{e_i - 1} \,|\, C' \implies P_i^{e_i - 1} \,|\, (A'C - A'C) \end{cases}$ .We see in either cases, any prime factor $P_i^{e_i - 1}$ divides W. Also that the polynomial are coprime, we can conclude that $\prod_{i=1}^{t} P_i^{e_i - 1} \,\Big|\, W$. □

Theorem 6.11 (Mason-Stother Theorem)**.** *Let $A, B, C \in K[t]$,if they are relatively prime, not all derivatives is 0, and $A + B = C$.Then,*

$$\max\{\deg A, \deg B, \deg C\} \leq \deg(\operatorname{rad}(ABC)) - 1$$

where $\mathrm{rad}(A)$ *is the product of distinct irreducible polynomial factors of* A*.*

Proof. Since

$$\prod_{i=1}^{t} P_i^{e_i-1} = \frac{\prod P_i^{e_i}}{\prod P_i} = \frac{ABC}{\mathrm{rad}(ABC)}$$

and that by the second lemma, $\frac{ABC}{\mathrm{rad}(ABC)} \Big| W \implies \deg(\frac{ABC}{\mathrm{rad}(ABC)}) \leq \deg W$.Then

$$\deg A + \deg B + \deg C - \deg(\mathrm{rad}(\mathrm{ABC})) \leq \deg W$$

Besides, since $W = (AB)'$ and we use the fact that if $F \in K[t], \deg F' \leq \deg F - 1$,

$$\deg W \leq \max\{\deg(A'B), \deg(AB')\} \leq \deg A + \deg B - 1$$

So,

$$\deg C \leq \deg(\mathrm{rad}(ABC)) - 1$$

By symmetry, and following the same step as previous proof, we conclude

$$\max\{\deg A, \deg B, \deg C\} \leq \deg(\mathrm{rad}(ABC)) - 1$$

□

Chapter 7

Squarefull Integers and Squarefull Polynomials

To investigate squarefull properties in polynomial, we shall first define basic definitions and interesting conjecture in integer ring, then raise it to function field.

7.1 Squarefull Integer

Definition 7.1. suppose that $n \in \mathbb{Z}_{>1}$, then n is said to be a squarefull integer if for any prime $p|n, v_p(n) \geq 2$.

Remark. The numbers that do not contain any squared prime factors are called squarefree.

Proposition 7.1. *Any squarefull number can be uniquely expressed in the form $n = a^2b^3$ where $a, b \in \mathbb{Z}_+$ and b is squarefree.*

Proof. Consider the prime decomposition of a squarefull integer $n : n = \prod_{j=1}^{t} p_i^{e_i}$ where $e_i \geq 2$ for all i.By division algorithm, if e_i is even, then $e_i = 2k_i, \exists k_i \geq 1$, if e_i is odd, $e_i = 2k_i + 3, \exists k_i \geq 0$. So every exponent can be written as $e_i = 2k_i + 3r_i$ where $k_i \in \mathbb{Z}_{\geq 0}, r_i = 0, 1$.Putting it back into the factorization of n, we obtained

$$n = \prod_{i=1}^{t} p_i^{2k_i+3r_i} = (\prod_i p_i^{k_i})^2 (\prod_i p_i^{r_i})^3$$

Since $r_i \in \{0, 1\}$, no prime in the factorization of b has an exponent greater than 1.Therefore, b is squarefree when setting $a = \prod_i p_i^{k_i}, b = \prod_i p_i^{r_i}$ □

Remark. The product of any two squarefull numbers is always a squarefull number, since for any $p|nm$, we must have $v_p(nm) \geq 2$, and note that if there exists identical prime in b, then the corresponding prime would be shifted into the prime factor of a, and the new b would also remains squarefree.

In the integer ring, we are interested in the squarefull pairs and also the consecutive squarefull triple. Using the Erdo Consecutive Squarefull Conjecture, we know that there

do not exist three consecutive squarefull integers. That is, the system: $n-1, n, n+1$ cannot all be squarefull. Where the pairs of consecutive squarefull numbers exist infinitely often like $(8,9),(288,289),(675,676)$.

Using the famous Pell's Equation: $x^2 - y^3 = 1 \implies x^2 - 1 = y^3$, consider the squarefull number be Dy^2, and the other be x^2, we would have the equation $x^2 - Dy^2 = 1$,which if uses the continued fractions method, would guaranteed to have infinitely many integer solutions (x_k, y_k).

However, there are no consecutive integers have ever been found to satisfy the consecutive triple:$n-1 = a_1^2b_1^3, n = a_2^2b_2^3, n+1 = a_3^2b_3^3 \implies (n+1)+n+(n-1) = 2n \implies a_3^2b_3^3 + a_1^2b_1^3 = 2(a_2^2b_2^3)$.

Note that this conjecture is readily connected to abc conjecture in integer ring: for any $\varepsilon > 0$, there exists a constant $C(\varepsilon)$ such that for any three coprime positive integers satisfying $A + B = C$, we have $C < C(\varepsilon) \cdot \mathrm{rad}(ABC)^{1+\varepsilon}$ where rad is the product of the distinct prime factors of N. Suppose that we want to investigate the consecutive triple $n-1, n, n+1$ where $n-1, n+1$ are consecutive squarefull number,using a trick, define $A = n-1, B = 2, C = n+1$, since $A+B = C, (n-1)+2 = n+1$, assuming that n is even,C odd, $\gcd(A,B,C) = 1$, and that $\mathrm{rad}(ABC) = \mathrm{rad}(2 \cdot (n-1) \cdot (n+1)) \leq 2\mathrm{rad}(n-1)\mathrm{rad}(n+1)$, since $n-1, n+1$ are squarefull, we would have all prime factor appears at least squared, so consider that $X = p_1^{e_1} \cdots p_k^{e_k}, e_i \geq 2, \mathrm{rad}(X) = p_1 \cdot p_k \implies \mathrm{rad}(X)^2 = p_1^2 \cdots p_k^2 \leq p_1^{e_1} \cdots p_k^{e_k} = X$. Hence $\mathrm{rad}(X) \leq X^{\frac{1}{2}}$ for X squarefull. Applying to the abc term, $\mathrm{rad}(n-1) \leq (n-1)^{\frac{1}{2}} < (n+1)^{\frac{1}{2}}, \mathrm{rad}(n+1) \leq (n+1)^{\frac{1}{2}} \implies \mathrm{rad}(ABC) \leq 2 \cdot (n+1) \implies n+1 < C(\varepsilon)(2(n+1))^{1+\varepsilon}$, meaning that if the abc conjecture is true, the number of such pairs grows slower than N^{ϵ} for any $\epsilon > 0$.

Remark. Note that by applying the standard abc conjecture, the presence of the constant $C(\varepsilon)$ and the growth rate exponent $1+\varepsilon$ only allows us to conclude that the number of such triplets is strictly finite.Unlike in function field, the integer abc conjecture cannot unconditionally rule out the existence of exceptional triplets, nor can it provide an effective bound to determine if any small examples exist.

7.2 Squarefull Polynomial

Definition 7.2.

Theorem 7.2 (Polynomial Erdos-Mollin-Walsh Conjecture in Characteristic 0)**.**

Proof. Suppose there exists consecutive squarefull triples ,let $f(t) \in K[t]$ be a non-constant polynomial. Let $A(t) = f(t) - 1, M(t) = f(t), C(t) = f(t) + 1, B(t) = 2$,using the same trick in the integer part, $A + B = f(t) - 1 + 2 = f(t) + 1$.Note that $\deg B = 0$, and that $f(t)$ is a non-constant, $\deg A = \deg f = \deg C$. They are clearly coprime since that any common polynomial factor: Suppose that $g|A, g|C \implies g|(C-A) \implies g|2$,

which would be a contradiction. Now for the radical part, if a non-constant polynomial g is squarefull, then all irreducible factor$P$ would have value $v_P(f) \geq 2$.So, by $\text{rad}(g)$ be the product of distinct irreducible polynomial of g, using same technique, that $\text{rad}(g)^2 = P_1^2 \cdots P_t^2 \leq P_1^{e_1} \cdots P_t^{e_t} = g \implies 2\deg(\text{rad(g)}) \leq \deg g \implies \deg(\text{rad(g)}) \leq \frac{1}{2}\deg g$. So now,

$$\text{rad}(ABC) = \text{rad}(2 \cdot (f(t)-1) \cdot (f(t)+1)) = \text{rad}(f-1) \cdot \text{rad}(f+1)$$

.Since in assumption that A, B, C are all squarefull polynomials, using above inequality, $\deg(\text{rad}(ABC)) \leq \deg \text{rad}(f-1) + \deg \text{rad}(f+1) \leq \frac{1}{2}\deg(f-1) + \frac{1}{2}\deg(f+1) = \frac{1}{2}\deg(f) + \frac{1}{2}\deg(f) = \deg f$, by Mason-Stother Theorem, $\max\{\deg A, \deg B, \deg C\} = \max\{\deg f, 0\} = \deg f$ .Hence

$$\deg f = \max\{\deg A, \deg B, \deg C\} \leq \deg(\text{rad}(ABC)) - 1 \leq \deg f - 1$$

Contradiction. Hence we show that there would be no consecutive squarefull triple in polynomial. □

Remark. However, on characteristic p, it behaves differently due to the recurrence property.

Note that for any polynomial of the form $f(t) = g(t)^p$, is automatically squarefull since $p \geq 2$.Hence, if we select $A(t) = (g(t)-1)^p, B(t) = g(t), C(t) = (g(t)+1)^p$ for arbitrary polynomial $g(t)$, we would have $A(t) = g(t)^p - 1 = B(t) - 1 = g(t)^p - 2 = C(t) - 2$ be consecutive polynomial of squarefull.

7.3 Squarefull Polynomial and Arithmetic Function in finite field

Notice that there are infinitely many Squarefull consecutive triple polynomial in the arbitrary characteristic p field, the next interesting area would be the counting problem regarding the squarefull polynomials and how they are distributed by using the arithmetic functions (d, σ, Φ) introduced in previous chapter.

Recall that in finite field $\mathbb{F}_q$, for the polynomial ring $\mathbb{F}_q[t]$, we also have infinitely many squarefull consecutive triple in it by the same construction. And that the three formula

$$\Phi(f) = |f| \prod_{P|f}(1 - |P|^{-1}) \ , \ d(f) = \prod_{i=1}^{t}(1+e_i) \ , \ \sigma(f) = \prod_{i=1}^{t} \frac{|P_i|^{e_i+1} - 1}{|P_i| - 1}$$

First, we define the useful lemma.

Lemma 7.3. *Suppose that $\mathbb{F}_q[T]$ is polynomial ring over finite field $\mathbb{F}_q$ with q elements. Any squarefull polynomial can be uniquely expressed in the form $f(T) = g(T)^2h(T)^3$ where $g, h \in \mathbb{F}_q[T]$ and $h(T)$ must be square-free polynomial. All f, g, h are monic.*

Proof. The proof follows the similar pattern as the one in the integer ring. Consider the prime decomposition of the polynomial $f(T) = P_1^{e_1} \cdots P_t^{e_t}$ where $P_1, ..., P_t$ are monic irreducible polynomials and $e_i \geq 2$ for all i. Similarly using division algorithm of polynomial, if e_i are even, then $e_i = 2k_i$, and if e_i is odd, then $e_i = 2k_i + 3$ where $k_i \geq 0$. Hence all polynomial exponent can be written out as $e_i = 2k_i + 3r_i$ where $k_i \geq 0, r_i = 0, 1 \implies f(T) = \prod_{j=1}^{t} P_j^{2k_j+3r_j} = (P_j^{k_j})^2(P_j^{r_j})^3$, treat $g(T) = \prod P_j^{k_j}, h(T) = \prod P_j^{r_j}$, we obtain the result. □

Proposition 7.4. *Define $S(n)$ to be the number of monic squarefull polynomial of degree n. Then*

$$S(n) = \sum_{\substack{2k+3l=n \\ l\geq 2}} q^{k+l}(1 - q^{-1}) + \sum_{\substack{2k+3l=n \\ l=0}} q^k + \sum_{\substack{2k+3l=n \\ l=1}} q^{k+1}$$

Proof. Recall that in previous chapters, we have found a formula for squarefree polynomial: suppose b_n is the number of square-free monic of degree n, then $b_n = q^n(1 - q^{-1})$ for $n > 1$ and $b_n = q$ for $n = 1$.Also note that there are total of q^n different polynomials of degree n. Using above lemma, suppose f is monic squarefull polynomial, $f = g^2h^3$ where h is square-free monic polynomial, g is monic polynomial. Then $\deg f = 2\deg g + 3\deg h$. Let $\deg f = n, \deg g = k, \deg h = l, (k, l \geq 1)$,we got $n = 2k + 3l$. So we should find all pair (k, l) such that it satisfies the equation $n = 2k + 3l$.Hence $S(n) = \sum_{\substack{2k+3l=n \\ l\geq 2}} q^k \cdot b_l + \sum_{\substack{2k+3l=n \\ l=1}} q^k \cdot q + \sum_{\substack{2k+3l=n \\ l=0}} q^k \cdot 1$ and the result follows immediately. □

Now, after we find our analytic function $S(n)$, to find the associated Dirichlet series,

Proposition 7.5. *Suppose that we define $h(f) = \begin{cases} 1 & \text{if } f \text{ is squarefull} \\ 0 & \text{otherwise} \end{cases}$, then $H(n) = S(n)$ where $H(n) := \sum_{\substack{f \text{ monic} \\ \deg f = n}} h(f)$*

Proof. Since

$$\begin{aligned} H(n) = \sum_{\substack{f \text{ monic} \\ \deg f = n}} h(f) &= \#\{f \in \mathbb{F}_q[T] | f \text{ monic and squarefull}\} \\ &= \#\{f \in \mathbb{F}_q[T] | f = g^2h^3, h \text{ is square-free monic}\} \\ &= \#\{(k,l) | n = 2k+3l, \deg g = k, \deg h = l, \deg f = n, h \text{ is square-free monic}\} \\ &= \sum_{2k+3l=n} \#\{(k,l) | \deg g = k, \deg h = l, \deg f = n, h \text{ is square-free monic}\} \\ &= S(n) \end{aligned}$$

if we count the element in the set. □

Proposition 7.6. *The associated Dirichlet Series of $h(f)$ is*

$$D_h(s) = \frac{\zeta_A(2s)\zeta_A(3s)}{\zeta_A(6s)}$$

Proof. First, it is clear that $h(f)$ is multiplicative: if f, g are coprime , they do not share any common prime factor, so $h(fg)$ depends on both $h(f)$ and also $h(g)$. Hence $h(fg) = h(f)h(g)$.Then by definition on the corresponding derichlet series of $h(f)$, $D_h(s) = \prod_P(\sum_{k=0}^{\infty} \frac{h(P^k)}{|P|^{ks}}) = \prod(\frac{h(P^0)}{|P|^0} + \frac{h(P)}{|P|^s} + \frac{h(P^2)}{|P|^{2s}} + \cdots) = \prod(1 + 0 + \frac{1}{|P|^{2s}} + \cdots) = \prod(1 + \frac{1}{|P|^{2s}}(1 + \frac{1}{|P|^s} + \frac{1}{|P|^{2s}} + \cdots)) = \prod(1 + \frac{|P|^{-2s}}{1-|P|^{-s}})$. Notice that the identity $1 + \frac{|P|^{-2s}}{1-|P|^{-s}} = \frac{1-|P|^{-s}+|P|^{-2s}}{1-|P|^{-s}}$. If we let $|P|^{-s} = a$, using simple algebra, $\frac{1-a+a^2}{1-a} = \frac{(1-a+a^2)(1+a)}{1-a^2} = \frac{1+a^3}{1-a^2}$. In order to use the formula $\prod_P(1 - |P|^{-s}) = \zeta_A(s)^{-1}$, we add one more term: $\frac{(1+a^3)(1-a^3)}{(1-a^2)(1-a^3)} = \frac{1-a^6}{(1-a^2)(1-a^3)}$. Hence we got

$$\begin{aligned} D_h(s) = \prod_P(\frac{1 - |P|^{-s} + |P|^{-2s}}{1 - |P|^{-s}}) &= \prod_P \frac{1 - |P|^{-6s}}{(1 - |P|^{-2s})(|P|^{-3s})} \\ &= \frac{\prod_P(1 - |P|^{-6s})}{(\prod_P(1 - |P|^{-2s}))(\prod_P(1 - |P|^{-3s}))} \\ &= \frac{\zeta_A(6s)^{-1}}{\zeta_A(2s)^{-1}\zeta_A(3s)^{-1}} \\ &= \frac{\zeta_A(2s)\zeta_A(3s)}{\zeta_A(6s)} \end{aligned}$$

□

Corollary 7.7. $\sum_{n=0}^{\infty} H(n)u^n = \frac{1-qu^6}{(1-qu^3)(1-qu^2)}$

Proof. Since by $\zeta_A(s) = \frac{1}{1-q^{1-s}}$. By using $u = q^{-s}$,$\zeta_A(2s) = \frac{1}{1-q^{1-2s}} = \frac{1}{1-qu^2}$. Similarly, we got $\zeta_A(3s) = \frac{1}{1-qu^3}, \zeta_A(6s) = \frac{1}{1-qu^6}$ and that $D_h(s) := \sum_{n=0}^{\infty} H(n)u^n = \frac{\zeta_A(2s)\zeta_A(3s)}{\zeta_A(6s)} = \frac{1-qu^6}{(1-qu^2)(1-qu^3)}$. □

Proposition 7.8.

$$\sum_{\substack{f \text{ monic} \\ \deg f=n}} h(f) := H(n) = \sum_{\substack{2k+3l=n \\ l\geq 2}} q^{k+l}(1-q^{-1}) + \sum_{\substack{2k+3l=n \\ l=0}} q^k + \sum_{\substack{2k+3l=n \\ l=1}} q^{k+1}$$

Proof. Since by above corollary we already have $\sum_{n=0}^{\infty} H(n)u^n = \frac{1-qu^6}{(1-qu^2)(1-qu^3)}$.

Now, since $\frac{1-qu^6}{(1-qu^2)(1-qu^3)} = (1-qu^6)(1-qu^2)^{-1}(1-qu^3)^{-1} = (1-qu^6)\sum_{k=0}^{\infty}(qu^2)^k \sum_{l=0}^{\infty}(qu^3)^l = \sum_{k,l\geq 0} q^{k+l}u^{2k+3l} - \sum_{k,l\geq 0} q^{k+l+1}u^{2k+3l+6}$. Hence we can write that

$$\begin{aligned}\sum_{n=0}^{\infty} H(n)u^n &= \sum_{k,l\geq 0} q^{k+l}u^{2k+3l} - \sum_{k,l\geq 0} q^{k+l+1}u^{2k+3l+6} \\ &= \sum_{\substack{2k+3l=n \\ k,l\geq 0}} q^{k+l}u^{2k+3l} - \sum_{\substack{2k+3l=n \\ l\geq 2}} q^{k+l-1}u^{2k+3(l)}\end{aligned}$$

Where here the second term we do transformation $l \to l-2$.Extract out only the coefficient, we have $H(n) = \sum_{\substack{2k+3l=n \\ l\geq 2}} \left(q^{k+l} - q^{k+l-1}\right) + \sum_{\substack{2k+3l=n \\ l=1}} q^{k+1} + \sum_{\substack{2k+3l=n \\ l=0}} q^k = S(n)$. □

Remark. From here, using Dirichlet Series, we can also get the general Sum formula for the squarefull polynomial, where we can then use average function to obtain the growth and also the average value of squarefull polynomial.

Onward, we would like to introduce some tools that distinguish the non-trivial squarefull polynomial from those infinitely many trivial polynomials.

Definition 7.3. if the squarefull polynomial $f(T)$ can be expressed as $f(T) = g(T)^p$, then we call it as trivial squarefull polynomials, otherwise, we say that it is non-trivial ones.

Proposition 7.9. *A monic polynomial $f(T) \in \mathbb{F}_q[T]$ is squarefull , then* $|\text{rad}(f)| \Big| \Phi(f)$

Proof. ($\implies$)

Suppose that $f(T)$ is a squarefull and that $f(T) = P_1^{e_1}\cdots P_t^{e_t}$ where $e_j \geq 2 \ \ \forall j = 1,2,...,t$.Then $\text{rad}(f) = P_1\cdots P_t$ and $|\text{rad}(f)| = |P_1\cdots P_t| = q^{\deg P_1}\cdots q^{\deg P_t} = |P_1|\cdots|P_t|$,$\Phi(f) = |P_1^{e_1}\cdots P_t^{e_t}|\prod_{j=1}^{t}|P_j|^{-1}(|P_j|-1) = |P_1|^{e_1-1}\cdots|P_t|^{e_t-1}\prod_{j=1}^{t}(|P_j|-1)$.Since $e_j \geq 2\,, \forall j = 1,...,t \implies e_j - 1 \geq 1 \forall j \implies |P_j| \Big| |P_j^{e_j-1}| \forall j = 1,...,t$, the result follows immediately.

□

Remark. Note the the other direction is not necessarily true:

Suppose that $|\mathrm{rad}(f)| \mid \Phi(f)$, then $|P_1|\cdots|P_t| \mid |P_1|^{e_1-1}\cdots|P_t|^{e_t-1}\prod_{j=1}^{t}(|P_j|-1)$, writing in other form, we have $q^{\sum \deg P_j} \mid q^{\sum(e_j-1)\deg P_j}\prod(q^{\deg P_j}-1)$. Notice that q is a prime number, then $q^{\deg P_j}-1$ must be coprime with q hence $q^{\sum \deg P_j}$, we have $\gcd(q^{\sum \deg P_j}, \prod_j(q^{\deg P_j}-1)) = 1$, then we can only have $q^{\sum \deg P_j} \mid q^{\sum(e_j-1)\deg P_j}$.which forces $\sum \deg P_j \leq \sum(e_j-1)\deg P_j$. However, we can see that e_j need not be greater than 2 for all j for the inequality to establish. Hence we establish a proposition with a more stronger bound.

Proposition 7.10. *Suppose that $\mathbb{F}_q[T]$ be finite field with characteristic p where $q = p^m$. If f is squarefull, then we would have*

$$v_p(\Phi(f)) = m(\deg(f) - \deg(\mathrm{rad}(f)))$$

Proof. Suppose that f is squarefull, considering the prime decomposition of $f = P_1^{e_1}\cdots P_t^{e_t}$, and where $\Phi(f) = |f|\prod_{j=1}^{t}(1-|P_j|^{-1}) = \prod_{j=1}^{t}|P_j|^{e_j-1}(|P_j|-1)$. So

$$\begin{aligned}
v_p(\Phi(f)) = v_p(\prod_{j=1}^{t}|P_j|^{e_j-1}(|P_j|-1)) &= \sum_{j=1}^{t} v_p(q^{(e_j-1)\deg(P_j)}) + \sum_{j=1}^{t} v_p(q^{\deg P_j}-1) \\
&= \sum_{j=1}^{t} v_p(p^{m(e_j-1)\deg(P_j)}) + \sum_{j=1}^{t} v_p(p^{m\deg P_j}-1) \\
&= \sum_{j=1}^{t} v_p(p^{m(e_j-1)\deg(P_j)}) \\
&= \sum_{j=1}^{t} m(e_j-1)\deg(P_j) \\
&= \sum_{j=1}^{t} me_j\deg P_j - \sum_{j=1}^{t} m\deg P_j \\
&= m(\deg f - \deg(\mathrm{rad}(f)))
\end{aligned}$$

□

Corollary 7.11. *Suppose that f is squarefull, then $v_p(\Phi(f)) \geq (m-1)\deg f$*

Proof. The result follows immediately when we apply $\deg \mathrm{rad}(f) \leq \deg f$. □

Proposition 7.12. *Let $g(T) \in \mathbb{F}_q[T]$ be an arbitrary monic polynomial, let $f = g^p$ be non-trivial squarefull polynomial. Then*

$$v_p(\Phi(f)) \geq m(p-1)\deg(g)$$

Proof. suppose that $f = g^p$ where g is arbitrary monic polynomial,let $g = Q_1^{h_1} \cdots Q_r^{h_r}$, since f is squarefull, by above proposition, $v_p(\Phi(f)) = m(\deg f - \deg(\mathrm{rad}(f))) = m(p \deg g - \deg \mathrm{rad}(g^p)) = m(p \deg g - \deg \mathrm{rad}(g))$. Now note that $\deg \mathrm{rad}(g) \leq \deg g \implies -\deg \mathrm{rad}(\mathrm{g}) \geq -\deg g \implies v_p(\Phi(f)) \geq m(p-1) \deg g$

□

Remark. Note that the above 2 propositions only help identify the squarefull polynomial (both trivial and non-trivial) in the finite field, but do no contribution from identifying non-trivial squarefull polynomial from the trivial ones.

Below, we introduce differentiation to help us further identify and filter out the trivial case.

Recall that if there are p-th power in the polynomial, the derivative goes 0.

Remark. Hence to test the non-trivial derivative, we can directly use derivative to check.

Note that the above Dirichlet Series involve the trivial squarefull polynomial too, to reduce it into only considering the non-trivial ones, we further define the function.

Definition 7.4. Define $\hbar(f) = \begin{cases} 1 & \text{if } f \text{ squarefull, and has no p-th power component} \\ 0 & \text{otherwise} \end{cases}$

Proposition 7.13. *The Dirichlet Series associated to $\hbar(f)$ is*

$$D_\hbar(s) = \frac{\zeta_A(2s)\zeta_A(3s)}{\zeta_A(6s)} \cdot \prod_P (1 - \frac{|P|^{-ps} + |P|^{-s(p+1)}}{1 + |P|^{-3s}})$$

Proof. Since it is clear that the function $\hbar$ is multiplicative, we then can use the Dirichlet series$D_\hbar(s) = \prod_P (\sum_{k=0}^\infty \frac{h(P^k)}{|P|^{ks}})$ again.

$$\begin{aligned} \sum_{k=0}^\infty \frac{h(P^k)}{|P|^{ks}} &= \frac{h(P^0)}{|P|^0} + \frac{h(P^1)}{|P|^s} + \frac{h(P^2)}{|P|^{2s}} + \cdots \frac{h(P^{p-1})}{|P|^{(p-1)s}} + \frac{h(P^p)}{|P|^{ps}} + \cdots \\ &= 1 + \frac{1}{|P|^{2s}} + \cdots + \frac{1}{|P|^{(p-1)s}} \\ &= 1 + \frac{1}{|P|^s} + \cdots + \frac{1}{|P|^{(p-1)s}} - \frac{1}{|P|^s} \\ &= \frac{1 - |P|^{-ps}}{1 - |P|^{-s}} - \frac{1}{|P|^s} \end{aligned}$$

Define $a = |P|^{-s}$ again,

$$\begin{aligned}\frac{1-a^p}{1-a} - a = \frac{1-a+a^2-a^p}{1-a} &= (\frac{1-a+a^2}{1-a}) \cdot (1 - \frac{a^p}{1-a+a^2}) \\ &= \frac{1+a^3}{1-a^2}(1 - \frac{a^p}{1-a+a^2}) \\ &= \frac{(1+a^3)(1-a^3)}{(1-a^2)(1-a^3)}(1 - \frac{a^p}{1-a+a^2}(\frac{1+a}{1+a})) \\ &= \frac{1-a^6}{(1-a^2)(1-a^3)} \cdot (1 - \frac{a^p+a^{p+1}}{1+a^3})\end{aligned}$$

Hence,

$$D_\hbar(s) = \prod_P \frac{1-a^6}{(1-a^2)(1-a^3)} \cdot \prod_P (1 - \frac{a^p+a^{p+1}}{1+a^3}) = \frac{\zeta_A(2s)\zeta_A(3s)}{\zeta_A(6s)} \cdot \prod_P (1 - \frac{|P|^{-ps} + |P|^{-s(p+1)}}{1+|P|^{-3s}})$$

□

Proposition 7.14. *The sum formula of $\hbar(f)$ can be expressed as*

$$H_\hbar(n) = \sum_{\substack{f \text{ monic} \\ \deg f = n}} h(f) = (\frac{G(\frac{1}{\sqrt{q}})}{2} + \frac{(-1)^n G(\frac{-1}{\sqrt{q}})}{2}) q^{\frac{n}{2}} + O(q^{n/p+\varepsilon})$$

Proof. Note that by proof in previous proposition, $D_\hbar(s) = \frac{\zeta_A(2s)\zeta_A(3s)}{\zeta_A(6s)} \cdot \prod_P (1 - \frac{|P|^{-ps}+|P|^{-s(p+1)}}{1+|P|^{-3s}})$, and that $\frac{\zeta_A(3s)}{\zeta_A(6s)} = \prod_P \frac{1-|P|^{-6s}}{1-|P|^{-3s}} = \prod_P (1 + |P|^{-3s})$, hence $D_\hbar(s) = \zeta_A(2s) \cdot \prod_P (1 + |P|^{-3s}) \cdot \prod_P (1 - \frac{|P|^{-ps}+|P|^{-s(p+1)}}{1+|P|^{-3s}}) = \zeta_A(2s) \cdot \prod_P [(1 + |P|^{-3s})(1 - \frac{|P|^{-ps}+|P|^{-s(p+1)}}{1+|P|^{-3s}})] = \zeta_A(2s) \cdot \prod_P (1 + |P|^{-3s} - |P|^{-ps} - |P|^{-s(p+1)})$.

We only focus on $p \geq 3$ case, since $p = 2$ is trivial.Set $u = q^{-s}$, then $|P|^{-s} = q^{-s \deg P} = u^{\deg P}$. Define $G(u) = \prod_P (1 + |P|^{-3s} - |P|^{-ps} - |P|^{-s(p+1)}) = \prod_P (1 + u)$, then

$$D_\hbar(u) = \frac{1}{1-qu^2} \cdot G(u)$$

Consider dividing both side by u^{n+1}, then we got

$$\frac{D(u)}{u^{n+1}} = \frac{H(0)}{u^{n+1}} + \frac{H(1)}{u^n} + \cdots + \frac{H(n)}{u} + H(n+1) + H(n+2)u + \cdots$$

Recall that by Cauchy's Residue Theorem, the integral of u^k around a loop around the origin is 0 for all exponent except $k = -1$, where when $k = 1$, the integral equals

$2\pi i$.Hence, define $\gamma = \{u \in \mathbb{C} || u| = \rho\}, 0 < \rho < q^{-1/2}$,anticlockwise.

$$\oint_\gamma \frac{D(u)}{u^{n+1}}du = 2\pi i \cdot H(n) \implies H(n) = \frac{1}{2\pi i} \cdot \oint_\gamma \frac{D(u)}{u^{n+1}}du$$

By Residue Theorem,

$$H(n) = -\sum \text{Res}_{u=\alpha} \frac{D(u)}{u^{n+1}} = -\sum \text{Res}_{u=\alpha} \frac{G(u)}{(1-qu^2)u^{n+1}}$$

Since 2 poles are $u_1 = \frac{1}{\sqrt{q}}, u_2 = \frac{-1}{\sqrt{q}}$.Now,

$$\text{Res}_{u=\frac{1}{\sqrt{q}}} \frac{G(u)}{(1-qu^2)u^{n+1}} = \lim_{u=\frac{1}{\sqrt{q}}} (u-\frac{1}{\sqrt{q}}) \frac{G(u)}{u^{n+1}(\frac{1}{\sqrt{q}}-u)(\frac{1}{\sqrt{q}}+u)q} = \frac{G(\frac{1}{\sqrt{q}})}{-q \cdot q^{-\frac{n+1}{2}} \cdot \frac{2}{\sqrt{q}}} = -G(\frac{1}{\sqrt{q}}) \cdot \frac{q^{\frac{n}{2}}}{2}$$

Similarly,

$$\text{Res}_{u=\frac{-1}{\sqrt{q}}} \frac{G(u)}{(1-qu^2)u^{n+1}} = \lim_{u=\frac{-1}{\sqrt{q}}} (u-\frac{-1}{\sqrt{q}}) \frac{G(u)}{u^{n+1}(\frac{1}{\sqrt{q}}-u)(\frac{1}{\sqrt{q}}+u)q} = \frac{G(\frac{-1}{\sqrt{q}})}{q \cdot (-q)^{-\frac{n+1}{2}} \cdot \frac{2}{\sqrt{q}}} = -(-1)^n G(\frac{-1}{\sqrt{q}})$$

Now suppose that if we enlarge the radius such that $q^{-1/2} < R < q^{-1/p}$,

$$H(n) = -\text{Res}_{u=\frac{1}{\sqrt{q}}} \frac{D(u)}{u^{n+1}} - \text{Res}_{u=u=\frac{-1}{\sqrt{q}}} \frac{D(u)}{u^{n+1}} + \frac{1}{2\pi i} \cdot \oint_{\gamma_R} \frac{D(u)}{u^{n+1}}du$$

Define $\varepsilon(R) = \frac{1}{2\pi i} \cdot \oint_{\gamma_R} \frac{D(u)}{u^{n+1}}du$. Since $|\varepsilon(R)| = \left| \frac{1}{2\pi i} \cdot \oint_{\gamma_R} \frac{D(u)}{u^{n+1}}du \right| \leq \frac{1}{2\pi} \cdot \max_{|u|=R} \left| \frac{D(u)}{u^{n+1}} \right| \cdot 2\pi R = \frac{\max_{|u|=R} \left| D(u) \right|}{R^n}$.Since on the circle $|u| = R, R > q^{-1/2}$, the term $|1-qu^2|$ is bounded away from zero by a constant M_1 , also,consider $a_P = -u^{pd} + u^{3d} - u^{(p+1)d}$,when $u \to q^{-1/p}, |a_P| \leq C_1 \cdot |u|^{pd}$ for some constant C_1.Since $\sum_P |a_P| = \sum_{d=1}^{\infty} N_q(d) \cdot |a_P| \leq \sum N_q(d) \cdot C_1 |u|^{pd}$ where $N_q(d)$ is defined to be the number of monic irreducible polynomials of degree d,by PNT,$N_q(d) \leq \frac{q^d}{d} \implies \sum_P |a_P| \leq C_1 \sum \frac{q^d}{d} |u|^{pd} = C_1 \sum \frac{1}{d} (q \cdot |u|^p)^d$,where the series converges if $q \cdot |u|^p < 1 \iff |u| < q^{-1/p}$,hence the product $G(u)$ converges absolutely for any complex number u where $|u| < q^{-1/p}$. So when we pick the radius $R = q^{-1/p-\varepsilon}$ for small $\varepsilon > 0$,$G(u)$ is analytic and bounded by some constant M_2 along the circle. Hence $\max_{|u|=R} \left| D(u) \right| \leq \frac{M_2}{M_1} = M$.Then $|\varepsilon(R)| \leq \frac{M}{R^n} = \frac{M}{(q^{-1/p-\varepsilon})^n} = M \cdot q^{\frac{n}{p}+\varepsilon n}$,choose new ε such that $|\varepsilon(R)| = O(q^{n/p+\varepsilon})$.Combining the results, we got

$$H(n) = (\frac{G(\frac{1}{\sqrt{q}})}{2} + \frac{(-1)^n G(\frac{-1}{\sqrt{q}})}{2}) q^{\frac{n}{2}} + \varepsilon(R) = (\frac{G(\frac{1}{\sqrt{q}})}{2} + \frac{(-1)^n G(\frac{-1}{\sqrt{q}})}{2}) q^{\frac{n}{2}} + O(q^{n/p+\varepsilon})$$

□

Chapter 8

Problems

8.1 Davenport's Inequality

Problem 8.1.1 (Davenport's inequality)**.** Let $f, g \in K[t]$ be non-constant polynomials such that $f^3 - g^2 \neq 0$ where K is characteristic 0. Then

$$\deg(f^3 - g^2) \geq \frac{1}{2}\deg f + 1$$

Proof. Before proving the davenport inequality, i should raise a lemma beforehand to shorten the proof.

Lemma 8.1. *Suppose that $p, q \in K[t]$ are two distinct no-constant polynomial, then*

$$\deg(p - q) \geq \deg p - N_0(p) - N_0(q)$$

Proof. Let $k(f)$ be leading coefficient of a polynomial f. If $\deg p \neq \deg q$ or $k(p) \neq k(q)$, then by mason stother theorem, $\deg(p-q) \geq \deg p \geq \deg p - N_0(p) - N_0(q) + 1$ since $N_0(p), N_0(q) \geq 1$. Next, suppose that $\deg p = \deg q, k(p) = k(q)$. If $(p, q) = 1$,since $p + p - q = q$ then by Mason Stothers again, $\deg p \leq N_0(pq(p-q)) - 1 = N_0(p) + N_0(q) + N_0(p-q) - 1 \leq \deg p + \deg q + \deg(p-q) - 1$. Otherwise, suppose that $(p, q) = r$, then $(\frac{p}{r}, \frac{q}{r}) = 1$, so by Mason Stothers again, $\deg(\frac{p}{r}) \leq N_0(\frac{p}{r} \cdot \frac{q}{r} \cdot \frac{p-q}{r}) - 1 \leq N_0(\frac{p}{r}) + N_0(\frac{q}{r}) + N_0(\frac{p-q}{r}) - 1 \leq N_0(\frac{p}{r}) + N_0(\frac{q}{r}) + \deg(\frac{p-q}{r}) - 1$. Hence we got $N_0(p) + N_0(q) \geq N_0(\frac{p}{r}) + N_0(\frac{q}{r}) + N_0(r) \geq N_0(\frac{p}{r}) + N_0(\frac{q}{r})$ □

Then, put $\deg f = m, \deg g = n$. If $3m \neq 2n$, then $\deg(f^3 - g^2) \geq \deg f^3 = 3m \geq \frac{1}{2}m + 1$.If $3m = 2n$, or $\exists r, m = 2r, n = 3r$, by Lemma, $\deg(f^3 - g^2) \geq \deg f^3 - N_0(f^3) - N_0(g^2) + 1 \geq 6r - 2r - 3r + 1 \geq r + 1 = \frac{1}{2}\deg f + 1$. □

8.2 $\mathbf{X}^m - Y^n = 1$

Problem 8.2.1. Consider the following equation

$$X^m - Y^n = 1$$

,where $\gcd(X, Y) = 1, X, Y$ are both non constant, $m, n \in \mathbb{Z}_+$,then $\frac{1}{m} + \frac{1}{n} > 1$ if equation has solution .

Proof. using the mason-stothers theorem, define $a = X^m, b = -Y^n$ directly we have

$$\max(\deg(X^m), \deg(-Y^n), 0) \leq N_0(-X^m Y^n) - 1$$

In other word, we have

$$m \deg(X) \leq N_0(-X^m Y^n) - 1$$

$$n \deg(Y) \leq N_0(-X^m Y^n) - 1$$

Now, since that $N_0(-X^m Y^n) = N_0(X^m Y^n) \leq \deg(X) + \deg(Y)$ Hence

$$m \deg(X) \leq \deg(X) + \deg(Y) - 1$$

$$n \deg(Y) \leq \deg(X) + \deg(Y) - 1$$

$\Longrightarrow$

$$m \deg(X) < \deg(X) + \deg(Y) \quad , \quad n \deg(Y) < \deg(X) + \deg(Y)$$

$$\frac{\deg(X)}{\deg(X) + \deg(Y)} < \frac{1}{m} \quad , \quad \frac{\deg(Y)}{\deg(X) + \deg(Y)} < \frac{1}{n}$$

So

$$\frac{1}{m} + \frac{1}{n} > 1$$

which shows that the equation have solution only when at least one of the exponent must be 1.Meaning that if $\frac{1}{m} + \frac{1}{n} \leq 1$, then there are no solutions in the equation.

For example, if $m = 1$, then we reduces into $X = Y^n + 1$ □

Remark. Note that if we consider function field of characteristic p instead, and that $m = n = p$ then the above equation is trivial:

$$X^p + Y^p = 1 \implies (X + Y)^p = 1 \implies X + Y = 1.$$

And same procedure for the equation

$$X^m - Y^n = Z^l$$

where if the solution exists ,then $\frac{1}{m} + \frac{1}{n} + \frac{1}{l} > 1$.

8.3 Pythagorean Equation

Problem 8.3.1. we focus on Pythagorean Theorem in function field char 0. Consider $f, g, h \in K[t]$, where K is a field char 0 .

$$X^2 + Y^2 = Z^2$$

Proof. Recall that in integer ring $\mathbb{Z}$, we can find the general solution of this diophantine equation:

Theorem 8.2. *All positive integer solutions of* $x^2 + y^2 = z^2$ *with* $\gcd(x, y, z) = 1$ *are given by*

$$x = n^2 - m^2, y = 2mn, z = n^2 + m^2$$

or

$$x = 2mn, y = n^2 - m^2, z = n^2 + m^2$$

for some integers n, m.

To investigate whether there are similar solution set or more, we should separate it into cases:

Case 1 (char 0):

Consider $X^2 + Y^2 = Z^2$ where $\gcd(X, Y, Z) = 1$.Suppose either X is monic polynomial. Then $X^2 = (Z-Y)(Z+Y)$. Note that we would have $\gcd(Z-Y, Z+Y) = 1$. Suppose not, let $Q \in K[t], Q = \gcd(Z-Y, Z+Y)$, then suppose that are monic irreducible $q \in K[t], q|Q$, then $q|Z-Y, q|Z+Y \implies q|2Z, q|2Y$,as $2 \in K^* \implies q|Z, Y$.Now, $q|(Z-Y), q|(Z+Y) \implies q^2|X^2. \implies q|X^2$.By $K[t]$ is an UFD, using Euclid Lemma for polynomial, $q|X$. Contradict that X, Y, Z are relatively coprime. Hence such q does not exist. Hence $\gcd(Z-Y, Z+Y) = 1$. Since $X^2 = (Z-Y)(Z+Y)$ and $\gcd(Z-Y, Z+Y) = 1$,using prime factorization for $X^2 = P_1^{2e_1} \cdots P_t^{2e_t}$ again, $Z-Y, Z+Y$ are correspond to different sub-partition primes$P_{i_1}^{2e_{i_1}} \cdots P_{i_l}^{2e_{i_l}}, P_{i_{l+1}}^{2e_{l+1}} \cdots P_{i_r}^{2e_{i_r}}$ where $\{1, ..., l, l+1, ..., r\} \subseteq \{1, ..., t\}$. So

$$Z - Y = c \cdot M^2 \ , \ Z + Y = c^{-1} \cdot N^2$$

where $c \in K^*, \gcd(M, N) = 1$. So we conclude that

$$Z = \frac{c \cdot M^2 + c^{-1} \cdot N^2}{2} \ , \ Y = \frac{c^{-1} \cdot N^2 - c \cdot M^2}{2} \ , \ X = MN$$

To show that the only solution set are above, we use stereographic projection over the field of rational functions $K(t)$.

First note that $Z \neq 0$. If $Z = 0, X^2 + Y^2 = 0 \implies X = Y = 0$ since characteristic 0 field. If $\sqrt{-1} \in K, (X - iY)(X + iY) = Z^2$(then by $\gcd(X, Y, Z) = 1$, this cannot happen.). So, dividing the original equation $X^2 + Y^2 = Z^2$ by Z^2 yields:

$$(\frac{X}{Z})^2 + (\frac{Y}{Z})^2 = 1$$

Let $x = \frac{X}{Z}, y = \frac{Y}{Z}$, where $x, y \in K(t)$. This defines the unit circle $x^2 + y^2 = 1$ over the field $K(t)$. We choose a fixed base point on this circle ,namely $P = (-1, 0)$. Any other point (x, y) on the circle can be uniquely determined by the straight line passing through P and (x, y).Let the slope of this line be $m \in K(t)$. The equation of the line is given by

$$y = m(x + 1)$$

Substituting this line equation into the circle equation $x^2 + y^2 = 1$ gives:

$$x^2 + m^2(x + 1)^2 = 1 \implies (x^2 - 1) + m^2(x + 1)^2 = 0$$

$$(x - 1)(x + 1) + m^2(x + 1)^2 = 0$$

Note that the intersecting point we choosing are distinct from $P : (x \neq 1)$. So $x + 1 \neq 0$:

$$(x - 1) + m^2(x + 1) = 0 \implies x(1 + m^2) = 1 - m^2$$

So we can express x :

$$x = \frac{1 - m^2}{1 + m^2}$$

Also y :

$$y = m(x + 1) = \frac{2m}{1 + m^2}$$

where $m \in K(t)$, so we can express it as a quotient of 2 coprime polynomial $m = \frac{M}{N}, M, N \in K[t], \gcd(M, N) = 1$. So in terms of polynomial,

$$x = \frac{1 - (\frac{M}{N})^2}{1 + (\frac{M}{N})^2} = \frac{N^2 - M^2}{N^2 + M^2}$$

$$y = \frac{2(\frac{M}{N})}{1 + (\frac{M}{N})^2} = \frac{2MN}{N^2 + M^2}$$

Now since that $\gcd(X, Y, Z) = 1$ and $\gcd(N^2 - M^2, 2MN, N^2 + M^2) = 1$,

define $A = N^2 - M^2, B = 2MN, C = N^2 + M^2$, by stereographic equation, we have $x = \frac{A}{C}, y = \frac{B}{C} \implies \frac{X}{Z} = \frac{A}{B}, \frac{Y}{Z} = \frac{B}{C}$.From above we already have $\gcd(A, C) = 1, \gcd(B, C) = 1$. So $X \cdot C = A \cdot Z \implies C|(A \cdot Z)$. But $K[t]$ is UFD, by Euclid Lemma , $C|A$ or $C|Z$, since $\gcd(A, C) = 1 \implies C|Z \implies Z = d \cdot C$. Hence by

$X \cdot C = A \cdot d \cdot C \implies X = d \cdot A$, and also from $Y \cdot C = B \cdot Z = Bd \cdot C \implies Y = d \cdot B$, we obtained

$$X = d(N^2 - M^2) \ , \ Y = d(2MN) \ , \ Z = d(N^2 + M^2)$$

Hence using stereographic parameterization, we know that all solutions from the Pythagorean Equation must be the solution to the line parametrized equation in the unit circle of the rational field of fraction, where the map $\Phi : \{(x, y)|x^2 + y^2 = 1\}\backslash\{(-1, 0)\} \longrightarrow K(t)$ by $\Phi(x, y) \mapsto \frac{y}{x+1}$ is a bijection.

Case 2 (char $p, p = 2$):
Note that

$$X^2 + Y^2 = (X + Y)^2 \implies (X + Y)^2 = Z^2 \implies X + Y = Z$$

Hence any choice of polynomials $x, y \in K[t]$ yields a valid solution.

Case 2 (char $p, p > 2$):
Note that using if $f \in K[t]$, $(f^p)^2 = (f^2)^p$, so

$$(X^p)^2 + (Y^p)^2 = (X^2)^p + (Y^2)^p = (X^2 + Y^2)^p = (Z^2)^p = (Z^p)^2$$

So if (x, y, z) is the solution of the Pythagorian equation, (x^p, y^p, z^p) would also be so.Besides,since for any arbitrary polynomial $g(t) \in K[t]$, $(g(t)^p)^2 = (g(t)^2)^p$, hence if the GCD requirement of X, Y, Z are dropped, there would be infinitely more solution by considering any arbitrary polynomials.

Remark. Note that the stereographic parameterization can also be applied into the characteristic p case, hence secure the existence of solutions.

□

8.4 $X^k + Y^k = P^n$

Problem 8.4.1. Consider K be a field and $P(t) \in K[t]$ be a fixed irreducible polynomial. Find all pair of integers (n, k) with $n \geq 1, k \geq 2$ for which there exist non-constant, relatively prime polynomial $X(t), Y(t) \in K[t]$ satisfying the equation

$$P(t)^n = X(t)^k + Y(t)^k$$

Proof.

Lemma 8.3 (Euclid Lemma for polynomial)**.** *Suppose that P is an irreducible polynomial such that $P|f(x)g(x)$, then P divides at least one of the factors.*

Case A

Note that we define $d = \max(\deg X, \deg Y)$. Since $X^k + Y^k = P^n$, P^n must have degree less than d. Hence

$$\deg(P^n) \leq \max(\deg X^k, \deg Y^k)$$

$$n \deg P \leq kd \implies \deg P \leq \frac{kd}{n}$$

Apply Mason-Stother Theorem to it,

$$\max(\deg X^k, \deg Y^k, \deg P^n) \leq N_0(XYP) - 1 \implies kd \leq N_0(XYP) - 1 \leq \deg X + \deg Y + \deg P - 1$$

Since $\deg X, \deg Y \leq d$, so

$$kd \leq 2d + \deg P - 1 \leq 2d + \frac{kd}{n} - 1$$

$$d(k - 2 - \frac{k}{n}) \leq -1$$

Consider for the case $\deg X > \deg Y$:

if char(K) = 0, then solution exists $\iff k = 2, n \geq 1$. Assume that $k \geq 3$, let $a = X^k, b = Y^k, c = P^n$. Since $\gcd(X, Y) = 1$, then the three polynomials are pairwise coprime. Using Mason-Stothers theorem:

$$\max(\deg(X^k), \deg(Y^k), \deg(P^n)) \leq N_0(X^k Y^k P^n) - 1 = N_0(XYP) - 1 \leq \deg(X) + \deg(Y) + \deg(P) - 1$$

Since that the number of distinct roots do not depend on the power and they are all pairwise coprime, with that number of roots cannot exceed the degree of polynomial.

Hence we get the following inequalities:

$$k \deg(X) \leq \deg(X) + \deg(Y) + \deg(P) - 1$$

Case 1

Suppose that for $k \geq 3$ even, by above, WLOG assume that $\deg X > \deg Y \implies \deg P^n = \deg(X^k + Y^k) = k \deg X$, Hence we have the equality $\deg P^n = \deg X^k \implies n \deg P = k \deg X$. So using above inequality derived from Mason- Stothers Theorem: $k \deg X \leq \deg X + \deg Y + \deg P - 1$, So, $k \deg X < 2 \deg X + \frac{k}{n} \deg X - 1 \implies (k - 2 - \frac{k}{n}) \deg X < -1$. Since $\deg X \geq 1$, then

$$k - 2 - \frac{k}{n} < -1 \implies k(1 - \frac{1}{n}) < 1$$

If $n \geq 2$, then clearly $k < 2$. So for $k \geq 3$, no valid solution.

If $n = 1$,

$$P(t) = X(t)^k + Y(t)^k$$

we would have $-2 \deg X < -1 \implies \deg X > \frac{1}{2} \implies \deg X \geq 1$. But note that in our assumption that we assume $\deg X > \deg Y$. Hence $\deg P = \deg(X^k + Y^k) = k \deg X \implies \deg P = k \deg X$

Case 2

Assume that $k \geq 3$ is odd, since $P^n = X^k + Y^k$ and k is odd,,then $P^n = (X + Y)(\sum_{j=0}^{k-1}(-1)^j X^{k-1-j} Y^j)$

,define $Q(X, Y) = \sum_{j=0}^{k-1}(-1)^j X^{k-1-j} Y^j$, since P is irreducible polynomial, by Euclid Lemma, since $P | P^n := (X + Y)Q(X, Y)$, then at least one could happen:$P|(X + Y)$ or $P|Q(X, Y)$.To show that only one could happens,we show $\gcd(X+Y, Q(X, Y)) = 1$:Consider $X \equiv -Y \mod (X + Y)$,

$$Q(-Y, Y) \equiv \sum_{j=0}^{k-1}(-1)^j(-1)^{k-1-j} Y^{k-1-j} Y^j = \sum_{j=0}^{k-1}(-1)^{k-1} Y^{k-1} = k(-1)^{k-1} Y^{k-1} \mod (X+Y)$$

Since k odd, $k - 1$ even, hence $(-1)^{k-1} = 1 \implies Q(-Y, Y) \equiv \sum_{j=0}^{k-1} Y^{k-1} = kY^{k-1} \mod (X + Y)$.If $d | \gcd(Q, X + Y) \implies d|Q, d|(X + Y) \implies d|kY^{k-1}$, so $\gcd(X + Y, Q(X, Y))|kY^{k-1}$.Now, suppose that $d|X + Y, Q$, then $d | \gcd(X + Y, Q) \implies d|kY^{k-1}$.Suppose that $X + Y, Q$ shares a non-constant polynomial $Q(t)$. Then $Q|d \implies Q|kY^{k-1}$.Since k is a non-zero constant in the field, $Q(t)|Y^{k-1}$.By Euclid Lemma for polynomial, $Q(t)|Y$.Note that we already know that $Q(t)|(X + Y)$,with $Q(t)|Y$, we would have the conclusion $Q(t)|X \implies \gcd(X, Y) > 1$.Contradiction. Hence we must have $\gcd(X + Y, Q(X, Y)) = 1$. As they are coprime, P can only divide one of them .Now, since X, Y are not constant ,$\deg(X + Y) \geq 1 \implies \deg P^n > \deg(Q(X, Y))$, we

cannot have $P^n|Q(X,Y)$.Here suppose $P^k|Q,\ k<n$, then since $P^n=(X+Y)Q \implies P^{n-k}|(X+Y) \implies P|(X+Y) \implies \gcd(X+Y,Q)\neq 1$, contradiction .So we only have $P|(X+Y)$. So by Lte in char 0, since $P|(X+Y), P\not|X, P\not|Y$, so

$$v_P(X^k+Y^k)=v_P(X+Y) \implies v_P(P^n)=v_P(X+Y) \implies v_P(X+Y)=n$$

meaning that $P^n||(X+Y) \implies \deg(X+Y)\geq \deg(P^n)=n\deg(P)$. Using the above equation $P^n=X^k+Y^k$, $n\deg P=\deg(X^k,Y^k)$, since for $k\geq 3$, $\deg(X+Y)<\deg(X^k+Y^k)$, then $n\deg P\leq \deg(X+Y)<\deg(X^k+Y^k)\leq n\deg P \implies n\deg P<n\deg P$ Contradiction arise.

So for $k\geq 3$, there are no valid solution.

Case 3

For $k=2$, using the inequality: $(k-2-\frac{k}{n})\deg(X)<-1 \implies \frac{2}{n}\deg X\geq 1 \implies \deg X>\frac{n}{2}$, So the solution exists as long as the degree of X is at least more than half of n.Hence the solution pair is (n,k) where $n<2\deg X$.

Case B

Consider the case of char p:

Suppose that $k=m\cdot p^a$ for some integer $m, \gcd(m,p)=1$, then

$$P(t)^n=X^{mp^a}+Y^{mp^a}=(X^m+Y^m)^{p^a}$$

Since $K[t]$ is a UFD and $P(t)$ is irreducible polynomial,by fundamental theorem of UFD, $A(t):=X^m+Y^m$ can be broken down into product of distinct irreducible polynomial factors, say $A(t)=c\cdot Q_1^{e_1}\cdots Q_t^{e_t}$. Since $A(t)^{p^a}=P(t)^n$, which forces $P(t)^n=c^{p^a}\cdot Q_1^{e_1p^a}\cdots Q_t^{e_tp^a}$, since P is irreducible, then $t=1$ and $P(t)^n=c^{p^a}Q_1^{p^ae_1}=Q_1^{p^ae_1} \implies P=Q_1, n=p^ae_1$,then using $\gcd(m,p)=1$,

$$X^m+Y^m=P^{e_1}$$

where e_1 is integer $\geq 1$and $n=n'\cdot p^a$ if we let $e_1=n'$ to obtain the result

$$X^m+Y^m=P^{n'}$$

Remark. Since for $(P^{n'})^{p^a}=A^{p^a} \implies (P^{n'})^{p^a}-A^{p^a}=0 \implies (P^{n'}-A)^{p^a}=0 \implies P^{n'}-A=0$by $K[t]$ is integer domain(no zero divisor).

Case 1

If we choose $m=1(k=p)$, then

$$P(t)^{n'}=X+Y$$

there are infinitely many coprime solution since we can pick any non-constant polynomial $Y(t)$ such that $\gcd(Y,P)=1$ and $\deg Y < \deg(P^{n'})$ and define $X(t)=P(t)^{n'}-Y(t) \implies \gcd(X,Y)=\gcd(P^{n'}-Y,Y)=\gcd(P^{n'},Y)=1$.

Case 2

Suppose that $m>1$,$\gcd(m,p)>1$, then let $m=m'p$, where $\gcd(m',p)=1$, then

$$X^{m'p}+Y^{m'p}=(X^{m'}+Y^{m'})^p=P^{n'}$$

By similar argument above, since $K[t]$ is UFD, and $P(t)$ is irreducible, then $X^{m'}+Y^{m'}$ must be a pure power of $P(t)$, hence let $n'=n''p$, it is again reduced to

$$X^{m'}+Y^{m'}=P^{n''}$$

Inductively, we can reduce the equation such that the residue exponent of m is coprime to $p(\gcd(m,p)=1)$.

Case 3

Now if $\gcd(m,p)=1, m>1$,

Case 3.1

If $m\geq 3$,there is no solution

First, it is clear that if $n'=m$, by Fermat Last Theorem, we can directly conclude that there are no solution.

for the case of $n'\neq m$,

we should first state the Mason-Stothers Theorem in characteristic p case:

Theorem 8.4. *Suppose that $f,g,h\in K[t]$ are pairwise relatively prime, and f,g,h cannot all have formal derivatives equal to zero. Then*

$$\max(\deg f,\deg g,\deg h)\leq N_0(fgh)-1$$

Remark. Note that a polynomial's derivative is zero in characteristic $p \iff$ it is a perfect p-th power.

Proof. ($\implies$) Assume $f'(t)=0$, let $f(t)=\sum a_n t^n$.By using the formal derivative rule, the derivative is $f'(t)=\sum n a_n t^{n-1}$. Since $f'(t)=0$,every coefficient in this new polynomial must be identically zero: $na_n=0,\ \forall n\geq 0$. In field of characteristic p, the integer $n \mod p$ is evaluated. Hence for each term $na_n=0$ one of 2 conditions must be satisfied:

1: The coefficient $a_n=0$

2: The integer n is a multiple of p. Meaning $n=kp,\ \exists k\in\mathbb{Z}$

Since any non-vanishing term must have an exponent n that is multiple of p, we can rewrite $f(t)$ by substituting $n = kp$: $f(t) = \sum a_{kp}t^{kp} = \sum a_{kp}(t^p)^k$.Define $g(t) = \sum a_{kp}t^k$,we have $f(t) = g(t^p) \implies f(t) \in K[t^p]$ is a perfect p-th power.

($\Longleftarrow$)Assume $f(t)$ is a perfect p-th power, meaning $f(t) = g(t^p)$ for some polynomial $g(t) = \sum b_k t^k$. Then, $f(t) = \sum b_k(t^p)^k = \sum b_k t^{pk}$, differentiating $f(t)$ with respect to t yields $f'(t) = \sum(pk)b_k t^{pk-1}$,since we are in characteristic p, the coefficient of each term $(pk)b_k = 0 \implies f'(t) = 0$ □

As m is no longer a multiple of the characteristic p, the algebraic derivatives of X^m, Y^m do not automatically vanish unless they are a perfect p-th powers. Suppose that X, Y are perfect p-th power,say $X = X_1^p, Y = Y_1^p$,then we can simply reduce the equation $X^m + Y^m = (X_1^m + Y_1^m)^p = P^{n'}$, which send us back into case 2. Suppose not, we assume without loss of generality that not all terms are p-th powers.Then, at least 1 derivative is non-zero by remark and we can apply the mason-stothers theorem. By Mason-Stothers Theorem, we obtained the same degree constraint as in characteristic 0(WLOG assume that $\deg X < \deg Y$):since $\deg X^m \leq N_0(XYP) - 1 \leq \deg X + \deg Y + \deg P - 1 < 2\deg X + \deg P - 1$. Since $P^{n'} = X^m + Y^m \implies \deg P^{n'} = \deg X^m + \deg Y^m = m\deg X \implies m\deg X < 2\deg X + \frac{m}{n'}\deg X - 1$.Hence we have

$$(m - 2 - \frac{m}{n'})\deg X < -1$$

For $n' = 1$, the equation reduces to

$$X^m + Y^m = P$$

Let $d = \max(\deg X, \deg Y) \geq 1$, by Mason Stother theorem,

$$\max(\deg X^m, \deg Y^m, \deg P) \leq N_0(XYP) - 1$$

Note that since $\deg P \leq md$, then $\max(\deg X^m, \deg Y^m, \deg P) = md$.So

$$md \leq \deg X + \deg Y + \deg P - 1 \leq 2d + \deg P - 1 \implies d(m-2) \leq \deg P - 1 \implies d \leq \frac{\deg P - 1}{m - 2}$$

Which shows that if $\deg P \leq m - 2$, no valid solutions can exist and solution could possibly exists only if the fixed irreducible polynomial P has sufficiently high degree relative to m.

.Since $\deg X \geq 1, n' \geq 1$, this inequality is false for all $m \geq 3$. So no solution for Case 3.1.

Case 3.2.1 if $m = 2, p \neq 2$, and K contains square root of -1.

then the equation becomes

$$X^2 + Y^2 = P^{n'}$$

If K contains the root $i, -i$, then we can factor it into

$$(X + iY)(X - iY) = P^{n'}$$

Since P is irreducible, then we first check the gcd of $(X + iY), (X - iY)$.Define $\gcd(X - iY, X + iY)$, then any divisor $d|g$,$d|(X + iY - X + iY), d|(X + iY + X - iY) \implies d|(2X), d|(2iY)$. Note that $p \neq 2$, then the constant $2, 2i$ are invertible, hence $d|X, d|Y \implies \gcd(X, Y) \neq 1$. Hence we conclude $\gcd(X + iY, X - iY) = 1$. Since their product is $P^{n'}$, and P is irreducible,the UFD of $K[t]$ makes that $P^{n'}|(X - iY)$ or $P^{n'}|(X + iY)$.Hence

$$X + iY = c_1 P^{n'}, X - iY = c_2, \;\; c_1 c_2 = 1, c_1, c_2 \in K^*$$

or

$$X + iY = c_2, X - iY = c_1 P^{n'}$$

which solving the linear system, we got $\begin{cases} X = \frac{c_1 P^{n'} + c_2}{2} \\ Y = \frac{c_1 P^{n'} - c_2}{2i} \end{cases}$ or $\begin{cases} X = \frac{c_1 P^{n'} + c_2}{2} \\ Y = \frac{c_2 - c_1 P^{n'}}{2i} \end{cases}$

(we can test that both $X(t), Y(t)$ are non-constant polynomial and also $\gcd(X, Y)$ $= \gcd(\frac{c_1 P^{n'} + c_2}{2}, \frac{c_1 P^{n'} - c_2}{2i}) = \gcd(c_1 P^{n'} + c_2, c_1 P^{n'} - c_2))$

Any common factor of these two expressions must divide their sum $(2c_1 P^{n'})$ and their difference $(2c_2)$. Because $2c_2$ is a non-zero element of the ground field K, the common factor is a constant.Thus, a valid coprime solution exists for every $n' \geq 1$. Translating this back to the original exponents n and k, we have $k = 2p$, and $n = n'p$. This yields valid solution pairs $(n, k) = (n'p, 2p)$ for any integer $n' \geq 1$.

Case 3.2.2

if $m = 2, p \neq 2$, and K does not contain square root of -1.Note that e $P(t)$ is irreducible in $K[t]$, it may either remain irreducible or split in $L[t]$,where the quadratic field extension $L = K(i)$, where $i^2 = -1$.

Case 3.2.2.1[$P(t)$ remains irreducible in $L[t]$]

Suppose that on UFD $L[t]$, we factor exactly in the above case

$$(X + iY)(X - iY) = P^{n'}$$

Since $X, Y \in K[t], \gcd(X, Y) = 1$ in $K[t]$,$\gcd(X, Y) = 1$ in $L[t]$ also. Since P remains

irreducible in $L[t]$, the unique factorization of UFD guarentees that

$$X + iY = c_1 P^{n'} \ , \ X - iY = c_2$$

for constant $c_1, c_2 \in L^*$. Expanding $X + iY = c_1 P^{n'}$, knowing that $P^{n'} \in K[t]$, the coefficient of polynomial $X + iY$ must match the scalar multiplication by c_1, in other word, suppose that $c_1 = a + ib, a, b \in K$. Then

$$X + iY = (a + bi)P^{n'} = aP^{n'} + ibP^{n'}$$

Hence $X = aP^{n'}, Y = bP^{n'}$. Note that $\gcd(X, Y) = \gcd(aP^{n'}, bP^{n'}) = P^{n'}$, contradict that $\gcd(X, Y) = 1$ since P is not constant. Hence we must need Y to be constant, which contradict that Y needs to be non-constant polynomial. Hence no solution exists if P remains irreducible.

Case 3.2.2.2[$P(t)$ is reducible in $L[t]$]

Since $P \in K[t]$, its factorization in the quadratic extension $L[t]$ must take the form of conjugate polynomials:

$$P(t) = Q(t)\bar{Q(t)}$$

where $Q(t) \in L[t]$ be irreducible and $\bar{Q(t)}$ is the complex conjugate: Note P can be factored in $L[t]$, hence take $Q|P$ in $L[t]$. Then, $P = Q \cdot H \implies \bar{P} = \bar{Q} \cdot \bar{H}$. But since $P \in K[t] \implies P = \bar{P}$in $L[t]$.Hence we obtained $P = \bar{Q} \cdot \bar{H} \implies \bar{Q}|P$.To show that $Q, \bar{Q}$ are distinct and also coprime, assume that $Q = \bar{Q}$, then $Q \in K[t]$ as all of the coefficient's conjugate must be itself.But we know that P si irreducible in $K[t]$, and Q is a factor of P in $K[t]$, contradiction. Also, they cannot have common factor d,the conjugate of that common factor $\bar{d}$is also a common factor of $Q, \bar{Q}$, hence $d \cdot \bar{d}|Q, d \cdot \bar{d}|\bar{Q} \implies d \cdot \bar{d}|\bar{Q}|P$ and $d \cdot \bar{d} \in K[t]$.So $Q, \bar{Q}$ is coprime polynomial. Now $Q\bar{Q}|P$ and since $\deg(Q\bar{Q}) = 2 \deg Q$,and P split into 2 irreducible factor in $L[t]$(since$[L : K] = 2$,has extension degree of 2),we can conclude that $Q\bar{Q} = P$.

Our factored equation in $L[t]$:

$$(X + iY)(X - iY) = Q^{n'}\bar{Q}^{n'}$$

By UFD of $L[t]$, since $\gcd(X + iY, X - iY) = 1$, the factors must partition the prime pieces. To make X, Y to evaluate back to real polynomials in $K[t]$,we must make

$$X + iY = c \cdot Q^{n'} \ , \ X - iY = \bar{c} \cdot \bar{Q}^{n'}$$

(or $X + iY = c_1 Q^{n'} \bar{Q}^{n'} = c_1 P^{n'}$, which fails by case 3.2.2.1). where $c \in L^*$. Hence,

$$X = \frac{cQ^{n'} + \bar{c}\bar{Q}^{n'}}{2} \ , \ Y = \frac{cQ^{n'} - \bar{c}\bar{Q}^{n'}}{2i}$$

To check that $X, Y \in K[t]$, we can consider the complex conjugate:

$$\bar{X} = \overline{(\frac{cQ^{n'} + \bar{c}\bar{Q}^{n'}}{2})} = \frac{cQ^{n'} + \bar{c}\bar{Q}^{n'}}{2} = X$$

$$\bar{Y} = \frac{\overline{cQ^{n'} - \bar{c}\bar{Q}^{n'}}}{2i} = \frac{cQ^{n'} - \bar{c}\bar{Q}^{n'}}{-2\bar{i}} = \frac{cQ^{n'} - \bar{c}\bar{Q}^{n'}}{2i} = Y$$

By construction, we know that $X(t), Y(t) \in K[t]$, also $n' \geq 1, \deg Q \geq 1$ to ensure X, Y are non-constant.

For the coprimality, suppose that $d \in K[t]$ is a common prime factor of X, Y, it must also divide both $X + iY = cQ^{n'}, X - iY = \bar{c}\bar{Q}^{n'}$ in $L[t]$(meaning that $d|cQ^{n'}, d|\bar{c}\bar{Q}^{n'}$). Since $Q, \bar{Q}$ are distinct and coprime, as d has coefficient in K, note that when $d|cQ^{n'}$ where Q is irreducible polynomial, then d must be build with only Q in it, with power less than or equal n' .So $Q|d$, and $\bar{Q}|\bar{d}$,by $d \in K[t]$, $d = \bar{d} \implies \bar{Q}|d \implies Q\bar{Q}|d \implies P|d$. Meaning that $P|d|(X + iY) := cQ^{n'} \implies P|cQ^{n'} \implies Q\bar{Q}|cQ^{n'} \implies \bar{Q}|cQ^{n'-1}$. However, $Q, \bar{Q}$ are coprime irreducible polynomials. Hence contradiction. So $P \nmid X+iY$. Which shows that X, Y are coprime for all integer $n' \geq 1$. So valid solution exit for every single $(n, k) = (n'p, 2p)$.

Case 4[$\text{char}(K) = 2$]

In the case where $\text{char}(K) = 2$, the number $2 = 1 + 1 = 0$, hence

$$(X + Y)^2 = X^2 + 2XY + Y^2 = X^2 + Y^2$$

and that

$$(X + Y)^{2^a} = X^{2^a} + Y^{2^a}$$

Hence consider $k = m \cdot 2^a$ where 2^a is the highest power of 2 divides k, m is odd.

$$P^n = X^{m \cdot 2^a} + Y^{m \cdot 2^a} = (X^m + Y^m)^{2^a}$$

Again, we can reduce it to $P^{n'} = X^m + Y^m$: Define $A(t) := X^m + Y^m \implies P^n = A^{2^a}$.Suppose that the prime factorization of $A(t) = c \cdot R_1^{n_1} \cdots R_t^{e_t}$,then $P^n = c^{2^a} R_1^{2^a e_1} \cdots R_t^{2^a e_t}$, since P is irreducible, then it forces $t = 1$ and $P = R_1 \implies c = 1, n = 2^a e_1$.So if we define $e_1 := n'$,we are left with $(P^{n'})^{2^a} = A^{2^a}$, and by $K[t]$ is integral domain again, we have $P^{n'} = A \implies P^{n'} = X^m + Y^m$.

Case 4.1[$m \geq 3$]

By $X^m + Y^m = (X+Y)Q(X,Y)$ again, $P^{n'} = (X+Y)Q(X,Y)$. Since X, Y are non-constant, coprime polynomials and $m \geq 3$, both factors $X+Y, Q$ are non-constant with degrees at least 1.Again by Euclid Lemma, P must divides at least 1 of the factors:

1. If $n' = 1$, $P = (X+Y)Q(X,Y)$, contradiction.
2. If $n' \geq 2$,again since $m \neq 0$ in $\text{char}(K) = 2$, $\gcd(X+Y, Q) = 1$, which would make $\gcd(X,Y) = P > 1$.

So no solution can exist for all $m \geq 3$.

Case 4.2$[m = 1]$ Then we have

$$P^{n'} = X + Y$$

Where there would be infinitely many coprime solutions if we set $X = P^{n'} - Y$.

Conclusion Summary

By combining the results gathered across the distinct characteristic branches, we completely classify all integer pairs (n, k) with $n \geq 1$ and $k \geq 2$ that admit non-constant, relatively prime polynomial solutions $X(t), Y(t) \in K[t]$ satisfying $P(t)^n = X(t)^k + Y(t)^k$:

1. **When** $\text{char}(K) = 0$**:** Valid solutions exist if and only if $k = 2$. For any integer $n \geq 1$, a coprime pair exists as long as the degree condition $\deg X \geq \frac{n}{2}$ is satisfied. Thus, the solution set is:

$$\{(n, 2) \mid, 2 \deg X > n \geq 1\}$$

2. **When** $\text{char}(K) = p > 2$**:** The solutions depend on the structure of the exponent k relative to the characteristic:
   - **Pure Characteristic Powers ($k = p^a$):** Valid solutions exist for all pairs of the form $(n, k) = (n'p^a, p^a)$ for any integer $n' \geq 1$ and $a \geq 1$.
   - **Twice the Characteristic ($k = 2p^a$):** Valid solutions exist for pairs of the form $(n, k) = (n'p^a, 2p^a)$ for any $n' \geq 1$ and $a \geq 1$, provided that $P(t)$ splits into two distinct conjugate factors over the quadratic field extension $K(\sqrt{-1})[t]$ (or if $\sqrt{-1} \in K$).
3. **When** $\text{char}(K) = 2$**:** The equation behaves under the rule $(X+Y)^2 = X^2 + Y^2$. The problem collapses entirely to linear sums raised to powers of 2. Valid solutions exist if and only if $k = 2^a$ for some integer $a \geq 1$. The corresponding solution set is:

$$\{(n'2^a, 2^a) \mid n' \geq 1, a \geq 1\}$$

No other pairs (n, k) yield valid polynomial solutions due to the rigid bounding limits established by the Mason-Stothers theorem.

□

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